

ZlibValidation

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1 ZlibValidation	1
1.1 Description	1
1.2 Usage	1
1.3 Development Diary	1
1.3.1 2025-01-27	1
1.3.2 2025-01-28	2
1.3.3 2025-02-01	2
1.3.4 2025-02-10	2
1.3.5 2025-02-11	2
1.3.6 2025-02-14	2
1.3.7 2025-02-15	3
1.3.8 2025-02-17	3
1.3.9 2025-02-18	3
1.3.10 2025-02-19	3
1.3.11 2025-02-20	3
1.3.12 2025-02-25	3
1.3.13 2025-02-26	3
1.3.14 2025-02-27	3
1.3.15 2025-02-28	4
1.3.16 2025-03-01	4
1.3.17 2025-03-07	4
1.3.18 2025-03-10	4
1.3.19 2025-03-14	4
1.3.20 2025-03-15	4
1.3.21 2025-03-16	5
1.3.22 2025-03-17	5
1.3.23 2025-03-18	5
1.3.24 2025-03-19	6
1.3.25 2025-03-21	6
1.3.26 2025-03-24	6
1.3.27 2025-03-25	6
1.3.28 2025-03-26	6
1.3.29 2025-03-27	7
1.3.30 2025-03-28	7
2 Hierarchical Index	9
2.1 Class Hierarchy	9
3 Class Index	11
3.1 Class List	11
4 File Index	13
4.1 File List	13

5 Class Documentation	15
5.1 AttributesIterator Class Reference	15
5.2 CellExtractor Class Reference	15
5.2.1 Detailed Description	15
5.2.2 Constructor & Destructor Documentation	16
5.2.2.1 CellExtractor()	16
5.2.3 Member Function Documentation	16
5.2.3.1 foundTargetCell()	16
5.2.3.2 handle()	16
5.2.4 Member Data Documentation	16
5.2.4.1 foundTarget_	16
5.2.4.2 targetCell_	17
5.3 CellPrinter Class Reference	17
5.3.1 Detailed Description	17
5.3.2 Constructor & Destructor Documentation	17
5.3.2.1 CellPrinter()	17
5.3.3 Member Function Documentation	18
5.3.3.1 handle()	18
5.3.4 Member Data Documentation	18
5.3.4.1 foundTarget_	18
5.3.4.2 out_	18
5.3.4.3 targetCell_	18
5.4 GroupsIterator Class Reference	18
5.4.1 Detailed Description	19
5.4.2 Constructor & Destructor Documentation	19
5.4.2.1 GroupsIterator()	19
5.4.2.2 ~GroupsIterator()	19
5.4.3 Member Function Documentation	19
5.4.3.1 end()	20
5.4.3.2 get()	20
5.4.3.3 next()	20
5.4.4 Member Data Documentation	20
5.4.4.1 err_	20
5.4.4.2 group_	20
5.4.4.3 groups_	21
5.5 LibAttribute Class Reference	21
5.5.1 Detailed Description	21
5.5.2 Constructor & Destructor Documentation	21
5.5.2.1 LibAttribute()	22
5.5.2.2 ~LibAttribute()	22
5.5.3 Member Function Documentation	22
5.5.3.1 getBoolean()	22

5.5.3.2 getFloat()	22
5.5.3.3 getInt()	22
5.5.3.4 getName()	23
5.5.3.5 getString()	23
5.5.3.6 getValues()	23
5.5.3.7 isComplex()	23
5.5.4 Member Data Documentation	23
5.5.4.1 attr_	23
5.5.4.2 err_	24
5.6 liberty_value_data Struct Reference	24
5.6.1 Detailed Description	24
5.6.2 Member Data Documentation	24
5.6.2.1 dim_sizes	24
5.6.2.2 dimensions	24
5.6.2.3 index_info	25
5.6.2.4 values	25
5.7 LibFile Class Reference	25
5.7.1 Detailed Description	26
5.7.2 Constructor & Destructor Documentation	26
5.7.2.1 LibFile()	26
5.7.2.2 ~LibFile()	27
5.7.3 Member Function Documentation	27
5.7.3.1 checkTimingArcMonotonicity()	27
5.7.3.2 modify()	28
5.7.3.3 mono()	28
5.7.3.4 parse()	28
5.7.3.5 read()	29
5.7.3.6 supercell()	30
5.7.3.7 verilog()	30
5.7.3.8 writeJsonToFile()	30
5.7.4 Member Data Documentation	31
5.7.4.1 basename_	31
5.7.4.2 err_	31
5.7.4.3 filename_	31
5.7.4.4 filepath_	31
5.7.4.5 jsonname_	32
5.7.4.6 lib_json_	32
5.7.4.7 libname_	32
5.7.4.8 logger_	32
5.7.4.9 loggename_	32
5.7.4.10 process_	32
5.7.4.11 temperature_	33

5.7.4.12 voltage_	33
5.8 LibGroup Class Reference	33
5.8.1 Detailed Description	33
5.8.2 Constructor & Destructor Documentation	33
5.8.2.1 LibGroup()	34
5.8.2.2 ~LibGroup()	34
5.8.3 Member Function Documentation	34
5.8.3.1 getAttrs()	34
5.8.3.2 getGroups()	34
5.8.3.3 getName()	34
5.8.3.4 getType()	35
5.8.4 Member Data Documentation	35
5.8.4.1 err_	35
5.8.4.2 group_	35
5.9 LibraryComparator Class Reference	35
5.9.1 Detailed Description	36
5.9.2 Constructor & Destructor Documentation	36
5.9.2.1 LibraryComparator()	36
5.9.3 Member Function Documentation	37
5.9.3.1 compareCell()	37
5.9.3.2 compareLut()	37
5.9.3.3 comparePin()	38
5.9.3.4 compareTimingArc()	39
5.9.3.5 generateReport()	39
5.9.4 Member Data Documentation	40
5.9.4.1 abstol_	40
5.9.4.2 comp_json_	40
5.9.4.3 comp_lib_path_	40
5.9.4.4 ref_json_	40
5.9.4.5 ref_lib_path_	41
5.9.4.6 reltol_	41
5.10 ModuleRewriter Class Reference	41
5.10.1 Detailed Description	42
5.10.2 Constructor & Destructor Documentation	42
5.10.2.1 ModuleRewriter()	42
5.10.3 Member Function Documentation	42
5.10.3.1 handle() [1/2]	42
5.10.3.2 handle() [2/2]	42
5.10.4 Member Data Documentation	42
5.10.4.1 cellName_	43
5.10.4.2 depth_	43
5.10.4.3 inputPins_	43

5.10.4.4 instance_count_	43
5.10.4.5 logger_	43
5.10.4.6 moduleName_	43
5.10.4.7 outputPins_	44
5.11 si2drExprT Struct Reference	44
5.11.1 Detailed Description	44
5.11.2 Member Data Documentation	44
5.11.2.1 b	44
5.11.2.2 d	45
5.11.2.3 i	45
5.11.2.4 left	45
5.11.2.5 right	45
5.11.2.6 s	45
5.11.2.7 type	45
5.11.2.8	46
5.11.2.9 valuetype	46
5.12 si2OID Struct Reference	46
5.12.1 Detailed Description	46
5.12.2 Member Data Documentation	46
5.12.2.1 v1	46
5.12.2.2 v2	47
5.13 ValuesIterator Class Reference	47
5.13.1 Detailed Description	47
5.13.2 Constructor & Destructor Documentation	47
5.13.2.1 ValuesIterator()	48
5.13.2.2 ~ValuesIterator()	48
5.13.3 Member Function Documentation	48
5.13.3.1 end()	48
5.13.3.2 next()	48
5.13.4 Member Data Documentation	48
5.13.4.1 bool_	48
5.13.4.2 err_	49
5.13.4.3 exprp_	49
5.13.4.4 float_	49
5.13.4.5 int_	49
5.13.4.6 str_	49
5.13.4.7 values_	49
5.13.4.8 vtype_	50
5.14 VerilogVisitor Class Reference	50
5.14.1 Detailed Description	50
5.14.2 Constructor & Destructor Documentation	50
5.14.2.1 VerilogVisitor()	51

5.14.3 Member Function Documentation	51
5.14.3.1 handle() [1/6]	51
5.14.3.2 handle() [2/6]	51
5.14.3.3 handle() [3/6]	51
5.14.3.4 handle() [4/6]	51
5.14.3.5 handle() [5/6]	52
5.14.3.6 handle() [6/6]	52
5.14.4 Member Data Documentation	52
5.14.4.1 depth_	52
5.14.4.2 inTargetModule_	52
5.14.4.3 targetCell_	52
6 File Documentation	53
6.1 include/compare.hpp File Reference	53
6.1.1 Typedef Documentation	53
6.1.1.1 json	53
6.2 compare.hpp	54
6.3 include/iterators.hpp File Reference	54
6.4 iterators.hpp	54
6.5 include/json_utils.hpp File Reference	55
6.5.1 Typedef Documentation	56
6.5.1.1 json	56
6.5.2 Function Documentation	56
6.5.2.1 generateCellJson()	56
6.5.2.2 generateLutJson()	57
6.5.2.3 generatePinJson()	57
6.5.2.4 generatePowerJson()	58
6.6 json_utils.hpp	59
6.7 include/lib_attribute.hpp File Reference	59
6.8 lib_attribute.hpp	59
6.9 include/lib_file.hpp File Reference	60
6.9.1 Typedef Documentation	60
6.9.1.1 json	60
6.10 lib_file.hpp	60
6.11 include/lib_group.hpp File Reference	61
6.12 lib_group.hpp	61
6.13 include/si2dr_liberty.h File Reference	61
6.13.1 Macro Definition Documentation	64
6.13.1.1 LONG_DOUBLE	65
6.13.1.2 SI2_ARGS	65
6.13.1.3 SI2_LONG_MAX	65
6.13.1.4 SI2_LONG_MIN	65

6.13.1.5 SI2_ULONG_MAX	65
6.13.1.6 SI2DR_FALSE	65
6.13.1.7 SI2DR_MAX_FLOAT32	66
6.13.1.8 SI2DR_MAX_FLOAT64	66
6.13.1.9 SI2DR_MAX_INT32	66
6.13.1.10 SI2DR_MAX_STRING_LEN	66
6.13.1.11 SI2DR_MIN_FLOAT32	66
6.13.1.12 SI2DR_MIN_FLOAT64	66
6.13.1.13 SI2DR_MIN_INT32	67
6.13.1.14 SI2DR_TRUE	67
6.13.2 Typedef Documentation	67
6.13.2.1 si2BooleanT	67
6.13.2.2 si2DoubleT	67
6.13.2.3 si2drAttrComplexValdT	67
6.13.2.4 si2drAttrIdT	67
6.13.2.5 si2drAttrSIdT	68
6.13.2.6 si2drAttrTypeT	68
6.13.2.7 si2drBooleanT	68
6.13.2.8 si2drDefinIdT	68
6.13.2.9 si2drDefinesIdT	68
6.13.2.10 si2drErrorT	68
6.13.2.11 si2drExprT	68
6.13.2.12 si2drExprTypeT	69
6.13.2.13 si2drFloat32T	69
6.13.2.14 si2drFloat64T	69
6.13.2.15 si2drGroupIdT	69
6.13.2.16 si2drGroupsIdT	69
6.13.2.17 si2drInt32T	69
6.13.2.18 si2drIterIdT	70
6.13.2.19 si2drMessageHandlerT	70
6.13.2.20 si2drNamesIdT	70
6.13.2.21 si2drObjectIdT	70
6.13.2.22 si2drObjectsIdT	70
6.13.2.23 si2drObjectTypeT	70
6.13.2.24 si2drSeverityT	71
6.13.2.25 si2drStringT	71
6.13.2.26 si2drValuesIdT	71
6.13.2.27 si2drValueTypeT	71
6.13.2.28 si2drVoidT	71
6.13.2.29 si2FloatT	71
6.13.2.30 si2IteratorIdT	72
6.13.2.31 si2LongT	72

6.13.2.32 si2ObjectIdT	72
6.13.2.33 si2StringT	72
6.13.2.34 si2TypeT	72
6.13.2.35 si2VoidT	72
6.13.3 Enumeration Type Documentation	72
6.13.3.1 si2BooleanEnum	72
6.13.3.2 si2drAttrTypeT	73
6.13.3.3 si2drErrorT	73
6.13.3.4 si2drExprTypeT	74
6.13.3.5 si2drObjectTypeT	74
6.13.3.6 si2drSeverityT	74
6.13.3.7 si2drValueTypeT	75
6.13.3.8 si2TypeEnum	75
6.13.4 Function Documentation	75
6.13.4.1 get_function()	75
6.13.4.2 liberty_destroy_value_data()	76
6.13.4.3 liberty_get_element()	76
6.13.4.4 liberty_get_values_data()	76
6.13.4.5 main_liberty_parser()	76
6.13.4.6 print_version()	76
6.13.4.7 SI2_ARGS() [1/41]	76
6.13.4.8 SI2_ARGS() [2/41]	76
6.13.4.9 SI2_ARGS() [3/41]	77
6.13.4.10 SI2_ARGS() [4/41]	77
6.13.4.11 SI2_ARGS() [5/41]	77
6.13.4.12 SI2_ARGS() [6/41]	77
6.13.4.13 SI2_ARGS() [7/41]	77
6.13.4.14 SI2_ARGS() [8/41]	77
6.13.4.15 SI2_ARGS() [9/41]	77
6.13.4.16 SI2_ARGS() [10/41]	78
6.13.4.17 SI2_ARGS() [11/41]	78
6.13.4.18 SI2_ARGS() [12/41]	78
6.13.4.19 SI2_ARGS() [13/41]	78
6.13.4.20 SI2_ARGS() [14/41]	78
6.13.4.21 SI2_ARGS() [15/41]	78
6.13.4.22 SI2_ARGS() [16/41]	78
6.13.4.23 SI2_ARGS() [17/41]	79
6.13.4.24 SI2_ARGS() [18/41]	79
6.13.4.25 SI2_ARGS() [19/41]	79
6.13.4.26 SI2_ARGS() [20/41]	79
6.13.4.27 SI2_ARGS() [21/41]	79
6.13.4.28 SI2_ARGS() [22/41]	79

6.13.4.29 SI2_ARGS() [23/41]	79
6.13.4.30 SI2_ARGS() [24/41]	80
6.13.4.31 SI2_ARGS() [25/41]	80
6.13.4.32 SI2_ARGS() [26/41]	80
6.13.4.33 SI2_ARGS() [27/41]	80
6.13.4.34 SI2_ARGS() [28/41]	80
6.13.4.35 SI2_ARGS() [29/41]	80
6.13.4.36 SI2_ARGS() [30/41]	80
6.13.4.37 SI2_ARGS() [31/41]	81
6.13.4.38 SI2_ARGS() [32/41]	81
6.13.4.39 SI2_ARGS() [33/41]	81
6.13.4.40 SI2_ARGS() [34/41]	81
6.13.4.41 SI2_ARGS() [35/41]	81
6.13.4.42 SI2_ARGS() [36/41]	81
6.13.4.43 SI2_ARGS() [37/41]	81
6.13.4.44 SI2_ARGS() [38/41]	82
6.13.4.45 SI2_ARGS() [39/41]	82
6.13.4.46 SI2_ARGS() [40/41]	82
6.13.4.47 SI2_ARGS() [41/41]	82
6.13.4.48 si2drDefaultMessageHandler()	82
6.13.4.49 si2drPIGetMessageHandler()	82
6.13.4.50 si2drPISetMessageHandler()	83
6.13.4.51 usage()	83
6.14 si2dr_liberty.h	83
6.15 include/verilog_utils.hpp File Reference	90
6.15.1 Function Documentation	91
6.15.1.1 getAST()	91
6.16 verilog_utils.hpp	91
6.17 include/version.h File Reference	92
6.17.1 Macro Definition Documentation	93
6.17.1.1 APP_AUTHOR	93
6.17.1.2 APP_CONTACT	93
6.17.1.3 APP_NAME	93
6.17.1.4 APP_VERSION	93
6.17.1.5 APP_VERSION_MAJOR	93
6.17.1.6 APP_VERSION_MINOR	94
6.17.1.7 APP_VERSION_PATCH	94
6.17.1.8 BUILD_TIMESTAMP	94
6.18 version.h	94
6.19 README.md File Reference	94
6.20 src/compare.cpp File Reference	94
6.21 compare.cpp	95

6.22 src/iterators.cpp File Reference	99
6.23 iterators.cpp	99
6.24 src/json_utils.cpp File Reference	100
6.24.1 Function Documentation	100
6.24.1.1 generateCellJson()	100
6.24.1.2 generateLutJson()	101
6.24.1.3 generatePinJson()	101
6.24.1.4 generatePowerJson()	102
6.24.1.5 generateTimingJson()	103
6.24.1.6 parseStringToVector()	103
6.25 json_utils.cpp	104
6.26 src/lib_attribute.cpp File Reference	107
6.27 lib_attribute.cpp	107
6.28 src/lib_file.cpp File Reference	107
6.29 lib_file.cpp	107
6.30 src/lib_group.cpp File Reference	115
6.31 lib_group.cpp	115
6.32 src/main.cpp File Reference	115
6.32.1 Function Documentation	116
6.32.1.1 compareLibFiles()	116
6.32.1.2 funcLibFile()	117
6.32.1.3 main()	117
6.32.1.4 monoCheckLibFile()	117
6.32.1.5 parseLibFile()	117
6.32.1.6 printInfo()	118
6.32.1.7 supercellLibFile()	119
6.32.1.8 verilogLibFile()	119
6.33 main.cpp	120
6.34 src/verilog_utils.cpp File Reference	125
6.34.1 Function Documentation	125
6.34.1.1 getAST()	126
6.35 verilog_utils.cpp	126

Chapter 1

ZlibValidation

1.1 Description

Command line tool to validate standard cell libraries in .lib format.

1.2 Usage

```
ZlibValidation
Usage: ./zlibvalidation [OPTIONS] [SUBCOMMAND]

Options:
  -h,--help          Print this help message and exit
  -v,--version       Display program version information and exit

Subcommands:
  parse              Parse the Liberty file and write JSON to a file
  mono              Check the monotonicity of timing arc values
  compare            Compare the comparison library against the reference library and report
                    differences
  supercell         Generate supercells for the given Liberty file
  zlibboost         ZlibBoost - Multi-threaded Library Processing Tool
  clear             Clear the log, JSON, map, markdown files in this directory
  verilog           Generate Verilog file for given Liberty file
  spice            Generate SPICE file for given Liberty file
  func             Check functional equivalence of two Liberty files or Verilog files
```

1.3 Development Diary

1.3.1 2025-01-27

- C++CMake libsi2dr_liberty.a
- CLI11
- spdlog

1.3.2 2025-01-28

- nlohmann/json JSON
- `version.h` CMake
- `--version --mode`
- `spdlog debug info`
- `LibFile` PVT
 - PVT
- C++ `#include`

1.3.3 2025-02-01

- `OOPCLibAttribute`
 - `char *std::string`
- `AttributesIterator`RAII

1.3.4 2025-02-10

- `GroupsIterator`
- `LibGroup`
- `srcincludeCMakeLists.txt`

1.3.5 2025-02-11

- `hppCMakeLists.txt`CMake
- `LibFile``jsonLibFile::writeJsonToFile``PVTcell_name``cell_footprint``cell_area`
- [x] voltage

1.3.6 2025-02-14

- `json_utils.cpp` JSON (lib) `generateCellJson`, `generatePinJson`, `generatePowerJson`, `generateLutJson` `hpp`
- [x] (timing)
- `parseStringToVector` Look Up Table (LUT)
- `LibFile` `lib_json_`(lib) JSON
- `ValuesIterator`
- `LibAttribute::isComplex()` `bool`
- `LibGroup::getName()` `std::string`
- `si2drGroupsIdT` `sub_groups` `lib_group.getGroups()``GroupsIterator`

1.3.7 2025-02-15

- generateTimingJson

1.3.8 2025-02-17

- begin()

1.3.9 2025-02-18

- LibFile::mono() values
- setupLogger()

1.3.10 2025-02-19

- mode == "mono" C++17 std::filesystem::exists() JSON JSON JSON
- LibFile::mono() cell_risecell_fallrise_transitionfall_transitiontiming LUT value

1.3.11 2025-02-20

- LibFile::mono() passed() failed() cell liberate_lv
- sl018_ff_3.96_-40.lib liberate_lv 205 out of 559 cells failed
- when

1.3.12 2025-02-25

- CLIL1 parsemono

1.3.13 2025-02-26

-
- Logger -h-v
- LibFile::mono() input_slew value tcbn65lpbc.lib liberate_lv

1.3.14 2025-02-27

- GDB
 - si2dr_liberty Liberty
 - si2dr_liberty
 - si2dr_liberty
 - strcmp

1.3.15 2025-02-28

- `LibFile` `readparsesi2dr` `parsemonosi2dr`
- [X] `si2drPIGetGroups` bug: WARNING: `si2drPIQuit`: `GetGroups` called 1 more times than `IterQuit`
- `CMakeLists.txt` `set (CMAKE_BUILD_TYPE Release)` GDB

1.3.16 2025-03-01

- `sub_groups_iter` `forsi2drPIQuit`
- `group_iter` `parseparseparse` `si2drPIQuit` `groups`
- `si2drPIQuit` `groups` `parse`
- `[]` `parse` `logPVT` `(parse 6s mono 1s)` `monolog`
- `LibFile`

1.3.17 2025-03-07

- `compare` JSON
- `[]`

1.3.18 2025-03-10

- `supercell` `LibFile::supercell`
- `zlibboost` `zlibboost`
-
- `clear` JSON

1.3.19 2025-03-14

- `tabulate/table` CMake `FetchContent`
- `compare.cpp` `compare.hpp` `LibraryComparator` JSON

1.3.20 2025-03-15

- `std::filesystem::path` `LibFile` `LibFile` `basename_filename_libname_↔`
`jsonname_ loggername_JSON`
- (scope) `LibFile::parse` `si2drPIQuit`
- `clear` `.map` `.md`
- `mono` JSON JSON
- `spdlog` `logger` `logger APP_NAME` `logger`
- `LibFile` `logger`
- `supercell` `mono`
- `LibraryComparator::generateReport()` `tabulate` `markdown`

1.3.21 2025-03-16

- `compare .md .txt .md`
- `LibraryComparator::generateReport()` `cell cell LibraryComparator::compareCell()`
`cell`
- `LibraryComparator::compareCell()` `cell cell cell comparePin cell`
- `LibraryComparator::comparePin()` `cell pin timing arc pin timing arc compareTimingArc`
`pin timing arc`
- `LibraryComparator::compareTimingArc()` `timing arc JSON timing arc "related_pin" timing`
`arc "cell_rise", "cell_fall", "rise_transition", "fall_transition" timing arc timing arc compareLut timing arc`
- `LibraryComparator::compareLut()` `compareValue JSON value JSON "index_1" "index_2"`
`LUT value (reltol) (abstol)`
- `cell_name pin_name timing_type related_pin arc_name`
- `LibraryComparator::compareValue()` `LibraryComparator::compareLut()`
-
- `abstol 0.002ns`

1.3.22 2025-03-17

- `verilog spice Verilog SPICE`
- `LibraryComparator`
 - `cell`
 - `cell Timing Delay`

1.3.23 2025-03-18

- | Cell Name | Data Type | Failed Count | Avg Diff | Avg Diff% | Max Diff | Max Diff% | Outliers |
- | Pin Name | Reference | Comparison | Diff | Diff % | Type | Arc Name | Row # | Index_1 | Column # | Index_2 | Note |
- `stdout`
- `si2drSimpleAttrGetBooleanValue`
- `LibFile::mono()` `min_pulse_width`
 - `"input_pins" cell`
 - `"timing_arcs"`
 - `"timing_type" "min_pulse_width" timing arc "rise_constraint" "fall_constraint" checkTimingArc↵`
`Monotonicity`
- `checkTimingArcMonotonicity when`
- `checkTimingArcMonotonicity`
- `checkTimingArcMonotonicity related_pin pin`

1.3.24 2025-03-19

- supercell1
- voltage std::round(voltage_ * 100) / 100.0

1.3.25 2025-03-21

- func Verilog

1.3.26 2025-03-24

- C++ Verilog Verilog slang (AST) CMakeLists slang API

1.3.27 2025-03-25

- verilog_utils.cpp verilog_utils.hpp VerilogVisitor Verilog AST
- slang
 - AND2D0 CMPE42D1 DFQD1
 - (UDP) tsmc_dff "Invalid instance declaration"
 - inputoutput
 -
 -

1.3.28 2025-03-26

- UDP tsmc_dff Entry
- slang::syntax::SyntaxRewriter CellExtractor slang::syntax::SyntaxPrinter::printFile() Verilog
- CellPrinter slang::syntax::SyntaxVisitor module.toString() Verilog
- LibFile::supercell() cell_names supercell
- supercell input_pins
- Doxygen Doxygen HTML LaTeX
- [] LaTeX

1.3.29 2025-03-27

- Git ****
- `LibFile::verilog`
 - `this->supercell()`
 - `.map std::pair<std::string, std::string> -> /`
 - `lib_json_ /`
 - Verilog instance 1 chain length
 - ANSI `fullModuleText`
 - `slang::syntax::SyntaxTree::fromText fullModuleText`
 - `ModuleRewriter` transform `handle()`
 - `slang::syntax::SyntaxPrinter::printFile(*tree)`
- `LibFile::verilog` ANSI

1.3.30 2025-03-28

-
- `ModuleRewriter::handle`(const `slang::syntax::ModuleDeclarationSyntax` &module) `insertAtBack(module.members, newDataNode) wire OP_i;`

Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

AttributesIterator	15
GroupsIterator	18
LibAttribute	21
liberty_value_data	24
LibFile	25
LibGroup	33
LibraryComparator	35
si2drExprT	44
si2OID	46
slang::syntax::SyntaxRewriter	
CellExtractor	15
ModuleRewriter	41
slang::syntax::SyntaxVisitor	
CellPrinter	17
VerilogVisitor	50
ValuesIterator	47

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

AttributesIterator	15
CellExtractor	15
CellPrinter	17
GroupsIterator	18
LibAttribute	21
liberty_value_data	24
LibFile	25
LibGroup	33
LibraryComparator	35
ModuleRewriter	41
si2drExprT	44
si2OID	46
ValuesIterator	47
VerilogVisitor	50

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

include/compare.hpp	53
include/iterators.hpp	54
include/json_utils.hpp	55
include/lib_attribute.hpp	59
include/lib_file.hpp	60
include/lib_group.hpp	61
include/si2dr_liberty.h	61
include/verilog_utils.hpp	90
include/version.h	92
src/compare.cpp	94
src/iterators.cpp	99
src/json_utils.cpp	100
src/lib_attribute.cpp	107
src/lib_file.cpp	107
src/lib_group.cpp	115
src/main.cpp	115
src/verilog_utils.cpp	125

Chapter 5

Class Documentation

5.1 AttributesIterator Class Reference

```
#include <iterators.hpp>
```

Collaboration diagram for AttributesIterator:

5.2 CellExtractor Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for CellExtractor:

Collaboration diagram for CellExtractor:

Public Member Functions

- [CellExtractor](#) (const std::string &targetCell)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)
- bool [foundTargetCell](#) () const

Private Attributes

- const std::string & [targetCell_](#)
- bool [foundTarget_](#)

5.2.1 Detailed Description

Definition at line 32 of file [verilog_utils.hpp](#).

5.2.2 Constructor & Destructor Documentation

5.2.2.1 CellExtractor()

```
CellExtractor::CellExtractor (
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 34 of file [verilog_utils.hpp](#).

5.2.3 Member Function Documentation

5.2.3.1 foundTargetCell()

```
bool CellExtractor::foundTargetCell ( ) const
```

Definition at line 364 of file [verilog_utils.cpp](#).

Here is the caller graph for this function:

5.2.3.2 handle()

```
void CellExtractor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 348 of file [verilog_utils.cpp](#).

5.2.4 Member Data Documentation

5.2.4.1 foundTarget_

```
bool CellExtractor::foundTarget_ [private]
```

Definition at line 40 of file [verilog_utils.hpp](#).

5.2.4.2 targetCell_

```
const std::string& CellExtractor::targetCell_ [private]
```

Definition at line 39 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- include/[verilog_utils.hpp](#)
- src/[verilog_utils.cpp](#)

5.3 CellPrinter Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for CellPrinter:

Collaboration diagram for CellPrinter:

Public Member Functions

- [CellPrinter](#) (const std::string &targetCell, std::ostream &out)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)

Private Attributes

- const std::string & [targetCell_](#)
- std::ostream & [out_](#)
- bool [foundTarget_](#)

5.3.1 Detailed Description

Definition at line 44 of file [verilog_utils.hpp](#).

5.3.2 Constructor & Destructor Documentation

5.3.2.1 CellPrinter()

```
CellPrinter::CellPrinter (
    const std::string & targetCell,
    std::ostream & out ) [inline], [explicit]
```

Definition at line 46 of file [verilog_utils.hpp](#).

5.3.3 Member Function Documentation

5.3.3.1 handle()

```
void CellPrinter::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 367 of file [verilog_utils.cpp](#).

5.3.4 Member Data Documentation

5.3.4.1 foundTarget_

```
bool CellPrinter::foundTarget_ [private]
```

Definition at line 53 of file [verilog_utils.hpp](#).

5.3.4.2 out_

```
std::ostream& CellPrinter::out_ [private]
```

Definition at line 52 of file [verilog_utils.hpp](#).

5.3.4.3 targetCell_

```
const std::string& CellPrinter::targetCell_ [private]
```

Definition at line 51 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- [include/verilog_utils.hpp](#)
- [src/verilog_utils.cpp](#)

5.4 GroupsIterator Class Reference

```
#include <iterators.hpp>
```

Collaboration diagram for GroupsIterator:

Public Member Functions

- [GroupsIterator](#) ([si2drGroupsIdT](#) groups, [si2drErrorT](#) &err)
- [~GroupsIterator](#) ()
- void [next](#) ()
- bool [end](#) ()
- [LibGroup](#) [get](#) ()

Private Attributes

- [si2drGroupsIdT](#) groups_
- [si2drGroupIdT](#) group_
- [si2drErrorT](#) & err_

5.4.1 Detailed Description

Definition at line 9 of file [iterators.hpp](#).

5.4.2 Constructor & Destructor Documentation

5.4.2.1 GroupsIterator()

```
GroupsIterator::GroupsIterator (  
    si2drGroupsIdT groups,  
    si2drErrorT & err )
```

Definition at line 3 of file [iterators.cpp](#).

5.4.2.2 ~GroupsIterator()

```
GroupsIterator::~GroupsIterator ( )
```

Definition at line 6 of file [iterators.cpp](#).

5.4.3 Member Function Documentation

5.4.3.1 end()

```
bool GroupsIterator::end ( )
```

Definition at line 9 of file [iterators.cpp](#).

Here is the caller graph for this function:

5.4.3.2 get()

```
LibGroup GroupsIterator::get ( )
```

Definition at line 11 of file [iterators.cpp](#).

Here is the caller graph for this function:

5.4.3.3 next()

```
void GroupsIterator::next ( )
```

Definition at line 8 of file [iterators.cpp](#).

Here is the caller graph for this function:

5.4.4 Member Data Documentation

5.4.4.1 err_

```
si2drErrorT& GroupsIterator::err_ [private]
```

Definition at line 20 of file [iterators.hpp](#).

5.4.4.2 group_

```
si2drGroupIdT GroupsIterator::group_ [private]
```

Definition at line 19 of file [iterators.hpp](#).

5.4.4.3 groups_

`si2drGroupsIdT GroupsIterator::groups_ [private]`

Definition at line 18 of file [iterators.hpp](#).

The documentation for this class was generated from the following files:

- [include/iterators.hpp](#)
- [src/iterators.cpp](#)

5.5 LibAttribute Class Reference

```
#include <lib_attribute.hpp>
```

Collaboration diagram for LibAttribute:

Public Member Functions

- [LibAttribute](#) ([si2drAttrIdT](#) attr, [si2drErrorT](#) &err)
- [~LibAttribute](#) ()
- `std::string` [getName](#) ()
- `bool` [isComplex](#) ()
- [si2drValuesIdT](#) [getValues](#) ()
- `long int` [getInt](#) ()
- `double` [getFloat](#) ()
- `std::string` [getString](#) ()
- `bool` [getBoolean](#) ()

Private Attributes

- [si2drAttrIdT](#) [attr_](#)
- [si2drErrorT](#) & [err_](#)

5.5.1 Detailed Description

Definition at line 8 of file [lib_attribute.hpp](#).

5.5.2 Constructor & Destructor Documentation

5.5.2.1 LibAttribute()

```
LibAttribute::LibAttribute (
    si2drAttrIdT attr,
    si2drErrorT & err )
```

Definition at line 3 of file [lib_attribute.cpp](#).

5.5.2.2 ~LibAttribute()

```
LibAttribute::~~LibAttribute ( )
```

Definition at line 5 of file [lib_attribute.cpp](#).

5.5.3 Member Function Documentation

5.5.3.1 getBoolean()

```
bool LibAttribute::getBoolean ( )
```

Definition at line 25 of file [lib_attribute.cpp](#).

Here is the caller graph for this function:

5.5.3.2 getFloat()

```
double LibAttribute::getFloat ( )
```

Definition at line 18 of file [lib_attribute.cpp](#).

Here is the caller graph for this function:

5.5.3.3 getInt()

```
long int LibAttribute::getInt ( )
```

Definition at line 16 of file [lib_attribute.cpp](#).

Here is the caller graph for this function:

5.5.3.4 getName()

```
std::string LibAttribute::getName ( )
```

Definition at line 7 of file [lib_attribute.cpp](#).

Here is the caller graph for this function:

5.5.3.5 getString()

```
std::string LibAttribute::getString ( )
```

Definition at line 20 of file [lib_attribute.cpp](#).

Here is the caller graph for this function:

5.5.3.6 getValues()

```
si2drValuesIdT LibAttribute::getValues ( )
```

Definition at line 14 of file [lib_attribute.cpp](#).

Here is the caller graph for this function:

5.5.3.7 isComplex()

```
bool LibAttribute::isComplex ( )
```

Definition at line 12 of file [lib_attribute.cpp](#).

5.5.4 Member Data Documentation

5.5.4.1 attr_

```
si2drAttrIdT LibAttribute::attr_ [private]
```

Definition at line 21 of file [lib_attribute.hpp](#).

5.5.4.2 err_

```
si2drErrorT& LibAttribute::err_ [private]
```

Definition at line 22 of file [lib_attribute.hpp](#).

The documentation for this class was generated from the following files:

- [include/lib_attribute.hpp](#)
- [src/lib_attribute.cpp](#)

5.6 liberty_value_data Struct Reference

```
#include <si2dr_liberty.h>
```

Public Attributes

- int [dimensions](#)
- int * [dim_sizes](#)
- LONG_DOUBLE ** [index_info](#)
- LONG_DOUBLE * [values](#)

5.6.1 Detailed Description

Definition at line 612 of file [si2dr_liberty.h](#).

5.6.2 Member Data Documentation

5.6.2.1 dim_sizes

```
int* liberty_value_data::dim_sizes
```

Definition at line 615 of file [si2dr_liberty.h](#).

5.6.2.2 dimensions

```
int liberty_value_data::dimensions
```

Definition at line 614 of file [si2dr_liberty.h](#).

5.6.2.3 index_info

`LONG_DOUBLE** liberty_value_data::index_info`

Definition at line 617 of file [si2dr_liberty.h](#).

5.6.2.4 values

`LONG_DOUBLE* liberty_value_data::values`

Definition at line 621 of file [si2dr_liberty.h](#).

The documentation for this struct was generated from the following file:

- [include/si2dr_liberty.h](#)

5.7 LibFile Class Reference

```
#include <lib_file.hpp>
```

Public Member Functions

- [LibFile](#) (const std::string &filepath, const std::string &loggername)
Constructs a [LibFile](#) object with specified file path and logger name.
- [~LibFile](#) ()
- void [writeJsonToFile](#) ()
Writes the JSON data stored in the `lib_json_` member to a file.
- void [parse](#) ()
Parses the Liberty file and extracts library information into a JSON structure.
- void [modify](#) ()
- void [mono](#) (const bool is_slew)
Validates that lookup tables are monotonically increasing with respect to the output load.
- void [supercell](#) (const int chain_length, const std::vector< std::string > &cell_names)
Creates supercells based on standard cells in the library file.
- void [verilog](#) (const int chain_length, const std::vector< std::string > &cell_names)

Public Attributes

- std::shared_ptr< spdlog::logger > [logger_](#)
- std::filesystem::path [filepath_](#)
- std::string [basename_](#)
- std::string [filename_](#)
- std::string [libname_](#) = ""
- std::string [jsonname_](#) = ""
- std::string [loggername_](#) = ""
- [json lib_json_](#) = json::object()

Private Member Functions

- void `read` ()
Reads the Liberty file specified by the `filename_` member variable.
- bool `checkTimingArcMonotonicity` (const `json` &cell, const `json` &pin, const `json` &arc, const std::string &type, bool is_slew)
Checks if the values in a timing arc are monotonically increasing.

Private Attributes

- `si2drErrorT err_`
- int `process_` = 0
- float `voltage_` = 0.0
- int `temperature_` = 0

5.7.1 Detailed Description

Definition at line 13 of file `lib_file.hpp`.

5.7.2 Constructor & Destructor Documentation

5.7.2.1 LibFile()

```
LibFile::LibFile (
    const std::string & filepath,
    const std::string & loggername )
```

Constructs a `LibFile` object with specified file path and logger name.

This constructor initializes a `LibFile` object with the given file path and creates a logger with the specified name. It sets up:

- Two logging sinks: a colored console sink and a file sink
- File path information including filename, basename, and a corresponding JSON filename
- Log levels (info for console, debug for file)

Parameters

<code>filepath</code>	The path to the file to be processed
<code>loggername</code>	The name for the logger and the log file

Definition at line 28 of file `lib_file.cpp`.

5.7.2.2 ~LibFile()

```
LibFile::~LibFile ( )
```

Definition at line 48 of file [lib_file.cpp](#).

5.7.3 Member Function Documentation

5.7.3.1 checkTimingArcMonotonicity()

```
bool LibFile::checkTimingArcMonotonicity (
    const json & cell,
    const json & pin,
    const json & arc,
    const std::string & timing_arc_name,
    bool is_slew ) [private]
```

Checks if the values in a timing arc are monotonically increasing.

This function examines the values in the specified timing arc to ensure they are monotonically increasing. It checks monotonicity in two dimensions:

1. Across rows (by output load capacitance)
2. Across columns (by input slew, only if `is_slew` is true)

The function logs detailed information about any non-monotonic values found, including cell name, pin names, and conditional statements (when clause) if present.

Parameters

<i>cell</i>	The JSON object representing the cell being checked
<i>pin</i>	The JSON object representing the pin being checked
<i>arc</i>	The JSON object representing the timing arc being checked
<i>timing_arc_name</i>	The name of the timing arc to check (e.g., "cell_rise", "cell_fall")
<i>is_slew</i>	Boolean flag indicating whether to check monotonicity across input slew values

Returns

true if all values in the timing arc are monotonic, false otherwise

Exceptions

<i>None, but</i>	logs errors or warnings for invalid data formats or non-monotonic values
------------------	--

Definition at line 219 of file [lib_file.cpp](#).

Here is the caller graph for this function:

5.7.3.2 modify()

```
void LibFile::modify ( )
```

Definition at line 196 of file [lib_file.cpp](#).

5.7.3.3 mono()

```
void LibFile::mono (
    const bool is_slew )
```

Validates that lookup tables are monotonically increasing with respect to the output load.

This function checks the following timing data in the current library to ensure monotonicity:

- cell_rise and retaining_rise
- cell_fall and retaining_fall
- rise_transition and retain_rise_slew
- fall_transition and retain_fall_slew
- min_pulse_width constraints (rise_constraint, fall_constraint)

The function first checks if a parsed JSON representation exists. If not, it parses the Liberty file and creates the JSON file. It then evaluates each timing arc in all cells and reports the pass/fail status at the end.

Parameters

<i>is_slew</i>	Boolean flag indicating whether to check slew values rather than delay values
----------------	---

The function outputs:

- Number of cells that passed/failed the monotonicity check
- List of cells that failed the check (if any)

Definition at line 340 of file [lib_file.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

5.7.3.4 parse()

```
void LibFile::parse ( )
```

Parses the Liberty file and extracts library information into a JSON structure.

This method handles the parsing of a Liberty file by:

1. Initializing the SI2 parser interface error handler
2. Reading the Liberty file contents
3. Extracting the library name
4. Processing second-level groups including:
 - Cell definitions, which are added to the JSON structure
 - Operating conditions, extracting PVT (Process, Voltage, Temperature) values

The parsed data is stored in the `lib_json_` member variable, with the following structure:

- "library_name": Name of the library
- "cells": Array of cell definitions
- "process": Process value
- "voltage": Voltage value (rounded to 2 decimal places)
- "temperature": Temperature value

Note

This method creates local scopes to ensure proper cleanup of SI2 iterators.

Definition at line 130 of file `lib_file.cpp`.

Here is the call graph for this function: Here is the caller graph for this function:

5.7.3.5 read()

```
void LibFile::read ( ) [private]
```

Reads the Liberty file specified by the `filename_` member variable.

This function attempts to read a Liberty file and logs the process. It measures the time taken to read the file and logs any errors encountered during the read operation. If an error occurs, the function logs the error and terminates the program.

- Logs the start of the read operation.
- Measures the duration of the read operation.
- Checks for specific errors (invalid name or syntax error) and logs them.
- Terminates the program with specific exit codes if errors are detected.
- Logs the completion of the read operation and the time taken.

Note

This function uses the `spdlog` library for logging and the `si2drReadLibertyFile` function for reading the Liberty file.

Exceptions

<i>This</i>	function will terminate the program with exit codes 301 or 401 if specific errors are encountered.
-------------	--

Definition at line 91 of file [lib_file.cpp](#).

Here is the caller graph for this function:

5.7.3.6 supercell()

```
void LibFile::supercell (
    const int chain_length,
    const std::vector< std::string > & cell_names )
```

Creates supercells based on standard cells in the library file.

This method creates supercells by combining standard cells in chains. For each input-output pin pair of a cell, it creates a supercell entry with naming format: <cellname>**X**<chain_length><input_pin>__<output_pin>. The results are written to a .map file.

For sequential cells (with clock pins), the chain length is always set to 1 regardless of the requested chain length.

Parameters

<i>chain_length</i>	The length of the chain for combinational cells
<i>cell_names</i>	Vector of specific cell names to process. If empty, all cells will be processed.

Note

Before creating supercells, this method checks for the existence of a JSON representation of the Liberty file and parses it if not found.

The method will report any requested cells that were not found in the library.

Definition at line 467 of file [lib_file.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

5.7.3.7 verilog()

```
void LibFile::verilog (
    const int chain_length,
    const std::vector< std::string > & cell_names )
```

Definition at line 596 of file [lib_file.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

5.7.3.8 writeJsonToFile()

```
void LibFile::writeJsonToFile ( )
```

Writes the JSON data stored in the `lib_json_` member to a file.

This function attempts to open the specified file for writing. If the file cannot be opened, an error message is logged. If the file is successfully opened, the JSON data is written to the file with an indentation of 2 spaces. After writing, the file is closed and a success message is logged.

Parameters

<i>jsonname</i> ↔ —	The name of the file to which the JSON data will be written.
------------------------	--

Definition at line 60 of file [lib_file.cpp](#).

Here is the caller graph for this function:

5.7.4 Member Data Documentation

5.7.4.1 `basename_`

```
std::string LibFile::basename_
```

Definition at line 19 of file [lib_file.hpp](#).

5.7.4.2 `err_`

```
si2drErrorT LibFile::err_ [private]
```

Definition at line 33 of file [lib_file.hpp](#).

5.7.4.3 `filename_`

```
std::string LibFile::filename_
```

Definition at line 20 of file [lib_file.hpp](#).

5.7.4.4 `filepath_`

```
std::filesystem::path LibFile::filepath_
```

Definition at line 18 of file [lib_file.hpp](#).

5.7.4.5 jsonname_

```
std::string LibFile::jsonname_ = ""
```

Definition at line 22 of file [lib_file.hpp](#).

5.7.4.6 lib_json_

```
json LibFile::lib_json_ = json::object()
```

Definition at line 24 of file [lib_file.hpp](#).

5.7.4.7 libname_

```
std::string LibFile::libname_ = ""
```

Definition at line 21 of file [lib_file.hpp](#).

5.7.4.8 logger_

```
std::shared_ptr<spdlog::logger> LibFile::logger_
```

Definition at line 17 of file [lib_file.hpp](#).

5.7.4.9 loggername_

```
std::string LibFile::loggername_ = ""
```

Definition at line 23 of file [lib_file.hpp](#).

5.7.4.10 process_

```
int LibFile::process_ = 0 [private]
```

Definition at line 34 of file [lib_file.hpp](#).

5.7.4.11 temperature_

```
int LibFile::temperature_ = 0 [private]
```

Definition at line 36 of file [lib_file.hpp](#).

5.7.4.12 voltage_

```
float LibFile::voltage_ = 0.0 [private]
```

Definition at line 35 of file [lib_file.hpp](#).

The documentation for this class was generated from the following files:

- [include/lib_file.hpp](#)
- [src/lib_file.cpp](#)

5.8 LibGroup Class Reference

```
#include <lib_group.hpp>
```

Collaboration diagram for LibGroup:

Public Member Functions

- [LibGroup](#) ([si2drGroupIdT](#) group, [si2drErrorT](#) &err)
- [~LibGroup](#) ()
- [std::string](#) [getName](#) ()
- [std::string](#) [getType](#) ()
- [si2drAttrIdT](#) [getAttrs](#) ()
- [si2drGroupsIdT](#) [getGroups](#) ()

Private Attributes

- [si2drGroupIdT](#) [group_](#)
- [si2drErrorT](#) & [err_](#)

5.8.1 Detailed Description

Definition at line 8 of file [lib_group.hpp](#).

5.8.2 Constructor & Destructor Documentation

5.8.2.1 LibGroup()

```
LibGroup::LibGroup (
    si2drGroupIdT group,
    si2drErrorT & err )
```

Definition at line 3 of file [lib_group.cpp](#).

5.8.2.2 ~LibGroup()

```
LibGroup::~LibGroup ( )
```

Definition at line 5 of file [lib_group.cpp](#).

5.8.3 Member Function Documentation

5.8.3.1 getAttrs()

```
si2drAttrsIdT LibGroup::getAttrs ( )
```

Definition at line 19 of file [lib_group.cpp](#).

Here is the caller graph for this function:

5.8.3.2 getGroups()

```
si2drGroupsIdT LibGroup::getGroups ( )
```

Definition at line 21 of file [lib_group.cpp](#).

Here is the caller graph for this function:

5.8.3.3 getName()

```
std::string LibGroup::getName ( )
```

Definition at line 7 of file [lib_group.cpp](#).

Here is the caller graph for this function:

5.8.3.4 getType()

```
std::string LibGroup::getType ( )
```

Definition at line 14 of file [lib_group.cpp](#).

Here is the caller graph for this function:

5.8.4 Member Data Documentation

5.8.4.1 err_

```
si2drErrorT& LibGroup::err_ [private]
```

Definition at line 19 of file [lib_group.hpp](#).

5.8.4.2 group_

```
si2drGroupIdT LibGroup::group_ [private]
```

Definition at line 18 of file [lib_group.hpp](#).

The documentation for this class was generated from the following files:

- [include/lib_group.hpp](#)
- [src/lib_group.cpp](#)

5.9 LibraryComparator Class Reference

```
#include <compare.hpp>
```

Public Member Functions

- [LibraryComparator](#) ([LibFile](#) &ref_libfile, [LibFile](#) &comp_libfile, double reltol, double abstol)
Constructs a [LibraryComparator](#) object.
- void [generateReport](#) (const std::string &output_file)
Generates a comparison report between the reference and comparison libraries.

Public Attributes

- std::filesystem::path [ref_lib_path_](#)
- std::filesystem::path [comp_lib_path_](#)

Private Member Functions

- void [compareCell](#) (const std::string &cell_name, const [json](#) &ref_cell, const [json](#) &comp_cell, Table &table)
Compares the output pins of a cell between a reference JSON and a comparison JSON.
- void [comparePin](#) (const std::string &cell_name, const std::string &pin_name, const [json](#) &ref_pin, const [json](#) &comp_pin, Table &table)
Compares the timing arcs of a given pin between a reference JSON and a comparison JSON.
- void [compareTimingArc](#) (const std::string &cell_name, const std::string &pin_name, const std::string &timing_type, const [json](#) &ref_timing_arc, const [json](#) &comp_timing_arc, Table &table)
Compares a timing arc between two JSON objects.
- void [compareLut](#) (const std::string &cell_name, const std::string &pin_name, const std::string &timing_type, const std::string &related_pin, const std::string &arc_name, const [json](#) &ref_lut, const [json](#) &comp_lut, Table &table)
Compares lookup tables (LUTs) between reference and comparison libraries for a specific timing arc.

Private Attributes

- [json](#) [ref_json_](#)
- [json](#) [comp_json_](#)
- double [reltol_](#)
- double [abstol_](#)

5.9.1 Detailed Description

Definition at line 12 of file [compare.hpp](#).

5.9.2 Constructor & Destructor Documentation

5.9.2.1 LibraryComparator()

```
LibraryComparator::LibraryComparator (
    LibFile & ref_libfile,
    LibFile & comp_libfile,
    double reltol,
    double abstol )
```

Constructs a [LibraryComparator](#) object.

This constructor initializes a [LibraryComparator](#) object by loading and parsing JSON files associated with the provided reference and comparison library files. If the JSON files do not exist, they are generated by parsing the library files.

Parameters

<i>ref_libfile</i>	Reference to the reference library file object.
<i>comp_libfile</i>	Reference to the comparison library file object.
<i>reltol</i>	Relative tolerance value for comparison.

Definition at line 23 of file [compare.cpp](#).

Here is the call graph for this function:

5.9.3 Member Function Documentation

5.9.3.1 compareCell()

```
void LibraryComparator::compareCell (
    const std::string & cell_name,
    const json & ref_cell,
    const json & comp_cell,
    Table & table ) [private]
```

Compares the output pins of a cell between a reference JSON and a comparison JSON.

This function iterates through the output pins of the comparison cell and checks if each pin exists in the reference cell. If a pin is found in both cells, it calls the comparePin function to compare the details of the pin. If a pin is not found in the reference cell, a warning is logged. If the comparison cell does not contain any output pins, an informational message is logged.

Parameters

<i>cell_name</i>	The name of the cell being compared.
<i>ref_cell</i>	The reference JSON object containing the cell data.
<i>comp_cell</i>	The comparison JSON object containing the cell data.
<i>table</i>	The table object where the comparison results are stored.

Definition at line 300 of file [compare.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

5.9.3.2 compareLut()

```
void LibraryComparator::compareLut (
    const std::string & cell_name,
    const std::string & pin_name,
    const std::string & timing_type,
    const std::string & related_pin,
    const std::string & arc_name,
    const json & ref_lut,
    const json & comp_lut,
    Table & table ) [private]
```

Compares lookup tables (LUTs) between reference and comparison libraries for a specific timing arc.

This function compares the lookup table values between reference and comparison libraries for a given timing arc, checking the consistency of indices and values. It verifies that the indices match between libraries and then compares each value in the LUT against relative and absolute tolerance thresholds. Any discrepancies that exceed the tolerance limits are recorded in the provided table.

Parameters

<i>cell_name</i>	The name of the cell containing the LUT.
<i>pin_name</i>	The name of the pin associated with the LUT.
<i>timing_type</i>	The timing type of the arc (e.g., "rise_transition", "fall_transition").
<i>related_pin</i>	The related pin for the timing arc.
<i>arc_name</i>	The name of the specific timing arc being compared.
<i>ref_lut</i>	The lookup table from the reference library (in JSON format).
<i>comp_lut</i>	The lookup table from the comparison library (in JSON format).
<i>table</i>	Table object to store comparison results for any mismatches found.

Note

The function logs various information/warning/error messages:

- Logs index mismatches as warnings
- Logs value mismatches as debug messages
- Logs format errors in LUT structure as errors
- Records values exceeding tolerance in the provided table

The comparison uses both relative tolerance (*reltol_*) and absolute tolerance (*abstol_*) to determine if values differ significantly.

Definition at line 99 of file [compare.cpp](#).

Here is the caller graph for this function:

5.9.3.3 comparePin()

```
void LibraryComparator::comparePin (
    const std::string & cell_name,
    const std::string & pin_name,
    const json & ref_pin,
    const json & comp_pin,
    Table & table ) [private]
```

Compares the timing arcs of a given pin between a reference JSON and a comparison JSON.

This function iterates through the timing arcs of the comparison pin and looks for matching timing types in the reference pin. If a matching timing type is found, it calls the `compareTimingArc` function to compare the timing arcs. If a timing type is not found in the reference JSON, a warning is logged.

Parameters

<i>cell_name</i>	The name of the cell containing the pin.
<i>pin_name</i>	The name of the pin to compare.
<i>ref_pin</i>	The reference JSON object containing the pin's data.
<i>comp_pin</i>	The comparison JSON object containing the pin's data.
<i>table</i>	A reference to a Table object where comparison results are stored.

Definition at line 261 of file [compare.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

5.9.3.4 compareTimingArc()

```
void LibraryComparator::compareTimingArc (
    const std::string & cell_name,
    const std::string & pin_name,
    const std::string & timing_type,
    const json & ref_timing_arc,
    const json & comp_timing_arc,
    Table & table ) [private]
```

Compares a timing arc between two JSON objects.

This function compares a specific timing arc (e.g., cell_rise, cell_fall, rise_transition, fall_transition) between a reference JSON object and a comparison JSON object. It extracts the related pin from the reference timing arc and iterates through a predefined list of arc names. If an arc name is found in the comparison timing arc, it checks if the same arc name exists in the reference timing arc. If both exist, it calls the compareLut function to compare the LUT data associated with the arc. If the arc name is found in the comparison JSON but not in the reference JSON, a warning message is logged.

Parameters

<i>cell_name</i>	The name of the cell.
<i>pin_name</i>	The name of the pin.
<i>timing_type</i>	The type of timing (e.g., "combinational", "sequential").
<i>ref_timing_arc</i>	The reference JSON object containing the timing arc information.
<i>comp_timing_arc</i>	The comparison JSON object containing the timing arc information.
<i>table</i>	The table to store comparison results.

Definition at line 221 of file [compare.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

5.9.3.5 generateReport()

```
void LibraryComparator::generateReport (
    const std::string & output_file )
```

Generates a comparison report between the reference and comparison libraries.

This function generates a detailed comparison report between the reference library and the comparison library. The report includes metadata such as the reference and comparison library paths, absolute and relative tolerances, and the application details. It also includes a legend explaining various symbols used in the report.

The function iterates through each cell in the comparison library, compares it with the corresponding cell in the reference library, and generates a table of differences. If there are any differences, it calculates summary statistics such as the average and maximum differences, and the number of outliers. The results are written to the specified output file in either Markdown or plain text format.

Parameters

<code>output_file</code>	The path to the output file where the report will be written.
--------------------------	---

Definition at line 342 of file [compare.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

5.9.4 Member Data Documentation

5.9.4.1 abstol_

```
double LibraryComparator::abstol_ [private]
```

Definition at line 23 of file [compare.hpp](#).

5.9.4.2 comp_json_

```
json LibraryComparator::comp_json_ [private]
```

Definition at line 21 of file [compare.hpp](#).

5.9.4.3 comp_lib_path_

```
std::filesystem::path LibraryComparator::comp_lib_path_
```

Definition at line 16 of file [compare.hpp](#).

5.9.4.4 ref_json_

```
json LibraryComparator::ref_json_ [private]
```

Definition at line 20 of file [compare.hpp](#).

5.9.4.5 ref_lib_path_

`std::filesystem::path LibraryComparator::ref_lib_path_`

Definition at line 15 of file [compare.hpp](#).

5.9.4.6 reltol_

`double LibraryComparator::reltol_ [private]`

Definition at line 22 of file [compare.hpp](#).

The documentation for this class was generated from the following files:

- [include/compare.hpp](#)
- [src/compare.cpp](#)

5.10 ModuleRewriter Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for ModuleRewriter:

Collaboration diagram for ModuleRewriter:

Public Member Functions

- [ModuleRewriter](#) (const std::vector< std::string > &inputPins, const std::vector< std::string > &outputPins, const std::pair< std::string, std::string > &supercell_entry, int instance_count, std::shared_ptr< spdlog::logger > logger)
- void [handle](#) (const slang::syntax::SyntaxNode &node)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)

Public Attributes

- std::shared_ptr< spdlog::logger > [logger_](#)

Private Attributes

- const std::vector< std::string > & [inputPins_](#)
- const std::vector< std::string > & [outputPins_](#)
- const std::string [cellName_](#)
- const std::string [moduleName_](#)
- int [depth_](#)
- int [instance_count_](#)

5.10.1 Detailed Description

Definition at line 57 of file [verilog_utils.hpp](#).

5.10.2 Constructor & Destructor Documentation

5.10.2.1 ModuleRewriter()

```
ModuleRewriter::ModuleRewriter (
    const std::vector< std::string > & inputPins,
    const std::vector< std::string > & outputPins,
    const std::pair< std::string, std::string > & supercell_entry,
    int instance_count,
    std::shared_ptr< spdlog::logger > logger ) [inline], [explicit]
```

Definition at line 59 of file [verilog_utils.hpp](#).

5.10.3 Member Function Documentation

5.10.3.1 handle() [1/2]

```
void ModuleRewriter::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 397 of file [verilog_utils.cpp](#).

5.10.3.2 handle() [2/2]

```
void ModuleRewriter::handle (
    const slang::syntax::SyntaxNode & node )
```

Definition at line 387 of file [verilog_utils.cpp](#).

5.10.4 Member Data Documentation

5.10.4.1 cellName_

```
const std::string ModuleRewriter::cellName_ [private]
```

Definition at line 73 of file [verilog_utils.hpp](#).

5.10.4.2 depth_

```
int ModuleRewriter::depth_ [private]
```

Definition at line 75 of file [verilog_utils.hpp](#).

5.10.4.3 inputPins_

```
const std::vector<std::string>& ModuleRewriter::inputPins_ [private]
```

Definition at line 71 of file [verilog_utils.hpp](#).

5.10.4.4 instance_count_

```
int ModuleRewriter::instance_count_ [private]
```

Definition at line 76 of file [verilog_utils.hpp](#).

5.10.4.5 logger_

```
std::shared_ptr<spdlog::logger> ModuleRewriter::logger_
```

Definition at line 66 of file [verilog_utils.hpp](#).

5.10.4.6 moduleName_

```
const std::string ModuleRewriter::moduleName_ [private]
```

Definition at line 74 of file [verilog_utils.hpp](#).

5.10.4.7 outputPins_

```
const std::vector<std::string>& ModuleRewriter::outputPins_ [private]
```

Definition at line 72 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- [include/verilog_utils.hpp](#)
- [src/verilog_utils.cpp](#)

5.11 si2drExprT Struct Reference

```
#include <si2dr_liberty.h>
```

Collaboration diagram for si2drExprT:

Public Attributes

- [si2drExprTypeT](#) type
- union {
 - [si2drInt32T](#) i
 - [si2drFloat64T](#) d
 - [si2drStringT](#) s
 - [si2drBooleanT](#) b
 } u
- [si2drValueTypeT](#) valuetype
- struct [si2drExprT](#) * left
- struct [si2drExprT](#) * right

5.11.1 Detailed Description

Definition at line 172 of file [si2dr_liberty.h](#).

5.11.2 Member Data Documentation

5.11.2.1 b

```
si2drBooleanT si2drExprT::b
```

Definition at line 180 of file [si2dr_liberty.h](#).

5.11.2.2 d

```
si2drFloat64T si2drExprT::d
```

Definition at line 178 of file [si2dr_liberty.h](#).

5.11.2.3 i

```
si2drInt32T si2drExprT::i
```

Definition at line 177 of file [si2dr_liberty.h](#).

5.11.2.4 left

```
struct si2drExprT* si2drExprT::left
```

Definition at line 185 of file [si2dr_liberty.h](#).

5.11.2.5 right

```
struct si2drExprT* si2drExprT::right
```

Definition at line 186 of file [si2dr_liberty.h](#).

5.11.2.6 s

```
si2drStringT si2drExprT::s
```

Definition at line 179 of file [si2dr_liberty.h](#).

5.11.2.7 type

```
si2drExprTypeT si2drExprT::type
```

Definition at line 174 of file [si2dr_liberty.h](#).

5.11.2.8

```
union { ... } si2drExprT::u
```

5.11.2.9 valuetype

```
si2drValueTypeT si2drExprT::valuetype
```

Definition at line 183 of file [si2dr_liberty.h](#).

The documentation for this struct was generated from the following file:

- [include/si2dr_liberty.h](#)

5.12 si2OID Struct Reference

```
#include <si2dr_liberty.h>
```

Public Attributes

- void * [v1](#)
- void * [v2](#)

5.12.1 Detailed Description

Definition at line 79 of file [si2dr_liberty.h](#).

5.12.2 Member Data Documentation

5.12.2.1 v1

```
void* si2OID::v1
```

Definition at line 79 of file [si2dr_liberty.h](#).

5.12.2.2 v2

```
void * si2OID::v2
```

Definition at line 79 of file [si2dr_liberty.h](#).

The documentation for this struct was generated from the following file:

- [include/si2dr_liberty.h](#)

5.13 ValuesIterator Class Reference

```
#include <iterators.hpp>
```

Collaboration diagram for ValuesIterator:

Public Member Functions

- [ValuesIterator](#) ([si2drValuesIdT](#) values, [si2drErrorT](#) &err)
- [~ValuesIterator](#) ()
- void [next](#) ()
- bool [end](#) ()

Public Attributes

- [si2drValuesIdT](#) values_
- [si2drValueTypeT](#) vtype_
- [si2drInt32T](#) int_
- [si2drFloat64T](#) float_
- [si2drStringT](#) str_
- [si2drBooleanT](#) bool_
- [si2drExprT](#) * exprp_
- [si2drErrorT](#) & err_

5.13.1 Detailed Description

Definition at line 37 of file [iterators.hpp](#).

5.13.2 Constructor & Destructor Documentation

5.13.2.1 ValuesIterator()

```
ValuesIterator::ValuesIterator (
    si2drValuesIdT values,
    si2drErrorT & err )
```

Definition at line 24 of file [iterators.cpp](#).

5.13.2.2 ~ValuesIterator()

```
ValuesIterator::~~ValuesIterator ( )
```

Definition at line 27 of file [iterators.cpp](#).

5.13.3 Member Function Documentation

5.13.3.1 end()

```
bool ValuesIterator::end ( )
```

Definition at line 32 of file [iterators.cpp](#).

Here is the caller graph for this function:

5.13.3.2 next()

```
void ValuesIterator::next ( )
```

Definition at line 29 of file [iterators.cpp](#).

Here is the caller graph for this function:

5.13.4 Member Data Documentation

5.13.4.1 bool_

```
si2drBooleanT ValuesIterator::bool_
```

Definition at line 49 of file [iterators.hpp](#).

5.13.4.2 err_

`si2drErrorT& ValuesIterator::err_`

Definition at line 51 of file [iterators.hpp](#).

5.13.4.3 exprp_

`si2drExprT* ValuesIterator::exprp_`

Definition at line 50 of file [iterators.hpp](#).

5.13.4.4 float_

`si2drFloat64T ValuesIterator::float_`

Definition at line 47 of file [iterators.hpp](#).

5.13.4.5 int_

`si2drInt32T ValuesIterator::int_`

Definition at line 46 of file [iterators.hpp](#).

5.13.4.6 str_

`si2drStringT ValuesIterator::str_`

Definition at line 48 of file [iterators.hpp](#).

5.13.4.7 values_

`si2drValuesIdT ValuesIterator::values_`

Definition at line 44 of file [iterators.hpp](#).

5.13.4.8 vtype_

`si2drValueType` `ValuesIterator::vtype_`

Definition at line 45 of file [iterators.hpp](#).

The documentation for this class was generated from the following files:

- [include/iterators.hpp](#)
- [src/iterators.cpp](#)

5.14 VerilogVisitor Class Reference

```
#include <verilog_utils.hpp>
```

Inheritance diagram for VerilogVisitor:

Collaboration diagram for VerilogVisitor:

Public Member Functions

- [VerilogVisitor](#) (const std::string &targetCell)
- void [handle](#) (const slang::syntax::SyntaxNode &node)
- void [handle](#) (const slang::syntax::ModuleDeclarationSyntax &module)
- void [handle](#) (const slang::syntax::PortDeclarationSyntax &portDecl)
- void [handle](#) (const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst)
- void [handle](#) (const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst)
- void [handle](#) (const slang::syntax::SpecifyBlockSyntax &specifyBlock)

Private Attributes

- const std::string & [targetCell_](#)
- int [depth_](#)
- bool [inTargetModule_](#)

5.14.1 Detailed Description

Definition at line 14 of file [verilog_utils.hpp](#).

5.14.2 Constructor & Destructor Documentation

5.14.2.1 VerilogVisitor()

```
VerilogVisitor::VerilogVisitor (
    const std::string & targetCell ) [inline], [explicit]
```

Definition at line 16 of file [verilog_utils.hpp](#).

5.14.3 Member Function Documentation

5.14.3.1 handle() [1/6]

```
void VerilogVisitor::handle (
    const slang::syntax::HierarchyInstantiationSyntax & hierarchyInst )
```

Definition at line 80 of file [verilog_utils.cpp](#).

5.14.3.2 handle() [2/6]

```
void VerilogVisitor::handle (
    const slang::syntax::ModuleDeclarationSyntax & module )
```

Definition at line 19 of file [verilog_utils.cpp](#).

5.14.3.3 handle() [3/6]

```
void VerilogVisitor::handle (
    const slang::syntax::PortDeclarationSyntax & portDecl )
```

Definition at line 48 of file [verilog_utils.cpp](#).

5.14.3.4 handle() [4/6]

```
void VerilogVisitor::handle (
    const slang::syntax::PrimitiveInstantiationSyntax & primitiveInst )
```

Definition at line 192 of file [verilog_utils.cpp](#).

5.14.3.5 `handle()` [5/6]

```
void VerilogVisitor::handle (
    const slang::syntax::SpecifyBlockSyntax & specifyBlock )
```

Definition at line 259 of file [verilog_utils.cpp](#).

Here is the call graph for this function:

5.14.3.6 `handle()` [6/6]

```
void VerilogVisitor::handle (
    const slang::syntax::SyntaxNode & node )
```

Definition at line 8 of file [verilog_utils.cpp](#).

Here is the caller graph for this function:

5.14.4 Member Data Documentation

5.14.4.1 `depth_`

```
int VerilogVisitor::depth_ [private]
```

Definition at line 27 of file [verilog_utils.hpp](#).

5.14.4.2 `inTargetModule_`

```
bool VerilogVisitor::inTargetModule_ [private]
```

Definition at line 28 of file [verilog_utils.hpp](#).

5.14.4.3 `targetCell_`

```
const std::string& VerilogVisitor::targetCell_ [private]
```

Definition at line 26 of file [verilog_utils.hpp](#).

The documentation for this class was generated from the following files:

- [include/verilog_utils.hpp](#)
- [src/verilog_utils.cpp](#)

Chapter 6

File Documentation

6.1 include/compare.hpp File Reference

```
#include "lib_file.hpp"
#include "nlohmann/json.hpp"
#include "tabulate/table.hpp"
```

Include dependency graph for compare.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibraryComparator](#)

Typedefs

- using [json](#) = nlohmann::json

6.1.1 Typedef Documentation

6.1.1.1 json

```
using json = nlohmann::json
```

Definition at line 9 of file [compare.hpp](#).

6.2 compare.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef COMPARE_HPP
00002 #define COMPARE_HPP
00003
00004 #include "lib_file.hpp"
00005
00006 #include "nlohmann/json.hpp"
00007 #include "tabulate/table.hpp"
00008
00009 using json = nlohmann::json;
00010 using namespace tabulate;
00011
00012 class LibraryComparator {
00013 public:
00014     LibraryComparator(LibFile &ref_libfile, LibFile &comp_libfile, double reltol, double abstol);
00015     std::filesystem::path ref_lib_path_;
00016     std::filesystem::path comp_lib_path_;
00017     void generateReport(const std::string &output_file);
00018
00019 private:
00020     json ref_json_;
00021     json comp_json_;
00022     double reltol_;
00023     double abstol_;
00024
00025     void compareCell(const std::string &cell_name, const json &ref_cell, const json &comp_cell,
00026                     Table &table);
00027     void comparePin(const std::string &cell_name, const std::string &pin_name, const json &ref_pin,
00028                    const json &comp_pin, Table &table);
00029     void compareTimingArc(const std::string &cell_name, const std::string &pin_name,
00030                           const std::string &timing_type, const json &ref_timing_arc,
00031                           const json &comp_timing_arc, Table &table);
00032     void compareLut(const std::string &cell_name, const std::string &pin_name,
00033                     const std::string &timing_type, const std::string &related_pin,
00034                     const std::string &arc_name, const json &ref_lut, const json &comp_lut, Table
00035                     &table);
00036     // void configureTableFormat(Table &table);
00037 };
00038 #endif // COMPARE_HPP
```

6.3 include/iterators.hpp File Reference

```
#include "si2dr_liberty.h"
#include "lib_attribute.hpp"
#include "lib_group.hpp"
```

Include dependency graph for iterators.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [GroupsIterator](#)
- class [AttributesIterator](#)
- class [ValuesIterator](#)

6.4 iterators.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef ITERATORS_H
00002 #define ITERATORS_H
00003
00004 #include "si2dr_liberty.h"
00005
00006 #include "lib_attribute.hpp"
00007 #include "lib_group.hpp"
```

```

00008
00009 class GroupsIterator {
00010 public:
00011     GroupsIterator(si2drGroupsIdT groups, si2drErrorT &err);
00012     ~GroupsIterator();
00013     void next();
00014     bool end();
00015     LibGroup get();
00016 private:
00017     si2drGroupsIdT groups_;
00018     si2drGroupIdT group_;
00019     si2drErrorT &err_;
00020 };
00021
00022
00023 class AttributesIterator {
00024 public:
00025     AttributesIterator(si2drAttrsIdT attrs, si2drErrorT &err);
00026     ~AttributesIterator();
00027     void next();
00028     bool end();
00029     LibAttribute get();
00030 private:
00031     si2drAttrsIdT attrs_;
00032     si2drAttrIdT attr_;
00033     si2drErrorT &err_;
00034 };
00035
00036
00037 class ValuesIterator {
00038 public:
00039     ValuesIterator(si2drValuesIdT values, si2drErrorT &err);
00040     ~ValuesIterator();
00041     void next();
00042     bool end();
00043 private:
00044     si2drValuesIdT values_;
00045     si2drValueTypeT vtype_;
00046     si2drInt32T int_;
00047     si2drFloat64T float_;
00048     si2drStringT str_;
00049     si2drBooleanT bool_;
00050     si2drExprT *exprp_;
00051     si2drErrorT &err_;
00052 };
00053
00054 #endif // ITERATORS_H

```

6.5 include/json_utils.hpp File Reference

```

#include <string>
#include <utility>
#include "nlohmann/json.hpp"
#include "si2dr_liberty.h"
#include "lib_group.hpp"

```

Include dependency graph for json_utils.hpp: This graph shows which files directly or indirectly include this file:

Typedefs

- using [json](#) = nlohmann::json

Functions

- [json generateLutJson](#) (LibGroup &lib_lut_group, si2drErrorT &err)
Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.
- [json generatePowerJson](#) (LibGroup &lib_power_group, si2drErrorT &err)
Converts a Liberty power group into a JSON representation.
- std::pair< std::string, [json](#) > [generatePinJson](#) (LibGroup &lib_pin_group, si2drErrorT &err)
Generates a JSON representation of a Liberty pin group.
- [json generateCellJson](#) (LibGroup &lib_cell_group, si2drErrorT &err)
Generates a JSON representation of a cell from a [LibGroup](#) object.

6.5.1 Typedef Documentation

6.5.1.1 json

```
using json = nlohmann::json
```

Definition at line 12 of file [json_utils.hpp](#).

6.5.2 Function Documentation

6.5.2.1 generateCellJson()

```
json generateCellJson (
    LibGroup & lib_cell_group,
    si2drErrorT & err )
```

Generates a JSON representation of a cell from a [LibGroup](#) object.

This function processes a [LibGroup](#) representing a cell and converts it into a JSON object. It extracts the cell name and processes specific attributes such as area, cell_leakage_power, and cell_footprint. It also processes pins within the cell by categorizing them based on their direction (input, output, internal, inout).

Parameters

<i>lib_cell_group</i>	The LibGroup object representing the cell
<i>err</i>	Reference to a si2drErrorT object for error handling

Returns

json A JSON object containing the cell's information with the following structure:

- "cell_name": string - name of the cell
- "area": float (optional) - cell area if present
- "cell_leakage_power": float (optional) - cell leakage power if present
- "cell_footprint": string (optional) - cell footprint if present
- "input_pins": array - list of input pins
- "output_pins": array - list of output pins
- "internal_pins": array - list of internal pins
- "inout_pins": array - list of inout pins

Definition at line 272 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.5.2.2 generateLutJson()

```
json generateLutJson (
    LibGroup & lib_lut_group,
    si2drErrorT & err )
```

Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.

This function iterates through the attributes of the provided [LibGroup](#) object representing a LUT and converts them into a JSON structure. It specifically handles the following attributes:

- "index_1": Converted to a vector and stored in the JSON
- "index_2": Converted to a vector and stored in the JSON
- "values": Each value is parsed into a vector and added to an array in the JSON

Any other attributes encountered will trigger a warning message.

Parameters

<i>lib_lut_group</i>	The LibGroup object containing the LUT data to be converted
<i>err</i>	Reference to an <code>si2drErrorT</code> object to track any errors during processing

Returns

json A JSON object representing the LUT data with keys for "index_1", "index_2", and "values"

Definition at line 62 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.5.2.3 generatePinJson()

```
std::pair< std::string, json > generatePinJson (
    LibGroup & lib_pin_group,
    si2drErrorT & err )
```

Generates a JSON representation of a Liberty pin group.

This function traverses a Liberty pin group, extracts relevant attributes and sub-groups, and converts them into a JSON object. It also determines the pin's direction.

Processes the following pin attributes:

- direction
- max_transition, capacitance, rise_capacitance, fall_capacitance, max_capacitance (float types)
- function, power_down_function, related_ground_pin, related_power_pin, three_state (string types)
- clock (boolean type)

Handles the following sub-groups:

- internal_power: Converted using [generatePowerJson\(\)](#)
- timing: Converted using [generateTimingJson\(\)](#)

Parameters

<i>lib_pin_group</i>	The Liberty pin group to process
<i>err</i>	Reference to an error object for tracking Liberty API errors

Returns

A pair containing the pin direction (string) and the JSON representation of the pin

Definition at line 206 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.5.2.4 generatePowerJson()

```
json generatePowerJson (
    LibGroup & lib_power_group,
    si2drErrorT & err )
```

Converts a Liberty power group into a JSON representation.

This function processes a Liberty power group and converts its attributes and subgroups into a JSON object. It handles attributes like "when", "related_pin", and "related_pg_pin", as well as "rise_power" and "fall_power" subgroups.

Parameters

<i>lib_power_group</i>	The Liberty power group to convert
<i>err</i>	Reference to an si2drErrorT object for error handling

Returns

json A JSON object representing the power group data

The function:

- Extracts string attributes (when, related_pin, related_pg_pin)
- Processes rise_power and fall_power subgroups by converting them to LUT JSON format
- Logs warnings for unknown power subgroup types

Definition at line 154 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.6 json_utils.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef JSON_UTILS_HPP
00002 #define JSON_UTILS_HPP
00003
00004 #include <string>
00005 #include <utility>
00006
00007 #include "nlohmann/json.hpp"
00008 #include "si2dr_liberty.h"
00009
00010 #include "lib_group.hpp"
00011
00012 using json = nlohmann::json;
00013
00014 json generateLutJson(LibGroup &lib_lut_group, si2drErrorT &err);
00015 json generatePowerJson(LibGroup &lib_power_group, si2drErrorT &err);
00016 std::pair<std::string, json> generatePinJson(LibGroup &lib_pin_group, si2drErrorT &err);
00017 json generateCellJson(LibGroup &lib_cell_group, si2drErrorT &err);
00018
00019 #endif // JSON_UTILS_HPP

```

6.7 include/lib_attribute.hpp File Reference

```

#include <string>
#include "si2dr_liberty.h"

```

Include dependency graph for lib_attribute.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibAttribute](#)

6.8 lib_attribute.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LIB_ATTRIBUTE_H
00002 #define LIB_ATTRIBUTE_H
00003
00004 #include <string>
00005
00006 #include "si2dr_liberty.h"
00007
00008 class LibAttribute {
00009 public:
00010     LibAttribute(si2drAttrIdT attr, si2drErrorT &err);
00011     ~LibAttribute();
00012     std::string getName();
00013     bool isComplex();
00014     si2drValuesIdT getValues();
00015     long int getInt();
00016     double getFloat();
00017     std::string getString();
00018     bool getBoolean();
00019
00020 private:
00021     si2drAttrIdT attr_;
00022     si2drErrorT &err_;
00023 };
00024
00025 #endif // LIB_ATTRIBUTE_H

```

6.9 include/lib_file.hpp File Reference

```
#include <filesystem>
#include <string>
#include <vector>
#include "nlohmann/json.hpp"
#include "si2dr_liberty.h"
```

Include dependency graph for lib_file.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibFile](#)

Typedefs

- using [json](#) = nlohmann::json

6.9.1 Typedef Documentation

6.9.1.1 json

```
using json = nlohmann::json
```

Definition at line 11 of file lib_file.hpp.

6.10 lib_file.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef LIB_FILE_H
00002 #define LIB_FILE_H
00003
00004 #include <filesystem>
00005 #include <string>
00006 #include <vector>
00007
00008 #include "nlohmann/json.hpp"
00009 #include "si2dr_liberty.h"
00010
00011 using json = nlohmann::json;
00012
00013 class LibFile {
00014 public:
00015     LibFile(const std::string &filepath, const std::string &loggername);
00016     ~LibFile();
00017     std::shared_ptr<spdlog::logger> logger_;
00018     std::filesystem::path filepath_; // full path to the file
00019     std::string basename_;          // file name without extension
00020     std::string filename_;          // full file name with extension
00021     std::string libname_ = "";      // library name obtained through parsing
00022     std::string jsonname_ = "";     // json file name to store parsed data
00023     std::string loggername_ = "";   // log file name
00024     json lib_json_ = json::object();
00025     void writeJsonToFile();
00026     void parse();
00027     void modify();
00028     void mono(const bool is_slew);
```



```

00029 void supercell(const int chain_length, const std::vector<std::string> &cell_names);
00030 void verilog(const int chain_length, const std::vector<std::string> &cell_names);
00031
00032 private:
00033     si2drErrorT err_;
00034     int process_ = 0;
00035     float voltage_ = 0.0;
00036     int temperature_ = 0;
00037     void read();
00038     bool checkTimingArcMonotonicity(const json &cell, const json &pin, const json &arc,
00039                                     const std::string &type, bool is_slew);
00040 };
00041
00042 #endif // LIB_FILE_H

```

6.11 include/lib_group.hpp File Reference

```

#include <string>
#include "si2dr_liberty.h"

```

Include dependency graph for lib_group.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [LibGroup](#)

6.12 lib_group.hpp

[Go to the documentation of this file.](#)

```

00001 #ifndef LIB_GROUP_H
00002 #define LIB_GROUP_H
00003
00004 #include <string>
00005
00006 #include "si2dr_liberty.h"
00007
00008 class LibGroup {
00009 public:
00010     LibGroup(si2drGroupIdT group, si2drErrorT &err);
00011     ~LibGroup();
00012     std::string getName();
00013     std::string getType();
00014     si2drAttrsIdT getAttrs();
00015     si2drGroupsIdT getGroups();
00016
00017 private:
00018     si2drGroupIdT group_;
00019     si2drErrorT &err_;
00020 };
00021
00022 #endif // LIB_GROUP_H

```

6.13 include/si2dr_liberty.h File Reference

This graph shows which files directly or indirectly include this file:

Classes

- struct [si2OID](#)
- struct [si2drExprT](#)
- struct [liberty_value_data](#)

Macros

- `#define SI2_ARGS(args) ()`
- `#define SI2_LONG_MAX 2147483647`
- `#define SI2_LONG_MIN -2147483647`
- `#define SI2_ULONG_MAX 4294967295`
- `#define SI2DR_MIN_FLOAT32 -1E+37`
- `#define SI2DR_MAX_FLOAT32 1E+37`
- `#define SI2DR_MIN_FLOAT64 -1.797693e+308`
- `#define SI2DR_MAX_FLOAT64 1.797693e+308`
- `#define SI2DR_MIN_INT32 -2147483647`
- `#define SI2DR_MAX_INT32 2147483647`
- `#define SI2DR_MAX_STRING_LEN 1048576 /*1024*1024*/`
- `#define SI2DR_TRUE SI2_TRUE`
- `#define SI2DR_FALSE SI2_FALSE`
- `#define LONG_DOUBLE long double`

Typedefs

- `typedef enum si2BooleanEnum si2BooleanT`
- `typedef long int si2LongT`
- `typedef float si2FloatT`
- `typedef double si2DoubleT`
- `typedef char * si2StringT`
- `typedef void si2VoidT`
- `typedef enum si2TypeEnum si2TypeT`
- `typedef struct si2OID si2ObjectIdT`
- `typedef void * si2IteratorIdT`
- `typedef si2BooleanT si2drBooleanT`
- `typedef si2LongT si2drInt32T`
- `typedef si2FloatT si2drFloat32T`
- `typedef si2DoubleT si2drFloat64T`
- `typedef si2StringT si2drStringT`
- `typedef si2VoidT si2drVoidT`
- `typedef si2IteratorIdT si2drIterIdT`
- `typedef si2ObjectIdT si2drObjectIdT`
- `typedef si2drObjectIdT si2drGroupIdT`
- `typedef si2drObjectIdT si2drAttrIdT`
- `typedef si2drObjectIdT si2drAttrComplexValIdT`
- `typedef si2drObjectIdT si2drDefineIdT`
- `typedef si2drIterIdT si2drObjectsIdT`
- `typedef si2drIterIdT si2drGroupsIdT`
- `typedef si2drIterIdT si2drAttrIdT`
- `typedef si2drIterIdT si2drDefinesIdT`
- `typedef si2drIterIdT si2drNamesIdT`
- `typedef si2drIterIdT si2drValuesIdT`
- `typedef enum si2drSeverityT si2drSeverityT`
- `typedef enum si2drObjectTypeT si2drObjectTypeT`
- `typedef enum si2drAttrTypeT si2drAttrTypeT`
- `typedef enum si2drValueTypeT si2drValueTypeT`
- `typedef enum si2drExprTypeT si2drExprTypeT`
- `typedef struct si2drExprT si2drExprT`
- `typedef enum si2drErrorT si2drErrorT`
- `typedef si2drVoidT(* si2drMessageHandlerT) (si2drSeverityT sev, si2drErrorT errToPrint, si2drStringT aux↵
Text, si2drErrorT *err)`

Enumerations

- enum `si2BooleanEnum` { `SI2_FALSE` = 0 , `SI2_TRUE` = 1 }
- enum `si2TypeEnum` {
`SI2_UNDEFINED_VALUETYPE` = 0 , `SI2_BOOLEAN` = 1 , `SI2_LONG` = 2 , `SI2_FLOAT` = 3 ,
`SI2_DOUBLE` = 4 , `SI2_STRING` = 5 , `SI2_VOID` = 6 , `SI2_OID` = 7 ,
`SI2_ITER` = 8 , `SI2_EXPR` = 9 , `SI2_MAX_VALUETYPE` = 10 }
- enum `si2drSeverityT` { `SI2DR_SEVERITY_NOTE` = 0 , `SI2DR_SEVERITY_WARN` = 1 , `SI2DR_SEVERITY_ERR` = 2 }
- enum `si2drObjectTypeT` {
`SI2DR_UNDEFINED_OBJECTTYPE` = 0 , `SI2DR_GROUP` = 1 , `SI2DR_ATTR` = 2 , `SI2DR_DEFINE` = 3 ,
`SI2DR_COMPLEXVAL` = 4 , `SI2DR_MAX_OBJECTTYPE` = 5 }
- enum `si2drAttrTypeT` { `SI2DR_SIMPLE` , `SI2DR_COMPLEX` }
- enum `si2drValueTypeT` {
`SI2DR_UNDEFINED_VALUETYPE` = `SI2_UNDEFINED_VALUETYPE` , `SI2DR_INT32` = `SI2_LONG` ,
`SI2DR_STRING` = `SI2_STRING` , `SI2DR_FLOAT64` = `SI2_DOUBLE` ,
`SI2DR_BOOLEAN` = `SI2_BOOLEAN` , `SI2DR_EXPR` = `SI2_EXPR` , `SI2DR_MAX_VALUETYPE` = `SI2_MAX_VALUETYPE` }
- enum `si2drExprTypeT` {
`SI2DR_EXPR_VAL` = 0 , `SI2DR_EXPR_OP_ADD` = 1 , `SI2DR_EXPR_OP_SUB` = 2 , `SI2DR_EXPR_OP_MUL` = 3 ,
`SI2DR_EXPR_OP_DIV` = 4 , `SI2DR_EXPR_OP_PAREN` = 5 , `SI2DR_EXPR_OP_LOG2` = 6 ,
`SI2DR_EXPR_OP_LOG10` = 7 ,
`SI2DR_EXPR_OP_EXP` = 8 }
- enum `si2drErrorT` {
`SI2DR_NO_ERROR` = 0 , `SI2DR_INTERNAL_SYSTEM_ERROR` = 1 , `SI2DR_INVALID_VALUE` = 4 ,
`SI2DR_INVALID_NAME` = 5 ,
`SI2DR_INVALID_OBJECTTYPE` = 6 , `SI2DR_INVALID_ATTRTYPE` = 10 , `SI2DR_UNUSABLE_OID` = 7 ,
`SI2DR_OBJECT_ALREADY_EXISTS` = 8 ,
`SI2DR_OBJECT_NOT_FOUND` = 9 , `SI2DR_SYNTAX_ERROR` = 2 , `SI2DR_TRACE_FILES_CANNOT_BE_OPENED` = 3 ,
`SI2DR_PIINIT_NOT_CALLED` = 11 ,
`SI2DR_SEMANTIC_ERROR` = 12 , `SI2DR_REFERENCE_ERROR` = 13 , `SI2DR_MAX_ERROR` = 14 }

Functions

- `si2drVoidT si2drDefaultMessageHandler` (`si2drSeverityT` sev, `si2drErrorT` errToPrint, `si2drStringT` auxText, `si2drErrorT` *err)
- `si2drMessageHandlerT si2drPIGetMessageHandler` (`si2drErrorT` *err)
- `si2drVoidT si2drPISetMessageHandler` (`si2drMessageHandlerT` MsgHandFuncPtr, `si2drErrorT` *err)
- `si2drGroupIdT si2drPICreateGroup` `SI2_ARGS` ((`si2drStringT` name, `si2drStringT` group_type, `si2drErrorT` *err))
- `si2drAttrIdT si2drGroupCreateAttr` `SI2_ARGS` ((`si2drGroupIdT` group, `si2drStringT` name, `si2drAttrTypeT` type, `si2drErrorT` *err))
- `si2drAttrTypeT si2drAttrGetAttrType` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddInt32Value` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drInt32T` intgr, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddStringValue` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drStringT` string, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddBooleanValue` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drBooleanT` boolval, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddFloat64Value` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drFloat64T` float64, `si2drErrorT` *err))
- `si2drVoidT si2drComplexAttrAddExprValue` `SI2_ARGS` ((`si2drAttrIdT` attr, `si2drExprT` *expr, `si2drErrorT` *err))
- `si2drVoidT si2drIterNextComplexValue` `SI2_ARGS` ((`si2drValuesIdT` iter, `si2drValueTypeT` *type, `si2drInt32T` *intgr, `si2drFloat64T` *float64, `si2drStringT` *string, `si2drBooleanT` *boolval, `si2drExprT` **expr, `si2drErrorT` *err))
- `si2drAttrComplexValIdT si2drIterNextComplex` `SI2_ARGS` ((`si2drValuesIdT` iter, `si2drErrorT` *err))

- `si2drVoidT si2drSimpleAttrSetBooleanValue SI2_ARGS ((si2drAttrIdT attr, si2drBooleanT integr, si2drErrorT *err))`
- `si2drVoidT si2drExprDestroy SI2_ARGS ((si2drExprT *expr, si2drErrorT *err))`
- `si2drExprT *si2drCreateExpr SI2_ARGS ((si2drExprTypeT type, si2drErrorT *err))`
- `si2drExprT *si2drCreateBooleanValExpr SI2_ARGS ((si2drBooleanT b, si2drErrorT *err))`
- `si2drExprT *si2drCreateDoubleValExpr SI2_ARGS ((si2drFloat64T d, si2drErrorT *err))`
- `si2drExprT *si2drCreateStringValExpr SI2_ARGS ((si2drStringT s, si2drErrorT *err))`
- `si2drExprT *si2drCreateIntValExpr SI2_ARGS ((si2drInt32T i, si2drErrorT *err))`
- `si2drExprT *si2drCreateBinaryOpExpr SI2_ARGS ((si2drExprT *left, si2drExprTypeT optype, si2drExprT *right, si2drErrorT *err))`
- `si2drExprT *si2drCreateUnaryOpExpr SI2_ARGS ((si2drExprTypeT optype, si2drExprT *expr, si2drErrorT *err))`
- `si2drDefinelDT si2drGroupCreateDefine SI2_ARGS ((si2drGroupIdT group, si2drStringT name, si2drStringT allowed_group_name, si2drValueTypeT valtype, si2drErrorT *err))`
- `si2drVoidT si2drDefineGetInfo SI2_ARGS ((si2drDefinelDT def, si2drStringT *name, si2drStringT *allowed_group_name, si2drValueTypeT *valtype, si2drErrorT *err))`
- `si2drStringT si2drDefineGetName SI2_ARGS ((si2drDefinelDT def, si2drErrorT *err))`
- `si2drGroupIdT si2drGroupCreateGroup SI2_ARGS ((si2drGroupIdT group, si2drStringT name, si2drStringT group_type, si2drErrorT *err))`
- `si2drStringT si2drGroupGetType SI2_ARGS ((si2drGroupIdT group, si2drErrorT *err))`
- `si2drVoidT si2drGroupSetComment SI2_ARGS ((si2drGroupIdT group, si2drStringT comment, si2drErrorT *err))`
- `si2drVoidT si2drAttrSetComment SI2_ARGS ((si2drAttrIdT attr, si2drStringT comment, si2drErrorT *err))`
- `si2drVoidT si2drDefineSetComment SI2_ARGS ((si2drDefinelDT def, si2drStringT comment, si2drErrorT *err))`
- `si2drVoidT si2drGroupAddName SI2_ARGS ((si2drGroupIdT group, si2drStringT name, si2drErrorT *err))`
- `si2drGroupIdT si2drPIFindGroupByName SI2_ARGS ((si2drStringT name, si2drStringT type, si2drErrorT *err))`
- `si2drGroupIdT si2drGroupFindGroupByName SI2_ARGS ((si2drGroupIdT group, si2drStringT name, si2drStringT type, si2drErrorT *err))`
- `si2drDefinelDT si2drPIFindDefineByName SI2_ARGS ((si2drStringT name, si2drErrorT *err))`
- `si2drGroupsIdT si2drPIGetGroups SI2_ARGS ((si2drErrorT *err))`
- `si2drVoidT si2drObjectDelete SI2_ARGS ((si2drObjectIdT object, si2drErrorT *err))`
- `si2drStringT si2drPIGetErrorText SI2_ARGS ((si2drErrorT errorCode, si2drErrorT *err))`
- `si2drBooleanT si2drObjectIsSame SI2_ARGS ((si2drObjectIdT object1, si2drObjectIdT object2, si2drErrorT *err))`
- `si2drVoidT si2drObjectSetFileName SI2_ARGS ((si2drObjectIdT object, si2drStringT filename, si2drErrorT *err))`
- `si2drVoidT si2drObjectSetLineNo SI2_ARGS ((si2drObjectIdT object, si2drInt32T lineno, si2drErrorT *err))`
- `si2drVoidT si2drReadLibertyFile SI2_ARGS ((char *filename, si2drErrorT *err))`
- `si2drVoidT si2drWriteLibertyFile SI2_ARGS ((char *filename, si2drGroupIdT group, char *cellname, si2drErrorT *err))`
- `si2drVoidT si2drPISetTraceMode SI2_ARGS ((si2drStringT fname, si2drErrorT *err))`
- `si2drVoidT si2drGroupMoveAfter SI2_ARGS ((si2drGroupIdT groupToMove, si2drGroupIdT targetGroup, si2drErrorT *err))`
- `LONG_DOUBLE liberty_get_element (struct liberty_value_data *vd,...)`
- `void liberty_destroy_value_data (struct liberty_value_data *vd)`
- `struct liberty_value_data * liberty_get_values_data (si2drGroupIdT table_group)`
- `void print_version ()`
- `void usage ()`
- `int main_liberty_parser (int argc, char *argv[ ])`
- `void get_function (char *filename)`

6.13.1 Macro Definition Documentation

6.13.1.1 LONG_DOUBLE

```
#define LONG_DOUBLE long double
```

Definition at line 609 of file [si2dr_liberty.h](#).

6.13.1.2 SI2_ARGS

```
#define SI2_ARGS(  
    args ) ()
```

Definition at line 34 of file [si2dr_liberty.h](#).

6.13.1.3 SI2_LONG_MAX

```
#define SI2_LONG_MAX 2147483647
```

Definition at line 74 of file [si2dr_liberty.h](#).

6.13.1.4 SI2_LONG_MIN

```
#define SI2_LONG_MIN -2147483647
```

Definition at line 75 of file [si2dr_liberty.h](#).

6.13.1.5 SI2_ULONG_MAX

```
#define SI2_ULONG_MAX 4294967295
```

Definition at line 76 of file [si2dr_liberty.h](#).

6.13.1.6 SI2DR_FALSE

```
#define SI2DR_FALSE SI2\_FALSE
```

Definition at line 96 of file [si2dr_liberty.h](#).

6.13.1.7 SI2DR_MAX_FLOAT32

```
#define SI2DR_MAX_FLOAT32 1E+37
```

Definition at line 87 of file [si2dr_liberty.h](#).

6.13.1.8 SI2DR_MAX_FLOAT64

```
#define SI2DR_MAX_FLOAT64 1.797693e+308
```

Definition at line 89 of file [si2dr_liberty.h](#).

6.13.1.9 SI2DR_MAX_INT32

```
#define SI2DR_MAX_INT32 2147483647
```

Definition at line 91 of file [si2dr_liberty.h](#).

6.13.1.10 SI2DR_MAX_STRING_LEN

```
#define SI2DR_MAX_STRING_LEN 1048576 /*1024*1024*/
```

Definition at line 93 of file [si2dr_liberty.h](#).

6.13.1.11 SI2DR_MIN_FLOAT32

```
#define SI2DR_MIN_FLOAT32 -1E+37
```

Definition at line 86 of file [si2dr_liberty.h](#).

6.13.1.12 SI2DR_MIN_FLOAT64

```
#define SI2DR_MIN_FLOAT64 -1.797693e+308
```

Definition at line 88 of file [si2dr_liberty.h](#).

6.13.1.13 SI2DR_MIN_INT32

```
#define SI2DR_MIN_INT32 -2147483647
```

Definition at line 90 of file [si2dr_liberty.h](#).

6.13.1.14 SI2DR_TRUE

```
#define SI2DR_TRUE SI2_TRUE
```

Definition at line 95 of file [si2dr_liberty.h](#).

6.13.2 Typedef Documentation

6.13.2.1 si2BooleanT

```
typedef enum si2BooleanEnum si2BooleanT
```

6.13.2.2 si2DoubleT

```
typedef double si2DoubleT
```

Definition at line 52 of file [si2dr_liberty.h](#).

6.13.2.3 si2drAttrComplexValIdT

```
typedef si2drObjectIdT si2drAttrComplexValIdT
```

Definition at line 111 of file [si2dr_liberty.h](#).

6.13.2.4 si2drAttrIdT

```
typedef si2drObjectIdT si2drAttrIdT
```

Definition at line 110 of file [si2dr_liberty.h](#).

6.13.2.5 si2drAttrsIdT

```
typedef si2drIterIdT si2drAttrsIdT
```

Definition at line 116 of file [si2dr_liberty.h](#).

6.13.2.6 si2drAttrTypeT

```
typedef enum si2drAttrTypeT si2drAttrTypeT
```

6.13.2.7 si2drBooleanT

```
typedef si2BooleanT si2drBooleanT
```

Definition at line 98 of file [si2dr_liberty.h](#).

6.13.2.8 si2drDefineIdT

```
typedef si2drObjectIdT si2drDefineIdT
```

Definition at line 112 of file [si2dr_liberty.h](#).

6.13.2.9 si2drDefinesIdT

```
typedef si2drIterIdT si2drDefinesIdT
```

Definition at line 117 of file [si2dr_liberty.h](#).

6.13.2.10 si2drErrorT

```
typedef enum si2drErrorT si2drErrorT
```

6.13.2.11 si2drExprT

```
typedef struct si2drExprT si2drExprT
```


6.13.2.12 si2drExprTypeT

```
typedef enum si2drExprTypeT si2drExprTypeT
```

6.13.2.13 si2drFloat32T

```
typedef si2FloatT si2drFloat32T
```

Definition at line 101 of file [si2dr_liberty.h](#).

6.13.2.14 si2drFloat64T

```
typedef si2DoubleT si2drFloat64T
```

Definition at line 102 of file [si2dr_liberty.h](#).

6.13.2.15 si2drGroupIdT

```
typedef si2drObjectIdT si2drGroupIdT
```

Definition at line 109 of file [si2dr_liberty.h](#).

6.13.2.16 si2drGroupsIdT

```
typedef si2drIterIdT si2drGroupsIdT
```

Definition at line 115 of file [si2dr_liberty.h](#).

6.13.2.17 si2drInt32T

```
typedef si2LongT si2drInt32T
```

Definition at line 100 of file [si2dr_liberty.h](#).

6.13.2.18 si2drIterIdT

```
typedef si2IteratorIdT si2drIterIdT
```

Definition at line 106 of file [si2dr_liberty.h](#).

6.13.2.19 si2drMessageHandlerT

```
typedef si2drVoidT(* si2drMessageHandlerT) (si2drSeverityT sev, si2drErrorT errToPrint, si2drStringT  
auxText, si2drErrorT *err)
```

Definition at line 215 of file [si2dr_liberty.h](#).

6.13.2.20 si2drNamesIdT

```
typedef si2drIterIdT si2drNamesIdT
```

Definition at line 118 of file [si2dr_liberty.h](#).

6.13.2.21 si2drObjectIdT

```
typedef si2ObjectIdT si2drObjectIdT
```

Definition at line 107 of file [si2dr_liberty.h](#).

6.13.2.22 si2drObjectsIdT

```
typedef si2drIterIdT si2drObjectsIdT
```

Definition at line 114 of file [si2dr_liberty.h](#).

6.13.2.23 si2drObjectTypeT

```
typedef enum si2drObjectTypeT si2drObjectTypeT
```

6.13.2.24 si2drSeverityT

```
typedef enum si2drSeverityT si2drSeverityT
```

6.13.2.25 si2drStringT

```
typedef si2StringT si2drStringT
```

Definition at line 103 of file [si2dr_liberty.h](#).

6.13.2.26 si2drValuesIdT

```
typedef si2drIterIdT si2drValuesIdT
```

Definition at line 119 of file [si2dr_liberty.h](#).

6.13.2.27 si2drValueTypeT

```
typedef enum si2drValueTypeT si2drValueTypeT
```

6.13.2.28 si2drVoidT

```
typedef si2VoidT si2drVoidT
```

Definition at line 104 of file [si2dr_liberty.h](#).

6.13.2.29 si2FloatT

```
typedef float si2FloatT
```

Definition at line 51 of file [si2dr_liberty.h](#).

6.13.2.30 si2IteratorIdT

```
typedef void* si2IteratorIdT
```

Definition at line 83 of file [si2dr_liberty.h](#).

6.13.2.31 si2LongT

```
typedef long int si2LongT
```

Definition at line 50 of file [si2dr_liberty.h](#).

6.13.2.32 si2ObjectIdT

```
typedef struct si2OID si2ObjectIdT
```

6.13.2.33 si2StringT

```
typedef char* si2StringT
```

Definition at line 53 of file [si2dr_liberty.h](#).

6.13.2.34 si2TypeT

```
typedef enum si2TypeEnum si2TypeT
```

6.13.2.35 si2VoidT

```
typedef void si2VoidT
```

Definition at line 54 of file [si2dr_liberty.h](#).

6.13.3 Enumeration Type Documentation

6.13.3.1 si2BooleanEnum

```
enum si2BooleanEnum
```

Enumerator

SI2_FALSE	
SI2_TRUE	

Definition at line 49 of file [si2dr_liberty.h](#).

6.13.3.2 si2drAttrTypeT

enum [si2drAttrTypeT](#)

Enumerator

SI2DR_SIMPLE	
SI2DR_COMPLEX	

Definition at line 139 of file [si2dr_liberty.h](#).

6.13.3.3 si2drErrorT

enum [si2drErrorT](#)

Enumerator

SI2DR_NO_ERROR	
SI2DR_INTERNAL_SYSTEM_ERROR	
SI2DR_INVALID_VALUE	
SI2DR_INVALID_NAME	
SI2DR_INVALID_OBJECTTYPE	
SI2DR_INVALID_ATTRTYPE	
SI2DR_UNUSABLE_OID	
SI2DR_OBJECT_ALREADY_EXISTS	
SI2DR_OBJECT_NOT_FOUND	
SI2DR_SYNTAX_ERROR	
SI2DR_TRACE_FILES_CANNOT_BE_OPENED	
SI2DR_PIINIT_NOT_CALLED	
SI2DR_SEMANTIC_ERROR	
SI2DR_REFERENCE_ERROR	
SI2DR_MAX_ERROR	

Definition at line 190 of file [si2dr_liberty.h](#).

6.13.3.4 si2drExprTypeT

enum [si2drExprTypeT](#)

Enumerator

SI2DR_EXPR_VAL	
SI2DR_EXPR_OP_ADD	
SI2DR_EXPR_OP_SUB	
SI2DR_EXPR_OP_MUL	
SI2DR_EXPR_OP_DIV	
SI2DR_EXPR_OP_PAREN	
SI2DR_EXPR_OP_LOG2	
SI2DR_EXPR_OP_LOG10	
SI2DR_EXPR_OP_EXP	

Definition at line [157](#) of file [si2dr_liberty.h](#).

6.13.3.5 si2drObjectTypeT

enum [si2drObjectTypeT](#)

Enumerator

SI2DR_UNDEFINED_OBJECTTYPE	
SI2DR_GROUP	
SI2DR_ATTR	
SI2DR_DEFINE	
SI2DR_COMPLEXVAL	
SI2DR_MAX_OBJECTTYPE	

Definition at line [129](#) of file [si2dr_liberty.h](#).

6.13.3.6 si2drSeverityT

enum [si2drSeverityT](#)

Enumerator

SI2DR_SEVERITY_NOTE	
SI2DR_SEVERITY_WARN	
SI2DR_SEVERITY_ERR	

Definition at line [121](#) of file [si2dr_liberty.h](#).

6.13.3.7 si2drValueTypeT

enum [si2drValueTypeT](#)

Enumerator

SI2DR_UNDEFINED_VALUETYPE	
SI2DR_INT32	
SI2DR_STRING	
SI2DR_FLOAT64	
SI2DR_BOOLEAN	
SI2DR_EXPR	
SI2DR_MAX_VALUETYPE	

Definition at line 146 of file [si2dr_liberty.h](#).

6.13.3.8 si2TypeEnum

enum [si2TypeEnum](#)

Enumerator

SI2_UNDEFINED_VALUETYPE	
SI2_BOOLEAN	
SI2_LONG	
SI2_FLOAT	
SI2_DOUBLE	
SI2_STRING	
SI2_VOID	
SI2_OID	
SI2_ITER	
SI2_EXPR	
SI2_MAX_VALUETYPE	

Definition at line 56 of file [si2dr_liberty.h](#).

6.13.4 Function Documentation

6.13.4.1 get_function()

```
void get_function (  
    char * filename )
```

6.13.4.2 liberty_destroy_value_data()

```
void liberty_destroy_value_data (
    struct liberty_value_data * vd )
```

6.13.4.3 liberty_get_element()

```
LONG_DOUBLE liberty_get_element (
    struct liberty_value_data * vd,
    ... )
```

6.13.4.4 liberty_get_values_data()

```
struct liberty_value_data * liberty_get_values_data (
    si2drGroupIdT table_group )
```

6.13.4.5 main_liberty_parser()

```
int main_liberty_parser (
    int argc,
    char * argv[] )
```

6.13.4.6 print_version()

```
void print_version ( )
```

6.13.4.7 SI2_ARGS() [1/41]

```
si2drVoidT si2drReadLibertyFile SI2_ARGS (
    (char *filename, si2drErrorT *err) )
```

6.13.4.8 SI2_ARGS() [2/41]

```
si2drVoidT si2drWriteLibertyFile SI2_ARGS (
    (char *filename, si2drGroupIdT group, char *cellname, si2drErrorT *err) )
```


6.13.4.9 SI2_ARGS() [3/41]

```
si2drVoidT si2drComplexAttrAddBooleanValue SI2_ARGS (  
    (si2drAttrIdT attr, si2drBooleanT boolval, si2drErrorT *err) )
```

6.13.4.10 SI2_ARGS() [4/41]

```
si2drVoidT si2drSimpleAttrSetBooleanValue SI2_ARGS (  
    (si2drAttrIdT attr, si2drBooleanT intgr, si2drErrorT *err) )
```

6.13.4.11 SI2_ARGS() [5/41]

```
si2drStringT si2drAttrGetComment SI2_ARGS (  
    (si2drAttrIdT attr, si2drErrorT *err) )
```

6.13.4.12 SI2_ARGS() [6/41]

```
si2drVoidT si2drSimpleAttrSetExprValue SI2_ARGS (  
    (si2drAttrIdT attr, si2drExprT *expr, si2drErrorT *err) )
```

6.13.4.13 SI2_ARGS() [7/41]

```
si2drVoidT si2drSimpleAttrSetFloat64Value SI2_ARGS (  
    (si2drAttrIdT attr, si2drFloat64T float64, si2drErrorT *err) )
```

6.13.4.14 SI2_ARGS() [8/41]

```
si2drVoidT si2drSimpleAttrSetInt32Value SI2_ARGS (  
    (si2drAttrIdT attr, si2drInt32T intgr, si2drErrorT *err) )
```

6.13.4.15 SI2_ARGS() [9/41]

```
si2drVoidT si2drAttrSetComment SI2_ARGS (  
    (si2drAttrIdT attr, si2drStringT comment, si2drErrorT *err) )
```

6.13.4.16 SI2_ARGS() [10/41]

```
si2drVoidT si2drSimpleAttrSetStringValue SI2_ARGS (  
    (si2drAttrIdT attr, si2drStringT string, si2drErrorT *err) )
```

6.13.4.17 SI2_ARGS() [11/41]

```
si2drExprT *si2drCreateBooleanValExpr SI2_ARGS (  
    (si2drBooleanT b, si2drErrorT *err) )
```

6.13.4.18 SI2_ARGS() [12/41]

```
si2drStringT si2drDefineGetComment SI2_ARGS (  
    (si2drDefineIdT def, si2drErrorT *err) )
```

6.13.4.19 SI2_ARGS() [13/41]

```
si2drVoidT si2drDefineGetInfo SI2_ARGS (  
    (si2drDefineIdT def, si2drStringT *name, si2drStringT *allowed_group_name, si2drValueTypeT  
    *valtype, si2drErrorT *err) )
```

6.13.4.20 SI2_ARGS() [14/41]

```
si2drVoidT si2drDefineSetComment SI2_ARGS (  
    (si2drDefineIdT def, si2drStringT comment, si2drErrorT *err) )
```

6.13.4.21 SI2_ARGS() [15/41]

```
si2drBooleanT si2drPIGetNocheckMode SI2_ARGS (  
    (si2drErrorT *err) )
```

6.13.4.22 SI2_ARGS() [16/41]

```
si2drStringT si2drPIGetErrorText SI2_ARGS (  
    (si2drErrorT errorCode, si2drErrorT *err) )
```

6.13.4.23 SI2_ARGS() [17/41]

```
si2drExprT *si2drOpExprGetRightExpr SI2_ARGS (  
    (si2drExprT *expr, si2drErrorT *err) )
```

6.13.4.24 SI2_ARGS() [18/41]

```
si2drExprT *si2drCreateBinaryOpExpr SI2_ARGS (  
    (si2drExprT *left, si2drExprTypeT optype, si2drExprT *right, si2drErrorT *err) )
```

6.13.4.25 SI2_ARGS() [19/41]

```
si2drExprT *si2drCreateUnaryOpExpr SI2_ARGS (  
    (si2drExprTypeT optype, si2drExprT *expr, si2drErrorT *err) )
```

6.13.4.26 SI2_ARGS() [20/41]

```
si2drExprT *si2drCreateExpr SI2_ARGS (  
    (si2drExprTypeT type, si2drErrorT *err) )
```

6.13.4.27 SI2_ARGS() [21/41]

```
si2drExprT *si2drCreateDoubleValExpr SI2_ARGS (  
    (si2drFloat64T d, si2drErrorT *err) )
```

6.13.4.28 SI2_ARGS() [22/41]

```
si2drVoidT si2drCheckLibertyLibrary SI2_ARGS (  
    (si2drGroupIdT group, si2drErrorT *err) )
```

6.13.4.29 SI2_ARGS() [23/41]

```
si2drVoidT si2drGroupSetComment SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT comment, si2drErrorT *err) )
```

6.13.4.30 SI2_ARGS() [24/41]

```
si2drAttrIdT si2drGroupCreateAttr SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drAttrTypeT type, si2drErrorT *err) )
```

6.13.4.31 SI2_ARGS() [25/41]

```
si2drDefineIdT si2drGroupFindDefineByName SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drErrorT *err) )
```

6.13.4.32 SI2_ARGS() [26/41]

```
si2drDefineIdT si2drGroupCreateDefine SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drStringT allowed_group_name, si2drValueTypeT  
    valtype, si2drErrorT *err) )
```

6.13.4.33 SI2_ARGS() [27/41]

```
si2drGroupIdT si2drGroupCreateGroup SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drStringT group_type, si2drErrorT  
    *err) )
```

6.13.4.34 SI2_ARGS() [28/41]

```
si2drGroupIdT si2drGroupFindGroupByName SI2_ARGS (  
    (si2drGroupIdT group, si2drStringT name, si2drStringT type, si2drErrorT *err) )
```

6.13.4.35 SI2_ARGS() [29/41]

```
si2drVoidT si2drGroupMoveBefore SI2_ARGS (  
    (si2drGroupIdT groupToMove, si2drGroupIdT targetGroup, si2drErrorT *err) )
```

6.13.4.36 SI2_ARGS() [30/41]

```
si2drExprT *si2drCreateIntValExpr SI2_ARGS (  
    (si2drInt32T i, si2drErrorT *err) )
```

6.13.4.37 SI2_ARGS() [31/41]

```
si2drStringT si2drObjectGetFileName SI2_ARGS (  
    (si2drObjectIdT object, si2drErrorT *err) )
```

6.13.4.38 SI2_ARGS() [32/41]

```
si2drVoidT si2drObjectSetLineNo SI2_ARGS (  
    (si2drObjectIdT object, si2drInt32T lineno, si2drErrorT *err) )
```

6.13.4.39 SI2_ARGS() [33/41]

```
si2drVoidT si2drObjectSetFileName SI2_ARGS (  
    (si2drObjectIdT object, si2drStringT filename, si2drErrorT *err) )
```

6.13.4.40 SI2_ARGS() [34/41]

```
si2drBooleanT si2drObjectIsSame SI2_ARGS (  
    (si2drObjectIdT object1, si2drObjectIdT object2, si2drErrorT *err) )
```

6.13.4.41 SI2_ARGS() [35/41]

```
si2drVoidT si2drPISetTraceMode SI2_ARGS (  
    (si2drStringT fname, si2drErrorT *err) )
```

6.13.4.42 SI2_ARGS() [36/41]

```
si2drDefineIdT si2drPIFindDefineByName SI2_ARGS (  
    (si2drStringT name, si2drErrorT *err) )
```

6.13.4.43 SI2_ARGS() [37/41]

```
si2drGroupIdT si2drPICreateGroup SI2_ARGS (  
    (si2drStringT name, si2drStringT group_type, si2drErrorT *err) )
```

6.13.4.44 SI2_ARGS() [38/41]

```
si2drGroupIdT si2drPIFindGroupByName SI2_ARGS (
    (si2drStringT name, si2drStringT type, si2drErrorT *err) )
```

6.13.4.45 SI2_ARGS() [39/41]

```
si2drExprT *si2drCreateStringValExpr SI2_ARGS (
    (si2drStringT s, si2drErrorT *err) )
```

6.13.4.46 SI2_ARGS() [40/41]

```
si2drVoidT si2drIterQuit SI2_ARGS (
    (si2drValuesIdT iter, si2drErrorT *err) )
```

6.13.4.47 SI2_ARGS() [41/41]

```
si2drVoidT si2drIterNextComplexValue SI2_ARGS (
    (si2drValuesIdT iter, si2drValueTypeT *type, si2drInt32T *intgr, si2drFloat64T
    *float64, si2drStringT *string, si2drBooleanT *boolval, si2drExprT **expr, si2drErrorT *err) )
```

6.13.4.48 si2drDefaultMessageHandler()

```
si2drVoidT si2drDefaultMessageHandler (
    si2drSeverityT sev,
    si2drErrorT errToPrint,
    si2drStringT auxText,
    si2drErrorT * err )
```

6.13.4.49 si2drPIGetMessageHandler()

```
si2drMessageHandlerT si2drPIGetMessageHandler (
    si2drErrorT * err )
```

6.13.4.50 si2drPISetMessageHandler()

```

si2drVoidT si2drPISetMessageHandler (
    si2drMessageHandlerT MsgHandFuncPtr,
    si2drErrorT * err )

```

6.13.4.51 usage()

```

void usage ( )

```

6.14 si2dr_liberty.h

[Go to the documentation of this file.](#)

```

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00016
00017 *****/
00018
00019 /* define a PI for manipulating the liberty data */
00020
00021
00022 /* basic data types */
00023
00024
00025 #ifndef SI2DR_H
00026 #define SI2DR_H
00027
00028 #ifdef SI2_ARGS
00029 #undef SI2_ARGS
00030 #endif
00031 #if defined(__STDC__) || defined(HPPA) || defined(ibmpoweraix) || defined(_MSC_VER) ||
    defined(__cplusplus)
00032 #define SI2_ARGS(args)  args
00033 #else
00034 #define SI2_ARGS(args)  ()
00035 #endif
00036
00037
00038 /* *****
00039     File contents : LIBERTY-DR type definitions.
00040     ***** */
00041 #ifdef __cplusplus
00042 extern "C" {
00043 #endif
00044
00045 /*
00046  * These primitive types are compatible with the ANSI C Standard (ANSI/
00047  * X3.159-1990 Programming Language C Standard.)
00048  */
00049 typedef enum si2BooleanEnum { SI2_FALSE = 0, SI2_TRUE = 1 } si2BooleanT;
00050 typedef long int si2LongT;
00051 typedef float si2FloatT;
00052 typedef double si2DoubleT;
00053 typedef char* si2StringT;
00054 typedef void si2VoidT;
00055
00056 typedef enum si2TypeEnum {
00057     SI2_UNDEFINED_VALUETYPE = 0,

```

```

00058     SI2_BOOLEAN = 1,
00059     SI2_LONG = 2,
00060     SI2_FLOAT = 3,
00061     SI2_DOUBLE = 4,
00062     SI2_STRING = 5,
00063     SI2_VOID = 6,
00064     SI2_OID = 7,
00065     SI2_ITER = 8,
00066     SI2_EXPR = 9,
00067     SI2_MAX_VALUETYPE = 10
00068 } si2TypeT;
00069
00070 /*
00071  * Size limits
00072  */
00073
00074 #define SI2_LONG_MAX    2147483647
00075 #define SI2_LONG_MIN    -2147483647
00076 #define SI2_ULONG_MAX   4294967295
00077
00078
00079 typedef struct si2OID {void *v1, *v2;} si2ObjectIdT;
00080 /*
00081  * Iterators are blind handles the size of a single pointer.
00082  */
00083 typedef void *si2IteratorIdT;
00084
00085
00086 #define SI2DR_MIN_FLOAT32    -1E+37
00087 #define SI2DR_MAX_FLOAT32    1E+37
00088 #define SI2DR_MIN_FLOAT64    -1.797693e+308
00089 #define SI2DR_MAX_FLOAT64    1.797693e+308
00090 #define SI2DR_MIN_INT32      -2147483647
00091 #define SI2DR_MAX_INT32      2147483647
00092
00093 #define SI2DR_MAX_STRING_LEN  1048576 /*1024*1024*/
00094
00095 #define SI2DR_TRUE SI2_TRUE
00096 #define SI2DR_FALSE SI2_FALSE
00097
00098 typedef si2BooleanT si2drBooleanT;
00099
00100 typedef si2LongT    si2drInt32T;
00101 typedef si2FloatT   si2drFloat32T;
00102 typedef si2DoubleT   si2drFloat64T;
00103 typedef si2StringT   si2drStringT;
00104 typedef si2VoidT     si2drVoidT;
00105
00106 typedef si2IteratorIdT    si2drIterIdT;
00107 typedef si2ObjectIdT      si2drObjectIdT;
00108
00109 typedef si2drObjectIdT si2drGroupIdT;
00110 typedef si2drObjectIdT si2drAttrIdT;
00111 typedef si2drObjectIdT si2drAttrComplexValIdT;
00112 typedef si2drObjectIdT si2drDefineIdT;
00113
00114 typedef si2drIterIdT si2drObjectsIdT;
00115 typedef si2drIterIdT si2drGroupsIdT;
00116 typedef si2drIterIdT si2drAttrsIdT;
00117 typedef si2drIterIdT si2drDefinesIdT;
00118 typedef si2drIterIdT si2drNamesIdT;
00119 typedef si2drIterIdT si2drValuesIdT;
00120
00121 typedef enum si2drSeverityT
00122 {
00123     SI2DR_SEVERITY_NOTE = 0,
00124     SI2DR_SEVERITY_WARN = 1,
00125     SI2DR_SEVERITY_ERR  = 2
00126 } si2drSeverityT;
00127
00128
00129 typedef enum si2drObjectTypeT
00130 {
00131     SI2DR_UNDEFINED_OBJECTTYPE = 0,
00132     SI2DR_GROUP                 = 1,
00133     SI2DR_ATTR                  = 2,
00134     SI2DR_DEFINE                = 3,
00135     SI2DR_COMPLEXVAL            = 4,
00136     SI2DR_MAX_OBJECTTYPE       = 5
00137 } si2drObjectTypeT;
00138
00139 typedef enum si2drAttrTypeT
00140 {
00141     SI2DR_SIMPLE,
00142     SI2DR_COMPLEX
00143 } si2drAttrTypeT;
00144

```



```

00145
00146 typedef enum si2drValueTypeT
00147 {
00148     SI2DR_UNDEFINED_VALUETYPE = SI2_UNDEFINED_VALUETYPE,
00149     SI2DR_INT32                = SI2_LONG,
00150     SI2DR_STRING               = SI2_STRING,
00151     SI2DR_FLOAT64              = SI2_DOUBLE,
00152     SI2DR_BOOLEAN              = SI2_BOOLEAN,
00153     SI2DR_EXPR                 = SI2_EXPR,
00154     SI2DR_MAX_VALUETYPE        = SI2_MAX_VALUETYPE
00155 } si2drValueTypeT;
00156
00157 typedef enum si2drExprTypeT
00158 {
00159     SI2DR_EXPR_VAL = 0, /* unary ops just have left arg set: add, sub, logs, paren */
00160     SI2DR_EXPR_OP_ADD = 1,
00161     SI2DR_EXPR_OP_SUB = 2,
00162     SI2DR_EXPR_OP_MUL = 3,
00163     SI2DR_EXPR_OP_DIV = 4,
00164     SI2DR_EXPR_OP_PAREN = 5,
00165     SI2DR_EXPR_OP_LOG2 = 6, /* pretty much flight of fancy from here on? */
00166     SI2DR_EXPR_OP_LOG10 = 7,
00167     SI2DR_EXPR_OP_EXP = 8, /* exponent? */
00168 } si2drExprTypeT;
00169
00170
00171
00172 typedef struct si2drExprT
00173 {
00174     si2drExprTypeT type;
00175     union
00176     {
00177         si2drInt32T i;
00178         si2drFloat64T d;
00179         si2drStringT s; /* most likely an identifier */
00180         si2drBooleanT b;
00181     } u;
00182     si2drValueTypeT valuetype; /* if the type is a fixed value */
00183     struct si2drExprT *left; /* the exprs form a classic binary tree rep of an arithmetic expression
00184     */
00185     struct si2drExprT *right;
00186 } si2drExprT;
00187
00188
00189
00190 typedef enum si2drErrorT
00191 {
00192     SI2DR_NO_ERROR = 0,
00193     SI2DR_INTERNAL_SYSTEM_ERROR = 1,
00194     SI2DR_INVALID_VALUE = 4,
00195     SI2DR_INVALID_NAME = 5,
00196     SI2DR_INVALID_OBJECTTYPE = 6,
00197     SI2DR_INVALID_ATTRTYPE = 10,
00198     SI2DR_UNUSABLE_OID = 7,
00199     SI2DR_OBJECT_ALREADY_EXISTS = 8,
00200     SI2DR_OBJECT_NOT_FOUND = 9,
00201     SI2DR_SYNTAX_ERROR = 2,
00202     SI2DR_TRACE_FILES_CANNOT_BE_OPENED = 3,
00203     SI2DR_PIINIT_NOT_CALLED = 11,
00204     SI2DR_SEMANTIC_ERROR = 12,
00205     SI2DR_REFERENCE_ERROR = 13,
00206     SI2DR_MAX_ERROR = 14
00207 } si2drErrorT;
00208
00209 si2drVoidT si2drDefaultMessageHandler(si2drSeverityT sev,
00210                                       si2drErrorT errToPrint,
00211                                       si2drStringT auxText,
00212                                       si2drErrorT *err);
00213
00214
00215     typedef si2drVoidT (*si2drMessageHandlerT)(si2drSeverityT sev,
00216                                                 si2drErrorT errToPrint,
00217                                                 si2drStringT auxText,
00218                                                 si2drErrorT *err);
00219
00220
00221 si2drMessageHandlerT si2drPIGetMessageHandler( si2drErrorT *err);
00222
00223
00224 si2drVoidT si2drPISetMessageHandler( si2drMessageHandlerT MsgHandFuncPtr,
00225                                       si2drErrorT *err);
00226
00227
00228     si2drGroupIdT si2drPICreateGroup SI2_ARGS(( si2drStringT name,
00229                                                  si2drStringT group_type,
00230                                                  si2drErrorT *err));

```

```

00231
00232
00233     si2drAttrIdT    si2drGroupCreateAttr    SI2_ARGS(( si2drGroupIdT    group,
00234                                                     si2drStringT        name,
00235                                                     si2drAttrTypeT     type,
00236                                                     si2drErrorT        *err));
00237
00238     si2drAttrTypeT si2drAttrGetAttrType    SI2_ARGS(( si2drAttrIdT    attr,
00239                                                     si2drErrorT        *err));
00240
00241     si2drStringT   si2drAttrGetName        SI2_ARGS(( si2drAttrIdT    attr,
00242                                                     si2drErrorT        *err));
00243
00244     si2drVoidT     si2drComplexAttrAddInt32Value    SI2_ARGS(( si2drAttrIdT    attr,
00245                                                     si2drInt32T        intgr,
00246                                                     si2drErrorT        *err ));
00247
00248     si2drVoidT     si2drComplexAttrAddStringValue    SI2_ARGS(( si2drAttrIdT    attr,
00249                                                     si2drStringT        string,
00250                                                     si2drErrorT        *err ));
00251
00252     si2drVoidT     si2drComplexAttrAddBooleanValue    SI2_ARGS(( si2drAttrIdT    attr,
00253                                                     si2drBooleanT        boolval,
00254                                                     si2drErrorT        *err ));
00255
00256     si2drVoidT     si2drComplexAttrAddFloat64Value    SI2_ARGS(( si2drAttrIdT    attr,
00257                                                     si2drFloat64T        float64,
00258                                                     si2drErrorT        *err ));
00259
00260     si2drVoidT     si2drComplexAttrAddExprValue    SI2_ARGS(( si2drAttrIdT    attr,
00261                                                     si2drExprT          *expr,
00262                                                     si2drErrorT        *err ));
00263
00264     si2drValuesIdT si2drComplexAttrGetValues    SI2_ARGS(( si2drAttrIdT    attr,
00265                                                     si2drErrorT        *err ));
00266
00267     si2drVoidT     si2drIterNextComplexValue    SI2_ARGS(( si2drValuesIdT    iter,
00268                                                     si2drValueTypeT     *type,
00269                                                     si2drInt32T          *intgr,
00270                                                     si2drFloat64T        *float64,
00271                                                     si2drStringT          *string,
00272                                                     si2drBooleanT        *boolval,
00273                                                     si2drExprT          **expr,
00274                                                     si2drErrorT        *err ));
00275
00276     si2drAttrComplexValIdT si2drIterNextComplex    SI2_ARGS(( si2drValuesIdT    iter,
00277                                                     si2drErrorT        *err ));
00278
00279     si2drValueTypeT si2drComplexValGetValueType    SI2_ARGS(( si2drAttrComplexValIdT    attr,
00280                                                     si2drErrorT        *err ));
00281
00282     si2drInt32T     si2drComplexValGetInt32Value    SI2_ARGS(( si2drAttrComplexValIdT    attr,
00283                                                     si2drErrorT        *err ));
00284
00285     si2drFloat64T   si2drComplexValGetFloat64Value    SI2_ARGS(( si2drAttrComplexValIdT    attr,
00286                                                     si2drErrorT        *err ));
00287
00288     si2drStringT     si2drComplexValGetStringValue    SI2_ARGS(( si2drAttrComplexValIdT    attr,
00289                                                     si2drErrorT        *err ));
00290
00291     si2drBooleanT    si2drComplexValGetBooleanValue    SI2_ARGS(( si2drAttrComplexValIdT    attr,
00292                                                     si2drErrorT        *err ));
00293
00294     si2drExprT       *si2drComplexValGetExprValue    SI2_ARGS(( si2drAttrComplexValIdT    attr,
00295                                                     si2drErrorT        *err ));
00296
00297
00298
00299
00300
00301
00302     si2drValueTypeT si2drSimpleAttrGetValueType    SI2_ARGS(( si2drAttrIdT    attr,
00303                                                     si2drErrorT        *err ));
00304
00305     si2drInt32T     si2drSimpleAttrGetInt32Value    SI2_ARGS(( si2drAttrIdT    attr,
00306                                                     si2drErrorT        *err ));
00307
00308     si2drFloat64T   si2drSimpleAttrGetFloat64Value    SI2_ARGS(( si2drAttrIdT    attr,
00309                                                     si2drErrorT        *err ));
00310
00311     si2drStringT     si2drSimpleAttrGetStringValue    SI2_ARGS(( si2drAttrIdT    attr,
00312                                                     si2drErrorT        *err ));
00313
00314     si2drBooleanT    si2drSimpleAttrGetBooleanValue    SI2_ARGS(( si2drAttrIdT    attr,
00315                                                     si2drErrorT        *err ));
00316
00317     si2drExprT       *si2drSimpleAttrGetExprValue    SI2_ARGS(( si2drAttrIdT    attr,

```

```

00318                                     si2drErrorT      *err ));
00319
00320
00321
00322     si2drVoidT      si2drSimpleAttrSetInt32Value      SI2_ARGS(( si2drAttrIdT attr,
00323                                     si2drInt32T      intgr,
00324                                     si2drErrorT      *err ));
00325
00326     si2drVoidT      si2drSimpleAttrSetBooleanValue    SI2_ARGS(( si2drAttrIdT attr,
00327                                     si2drBooleanT    intgr,
00328                                     si2drErrorT      *err ));
00329
00330     si2drVoidT      si2drSimpleAttrSetFloat64Value    SI2_ARGS(( si2drAttrIdT attr,
00331                                     si2drFloat64T    float64,
00332                                     si2drErrorT      *err ));
00333
00334     si2drVoidT      si2drSimpleAttrSetStringValue      SI2_ARGS(( si2drAttrIdT attr,
00335                                     si2drStringT      string,
00336                                     si2drErrorT      *err ));
00337
00338     si2drVoidT      si2drSimpleAttrSetExprValue        SI2_ARGS(( si2drAttrIdT attr,
00339                                     si2drExprT        *expr,
00340                                     si2drErrorT      *err ));
00341
00342     si2drBooleanT    si2drSimpleAttrGetIsVar            SI2_ARGS(( si2drAttrIdT attr,
00343                                     si2drErrorT      *err ));
00344
00345     si2drVoidT      si2drSimpleAttrSetIsVar            SI2_ARGS(( si2drAttrIdT attr,
00346                                     si2drErrorT      *err ));
00347
00348     /* helper functions for expr creation, building, and destruction */
00349
00350     si2drVoidT      si2drExprDestroy                  SI2_ARGS(( si2drExprT *expr, /* recursively
free the structures */
00351                                     si2drErrorT *err));
00352
00353     si2drExprT      *si2drCreateExpr                  SI2_ARGS(( si2drExprTypeT type, /* malloc an
Expr and return it */
00354                                     si2drErrorT *err));
00355
00356     si2drExprT      *si2drCreateBooleanValExpr        SI2_ARGS(( si2drBooleanT b,
00357                                     si2drErrorT *err ));
00358
00359     si2drExprT      *si2drCreateDoubleValExpr         SI2_ARGS(( si2drFloat64T d,
00360                                     si2drErrorT *err ));
00361
00362     si2drExprT      *si2drCreateStringValExpr         SI2_ARGS(( si2drStringT s,
00363                                     si2drErrorT *err ));
00364
00365     si2drExprT      *si2drCreateIntValExpr            SI2_ARGS(( si2drInt32T i,
00366                                     si2drErrorT *err ));
00367
00368     si2drExprT      *si2drCreateBinaryOpExpr          SI2_ARGS(( si2drExprT *left,
00369                                     si2drExprTypeT optype,
00370                                     si2drExprT *right,
00371                                     si2drErrorT *err ));
00372
00373     si2drExprT      *si2drCreateUnaryOpExpr           SI2_ARGS(( si2drExprTypeT optype,
00374                                     si2drExprT *expr,
00375                                     si2drErrorT *err ));
00376
00377     si2drStringT      si2drExprToString                SI2_ARGS(( si2drExprT *expr,
00378                                     si2drErrorT *err ));
00379
00380     si2drExprTypeT    si2drExprGetType                SI2_ARGS(( si2drExprT *expr,
00381                                     si2drErrorT *err ));
00382
00383     si2drValueTypeT    si2drValExprGetValueType        SI2_ARGS(( si2drExprT *expr,
00384                                     si2drErrorT *err ));
00385
00386     si2drInt32T      si2drIntValExprGetInt            SI2_ARGS(( si2drExprT *expr,
00387                                     si2drErrorT *err ));
00388
00389     si2drFloat64T      si2drDoubleValExprGetDouble    SI2_ARGS(( si2drExprT *expr,
00390                                     si2drErrorT *err ));
00391
00392     si2drBooleanT      si2drBooleanValExprGetBoolean  SI2_ARGS(( si2drExprT *expr,
00393                                     si2drErrorT *err ));
00394
00395     si2drStringT      si2drStringValExprGetString     SI2_ARGS(( si2drExprT *expr,
00396                                     si2drErrorT *err ));
00397
00398     si2drExprT      *si2drOpExprGetLeftExpr          SI2_ARGS(( si2drExprT *expr, /* will apply to Unary & binary
Ops */
00399                                     si2drErrorT *err ));
00400
00401     si2drExprT      *si2drOpExprGetRightExpr          SI2_ARGS(( si2drExprT *expr,

```

```

00402                                     si2drErrorT *err ));
00403
00404 /* Define related funcs */
00405
00406     si2drDefineIdT si2drGroupCreateDefine SI2_ARGS(( si2drGroupIdT group,
00407                                                       si2drStringT name,
00408                                                       si2drStringT allowed_group_name,
00409                                                       si2drValueTypeT valtype,
00410                                                       si2drErrorT *err));
00411
00412     si2drVoidT      si2drDefineGetInfo      SI2_ARGS(( si2drDefineIdT def,
00413                                                         si2drStringT *name,
00414                                                         si2drStringT *allowed_group_name,
00415                                                         si2drValueTypeT *valtype,
00416                                                         si2drErrorT *err));
00417
00418     si2drStringT    si2drDefineGetName      SI2_ARGS(( si2drDefineIdT def,
00419                                                         si2drErrorT *err));
00420
00421     si2drStringT    si2drDefineGetAllowedGroupName SI2_ARGS(( si2drDefineIdT def,
00422                                                                si2drErrorT *err));
00423
00424     si2drValueTypeT si2drDefineGetValueType SI2_ARGS(( si2drDefineIdT def,
00425                                                         si2drErrorT *err));
00426
00427     si2drGroupIdT  si2drGroupCreateGroup SI2_ARGS(( si2drGroupIdT group,
00428                                                       si2drStringT name,
00429                                                       si2drStringT group_type,
00430                                                       si2drErrorT *err));
00431
00432     si2drStringT    si2drGroupGetGroupType SI2_ARGS(( si2drGroupIdT group,
00433                                                         si2drErrorT *err));
00434
00435     si2drStringT    si2drGroupGetComment SI2_ARGS(( si2drGroupIdT group,
00436                                                       si2drErrorT *err));
00437
00438     si2drVoidT      si2drGroupSetComment SI2_ARGS(( si2drGroupIdT group,
00439                                                       si2drStringT comment,
00440                                                       si2drErrorT *err));
00441
00442     si2drStringT    si2drAttrGetComment SI2_ARGS(( si2drAttrIdT attr,
00443                                                       si2drErrorT *err));
00444
00445     si2drVoidT      si2drAttrSetComment SI2_ARGS(( si2drAttrIdT attr,
00446                                                       si2drStringT comment,
00447                                                       si2drErrorT *err));
00448
00449     si2drStringT    si2drDefineGetComment SI2_ARGS(( si2drDefineIdT def,
00450                                                       si2drErrorT *err));
00451
00452     si2drVoidT      si2drDefineSetComment SI2_ARGS(( si2drDefineIdT def,
00453                                                       si2drStringT comment,
00454                                                       si2drErrorT *err));
00455
00456     si2drVoidT      si2drGroupAddName SI2_ARGS(( si2drGroupIdT group,
00457                                                       si2drStringT name,
00458                                                       si2drErrorT *err));
00459
00460     si2drVoidT      si2drGroupDeleteName SI2_ARGS(( si2drGroupIdT group,
00461                                                       si2drStringT name,
00462                                                       si2drErrorT *err));
00463
00464
00465     si2drGroupIdT  si2drPIFindGroupByName SI2_ARGS(( si2drStringT name,
00466                                                       si2drStringT type,
00467                                                       si2drErrorT *err));
00468
00469     si2drGroupIdT  si2drGroupFindGroupByName SI2_ARGS(( si2drGroupIdT group,
00470                                                       si2drStringT name,
00471                                                       si2drStringT type,
00472                                                       si2drErrorT *err));
00473
00474     si2drAttrIdT   si2drGroupFindAttrByName SI2_ARGS(( si2drGroupIdT group,
00475                                                       si2drStringT name,
00476                                                       si2drErrorT *err));
00477
00478     si2drDefineIdT si2drGroupFindDefineByName SI2_ARGS(( si2drGroupIdT group,
00479                                                       si2drStringT name,
00480                                                       si2drErrorT *err));
00481
00482
00483     si2drDefineIdT si2drPIFindDefineByName SI2_ARGS(( si2drStringT name,
00484                                                       si2drErrorT *err));
00485
00486
00487
00488     si2drGroupsIdT si2drPIGetGroups SI2_ARGS(( si2drErrorT *err));

```

```

00489
00490
00491     si2drGroupsIdT    si2drGroupGetGroups    SI2_ARGS(( si2drGroupIdT group,
00492             si2drErrorT *err));
00493
00494     si2drNamesIdT     si2drGroupGetNames     SI2_ARGS(( si2drGroupIdT group,
00495             si2drErrorT *err));
00496
00497     si2drAttrsIdT     si2drGroupGetAttrs     SI2_ARGS(( si2drGroupIdT group,
00498             si2drErrorT *err));
00499
00500     si2drDefinesIdT   si2drGroupGetDefines   SI2_ARGS(( si2drGroupIdT group,
00501             si2drErrorT *err));
00502
00503     si2drGroupIdT     si2drIterNextGroup     SI2_ARGS(( si2drGroupsIdT iter,
00504             si2drErrorT *err));
00505     si2drStringT      si2drIterNextName     SI2_ARGS(( si2drNamesIdT iter,
00506             si2drErrorT *err));
00507     si2drAttrIdT      si2drIterNextAttr     SI2_ARGS(( si2drAttrsIdT iter,
00508             si2drErrorT *err));
00509     si2drDefineIdT    si2drIterNextDefine    SI2_ARGS(( si2drDefinesIdT iter,
00510             si2drErrorT *err));
00511     si2drVoidT        si2drIterQuit         SI2_ARGS(( si2drIterIdT iter,
00512             si2drErrorT *err));
00513
00514
00515
00516     si2drVoidT        si2drObjectDelete     SI2_ARGS(( si2drObjectIdT object,
00517             si2drErrorT *err));
00518
00519
00520
00521     si2drStringT      si2drPIGetErrorText    SI2_ARGS(( si2drErrorT errorCode,
00522             si2drErrorT *err));
00523
00524     si2drObjectIdT    si2drPIGetNullId       SI2_ARGS(( si2drErrorT *err));
00525
00526     si2drVoidT        si2drPIInit            SI2_ARGS(( si2drErrorT *err));
00527
00528     si2drVoidT        si2drPIQuit           SI2_ARGS(( si2drErrorT *err));
00529
00530     si2drObjectTypeT  si2drObjectGetObjectType SI2_ARGS(( si2drObjectIdT object,
00531             si2drErrorT *err));
00532
00533     si2drObjectIdT    si2drObjectGetOwner     SI2_ARGS(( si2drObjectIdT object,
00534             si2drErrorT *err));
00535
00536     si2drBooleanT     si2drObjectIsNull      SI2_ARGS(( si2drObjectIdT object,
00537             si2drErrorT *err));
00538
00539     si2drBooleanT     si2drObjectIsSame      SI2_ARGS(( si2drObjectIdT object1,
00540             si2drObjectIdT object2,
00541             si2drErrorT *err));
00542
00543     si2drBooleanT     si2drObjectIsUsable    SI2_ARGS(( si2drObjectIdT object,
00544             si2drErrorT *err));
00545
00546     si2drVoidT        si2drObjectSetFileName SI2_ARGS(( si2drObjectIdT object,
00547             si2drStringT filename,
00548             si2drErrorT *err));
00549
00550     si2drVoidT        si2drObjectSetLineNo   SI2_ARGS(( si2drObjectIdT object,
00551             si2drInt32T lineno,
00552             si2drErrorT *err));
00553
00554     si2drInt32T       si2drObjectGetLineNo    SI2_ARGS(( si2drObjectIdT object,
00555             si2drErrorT *err));
00556
00557     si2drStringT      si2drObjectGetFileName SI2_ARGS(( si2drObjectIdT object,
00558             si2drErrorT *err));
00559
00560     si2drVoidT        si2drReadLibertyFile   SI2_ARGS(( char *filename,
00561             si2drErrorT *err));
00562
00563     si2drVoidT        si2drWriteLibertyFile  SI2_ARGS(( char *filename,
00564             si2drGroupIdT group,
00565             char* cellname,
00566             si2drErrorT *err ));
00567
00568     si2drVoidT        si2drCheckLibertyLibrary SI2_ARGS(( si2drGroupIdT group,
00569             si2drErrorT *err));
00570
00571
00572     si2drBooleanT     si2drPIGetTraceMode    SI2_ARGS(( si2drErrorT *err));
00573
00574     si2drVoidT        si2drPIUnSetTraceMode  SI2_ARGS(( si2drErrorT *err));
00575

```

```

00576     si2drVoidT      si2drPISetTraceMode      SI2_ARGS((si2drStringT fname,
00577                                           si2drErrorT *err));
00578
00579
00580     si2drVoidT      si2drPISetDebugMode      SI2_ARGS((si2drErrorT *err));
00581
00582     si2drVoidT      si2drPIUnSetDebugMode    SI2_ARGS((si2drErrorT *err));
00583
00584     si2drBooleanT   si2drPIGetDebugMode      SI2_ARGS((si2drErrorT *err));
00585
00586
00587     si2drVoidT      si2drPISetNocheckMode    SI2_ARGS((si2drErrorT *err));
00588
00589     si2drVoidT      si2drPIUnSetNocheckMode  SI2_ARGS((si2drErrorT *err));
00590
00591     si2drBooleanT   si2drPIGetNocheckMode    SI2_ARGS((si2drErrorT *err));
00592
00593
00594     si2drVoidT      si2drGroupMoveAfter SI2_ARGS((si2drGroupIdT groupToMove,
00595                                           si2drGroupIdT targetGroup,
00596                                           si2drErrorT *err));
00597
00598     si2drVoidT      si2drGroupMoveBefore SI2_ARGS((si2drGroupIdT groupToMove,
00599                                           si2drGroupIdT targetGroup,
00600                                           si2drErrorT *err));
00601
00602
00603 /* HELPER FUNCTIONS/STRUCTURES */
00604
00605 #ifdef NO_LONG_DOUBLE
00606 #define LONG_DOUBLE double
00607 #define strtold strtod
00608 #else
00609 #define LONG_DOUBLE long double
00610 #endif
00611
00612 struct liberty_value_data
00613 {
00614     int dimensions;
00615     int *dim_sizes; /* a malloc'd array of <dimensions> numbers,
00616                     each number is the size of the array in that dim.*/
00617     LONG_DOUBLE **index_info; /* a Ptr to a malloc'd array of size <dimensions>, ptrs to
00618                               malloc'd arrays of long double index values.
00619                               Each array of long double is of length set by
00620                               the corresponding dim_sizes array values */
00621     LONG_DOUBLE *values; /* a ptr to a malloc'd array of long doubles,
00622                           starting with [0,0,...,0,0], and ending with
00623                           [z-1,y-1,...,b-1,a-1], where a-z are the max
00624                           number of elements in each dimension. */
00625 };
00626
00627 LONG_DOUBLE liberty_get_element(struct liberty_value_data *vd, ...); /* returns NaN if bounds
exceeded */
00628
00629 void liberty_destroy_value_data(struct liberty_value_data *vd);
00630
00631 struct liberty_value_data *liberty_get_values_data( si2drGroupIdT table_group);
00632
00633 // directly call the main function of liberty parser
00634 extern void print_version();
00635 extern void usage();
00636 extern int main_liberty_parser(int argc, char* argv[]);
00637 extern void get_function(char* filename);
00638
00639 #ifdef __cplusplus
00640 }
00641 #endif
00642
00643 #endif
00644

```

6.15 include/verilog_utils.hpp File Reference

```

#include <fstream>
#include <iostream>
#include <string>
#include <unordered_set>
#include "slang/syntax/SyntaxPrinter.h"

```

```
#include "slang/syntax/SyntaxTree.h"
#include "slang/syntax/SyntaxVisitor.h"
```

Include dependency graph for verilog_utils.hpp: This graph shows which files directly or indirectly include this file:

Classes

- class [VerilogVisitor](#)
- class [CellExtractor](#)
- class [CellPrinter](#)
- class [ModuleRewriter](#)

Functions

- void [getAST](#) (const std::string &verilog_file, const std::string &cell)

6.15.1 Function Documentation

6.15.1.1 getAST()

```
void getAST (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 408 of file [verilog_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.16 verilog_utils.hpp

[Go to the documentation of this file.](#)

```
00001 #ifndef VERILOG_UTILS_H
00002 #define VERILOG_UTILS_H
00003
00004 #include <fstream>
00005 #include <iostream> // For std::ostream
00006 #include <string>
00007 #include <unordered_set>
00008
00009 #include "slang/syntax/SyntaxPrinter.h"
00010 #include "slang/syntax/SyntaxTree.h"
00011 #include "slang/syntax/SyntaxVisitor.h"
00012
00013 // Creating custom visitor class
00014 class VerilogVisitor : public slang::syntax::SyntaxVisitor<VerilogVisitor> {
00015 public:
00016     explicit VerilogVisitor(const std::string &targetCell)
00017         : targetCell_(targetCell), depth_(0), inTargetModule_(false) {}
00018     void handle(const slang::syntax::SyntaxNode &node);
00019     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00020     void handle(const slang::syntax::PortDeclarationSyntax &portDecl);
00021     void handle(const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst);
00022     void handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst);
00023     void handle(const slang::syntax::SpecifyBlockSyntax &specifyBlock);
00024
00025 private:
00026     const std::string &targetCell_;
```

```

00027     int depth_;
00028     bool inTargetModule_;
00029 };
00030
00031 // Creating a Rewriter to extract a specific cell
00032 class CellExtractor : public slang::syntax::SyntaxRewriter<CellExtractor> {
00033 public:
00034     explicit CellExtractor(const std::string &targetCell) : targetCell_(targetCell), foundTarget_(false)
00035     {}
00036     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00037     bool foundTargetCell() const;
00038 private:
00039     const std::string &targetCell_;
00040     bool foundTarget_;
00041 };
00042
00043 // Print specific cell when visiting SyntaxTree
00044 class CellPrinter : public slang::syntax::SyntaxVisitor<CellPrinter> {
00045 public:
00046     explicit CellPrinter(const std::string &targetCell, std::ostream &out)
00047         : targetCell_(targetCell), out_(out), foundTarget_(false) {}
00048     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00049 private:
00050     const std::string &targetCell_;
00051     std::ostream &out_;
00052     bool foundTarget_;
00053 };
00054
00055 // Comprehensive module rewriter for adding ports, instances, and connections
00056 class ModuleRewriter : public slang::syntax::SyntaxRewriter<ModuleRewriter> {
00057 public:
00058     explicit ModuleRewriter(const std::vector<std::string> &inputPins,
00059                             const std::vector<std::string> &outputPins,
00060                             const std::pair<std::string, std::string> &superCellEntry,
00061                             int instanceCount,
00062                             std::shared_ptr<spdlog::logger> logger)
00063         : inputPins_(inputPins), outputPins_(outputPins), cellName_(superCellEntry.first),
00064           moduleName_(superCellEntry.second), logger_(logger), depth_(0),
00065           instanceCount_(instanceCount) {}
00066     std::shared_ptr<spdlog::logger> logger_;
00067     void handle(const slang::syntax::SyntaxNode &node);
00068     void handle(const slang::syntax::ModuleDeclarationSyntax &module);
00069 private:
00070     const std::vector<std::string> &inputPins_;
00071     const std::vector<std::string> &outputPins_;
00072     const std::string cellName_;
00073     const std::string moduleName_;
00074     int depth_;
00075     int instanceCount_;
00076 };
00077
00078 void getAST(const std::string &verilog_file, const std::string &cell);
00079
00080 #endif // VERILOG_UTILS_H

```

6.17 include/version.h File Reference

This graph shows which files directly or indirectly include this file:

Macros

- #define APP_NAME "ZlibValidation"
- #define APP_VERSION_MAJOR 0
- #define APP_VERSION_MINOR 1
- #define APP_VERSION_PATCH 0
- #define APP_VERSION "0.1.0"
- #define APP_AUTHOR "Song Zixuan"
- #define APP_CONTACT "cedar@zju.edu.cn"
- #define BUILD_TIMESTAMP "2025-03-28 09:43:58"

6.17.1 Macro Definition Documentation

6.17.1.1 APP_AUTHOR

```
#define APP_AUTHOR "Song Zixuan"
```

Definition at line 10 of file [version.h](#).

6.17.1.2 APP_CONTACT

```
#define APP_CONTACT "cedar@zju.edu.cn"
```

Definition at line 11 of file [version.h](#).

6.17.1.3 APP_NAME

```
#define APP_NAME "ZlibValidation"
```

Definition at line 5 of file [version.h](#).

6.17.1.4 APP_VERSION

```
#define APP_VERSION "0.1.0"
```

Definition at line 9 of file [version.h](#).

6.17.1.5 APP_VERSION_MAJOR

```
#define APP_VERSION_MAJOR 0
```

Definition at line 6 of file [version.h](#).

6.17.1.6 APP_VERSION_MINOR

```
#define APP_VERSION_MINOR 1
```

Definition at line 7 of file [version.h](#).

6.17.1.7 APP_VERSION_PATCH

```
#define APP_VERSION_PATCH 0
```

Definition at line 8 of file [version.h](#).

6.17.1.8 BUILD_TIMESTAMP

```
#define BUILD_TIMESTAMP "2025-03-28 09:43:58"
```

Definition at line 12 of file [version.h](#).

6.18 version.h

[Go to the documentation of this file.](#)

```
00001 // version.h.in
00002 #pragma once
00003
00004 // Auto-generated by CMake - DO NOT EDIT
00005 #define APP_NAME "ZlibValidation"
00006 #define APP_VERSION_MAJOR 0
00007 #define APP_VERSION_MINOR 1
00008 #define APP_VERSION_PATCH 0
00009 #define APP_VERSION "0.1.0"
00010 #define APP_AUTHOR "Song Zixuan"
00011 #define APP_CONTACT "cedar@zju.edu.cn"
00012 #define BUILD_TIMESTAMP "2025-03-28 09:43:58"
```

6.19 README.md File Reference

6.20 src/compare.cpp File Reference

```
#include <chrono>
#include <filesystem>
#include <fstream>
#include <iostream>
#include "spdlog/spdlog.h"
#include <tabulate/markdown_exporter.hpp>
#include "compare.hpp"
#include "version.h"
Include dependency graph for compare.cpp:
```

6.21 compare.cpp

[Go to the documentation of this file.](#)

```

00001 #include <chrono>
00002 #include <filesystem>
00003 #include <fstream>
00004 #include <iostream>
00005
00006 #include "spdlog/spdlog.h"
00007 #include <tabulate/markdown_exporter.hpp>
00008
00009 #include "compare.hpp"
00010 #include "version.h"
00011
00023 LibraryComparator::LibraryComparator(LibFile &ref_libfile, LibFile &comp_libfile, double reltol,
00024                                     double abstol) {
00025     : reltol_(reltol), abstol_(abstol) {
00026     std::string ref_json_name = ref_libfile.basename_ + ".json";
00027     if (!std::filesystem::exists(ref_json_name)) {
00028         spdlog::info("Reference JSON file {} not found. Parsing first.", ref_json_name);
00029         ref_libfile.parse();
00030         ref_libfile.writeJsonToFile();
00031         ref_json_ = ref_libfile.lib_json_;
00032     } else {
00033         // Read the JSON file into json object
00034         std::ifstream ref_in(ref_json_name);
00035         if (!ref_in.is_open()) {
00036             spdlog::error("Could not open file '{}' for reading", ref_json_name);
00037             return;
00038         }
00039         try {
00040             ref_json_ = json::parse(ref_in);
00041         } catch (const json::parse_error &e) {
00042             spdlog::error("Error parsing reference JSON file '{}': {}", ref_json_name, e.what());
00043             return;
00044         }
00045     }
00046     std::string comp_json_name = comp_libfile.basename_ + ".json";
00047     if (!std::filesystem::exists(comp_json_name)) {
00048         spdlog::info("Comparison JSON file {} not found. Parsing first.", comp_json_name);
00049         comp_libfile.parse();
00050         comp_libfile.writeJsonToFile();
00051         comp_json_ = comp_libfile.lib_json_;
00052     } else {
00053         // Read the JSON file into json object
00054         std::ifstream comp_in(comp_json_name);
00055         if (!comp_in.is_open()) {
00056             spdlog::error("Could not open file '{}' for reading", comp_json_name);
00057             return;
00058         }
00059         try {
00060             comp_json_ = json::parse(comp_in);
00061         } catch (const json::parse_error &e) {
00062             spdlog::error("Error parsing comparison JSON file '{}': {}", comp_json_name, e.what());
00063             return;
00064         }
00065     }
00066
00067     ref_lib_path_ = ref_libfile.filepath_;
00068     comp_lib_path_ = comp_libfile.filepath_;
00069     spdlog::info("Successfully loaded JSON files for comparison");
00070 }
00071
00079 void LibraryComparator::compareLut(const std::string &cell_name, const std::string &pin_name,
00100                                const std::string &timing_type, const std::string &related_pin,
00101                                const std::string &arc_name, const json &ref_lut,
00102                                const json &comp_lut, Table &table) {
00103     spdlog::info("Comparing LUT of '{}' ...", arc_name);
00104     std::vector<double> ref_index_1 = ref_lut["index_1"].get<std::vector<double>>();
00105     std::vector<double> ref_index_2 = ref_lut["index_2"].get<std::vector<double>>();
00106     std::vector<double> comp_index_1 = comp_lut["index_1"].get<std::vector<double>>();
00107     std::vector<double> comp_index_2 = comp_lut["index_2"].get<std::vector<double>>();
00108     if (ref_index_1 != comp_index_1 || ref_index_2 != comp_index_2) {
00109         spdlog::warn("Mismatch in LUT indices for '{}' in cell: '{}', pin: '{}', related_pin: '{}', "
00110                     "timing_type: '{}', "
00111                     "arc_name, cell_name, pin_name, related_pin, timing_type);
00112         return;
00113     } else {
00114         spdlog::debug("LUT indices match for '{}'", arc_name);
00115
00116         std::vector<std::vector<double>> comp_value_matrix;
00117         // Generate a matrix of values from the JSON array for comparison library
00118         for (const auto &row_val : comp_lut["values"]) {
00119             std::vector<double> row_data;
00120             // Check if the row value is an array

```

```

00121         if (!row_val.is_array()) {
00122             spdlog::error("Invalid LUT format: '{}' values should be an array of arrays in comp_lib '{}',
"
00123                 "cell '{}', pin '{}', "
00124                 "related_pin '{}', timing_type '{}'",
00125                 arc_name, this->comp_lib_path_.string(), cell_name, pin_name, related_pin,
00126                 timing_type);
00127             return;
00128         }
00129         // Check if non-numeric values
00130         for (const auto &val : row_val) {
00131             if (val.is_number()) {
00132                 row_data.push_back(val.get<double>());
00133             } else {
00134                 spdlog::error(
00135                     "Non-numeric value found in '{}'.values, skipping value: {} in comp_lib '{}', cell '{}',
"
00136                     "pin '{}', related_pin '{}', timing_type '{}'",
00137                     arc_name, val.dump(), this->comp_lib_path_.string(), cell_name, pin_name, related_pin,
00138                     timing_type);
00139                 return;
00140             }
00141         }
00142         comp_value_matrix.push_back(row_data);
00143     }
00144     // Generate a matrix of values from the JSON array for reference library
00145     std::vector<std::vector<double>> ref_value_matrix;
00146     for (const auto &row_val : ref_lut["values"]) {
00147         std::vector<double> row_data;
00148         // Check if the row value is an array
00149         if (!row_val.is_array()) {
00150             spdlog::error("Invalid LUT format: '{}' values should be an array of arrays in ref_lib '{}', "
00151                 "cell '{}', pin '{}', "
00152                 "related_pin '{}', timing_type '{}'",
00153                 arc_name, this->ref_lib_path_.string(), cell_name, pin_name, related_pin,
00154                 timing_type);
00155             return;
00156         }
00157         // Check if non-numeric values
00158         for (const auto &val : row_val) {
00159             if (val.is_number()) {
00160                 row_data.push_back(val.get<double>());
00161             } else {
00162                 spdlog::error(
00163                     "Non-numeric value found in '{}'.values, skipping value: {} in ref_lib '{}', cell '{}',
"
00164                     "pin '{}', related_pin '{}', timing_type '{}'",
00165                     arc_name, val.dump(), this->ref_lib_path_.string(), cell_name, pin_name, related_pin,
00166                     timing_type);
00167                 return;
00168             }
00169         }
00170         ref_value_matrix.push_back(row_data);
00171     }
00172     // Compare the matrix for relative tolerance
00173     if (!comp_value_matrix.empty() && !comp_value_matrix[0].empty()) {
00174         for (size_t i = 0; i < comp_value_matrix.size(); ++i) {
00175             for (size_t j = 0; j < comp_value_matrix[i].size(); ++j) {
00176                 double ref_val = ref_value_matrix[i][j];
00177                 double comp_val = comp_value_matrix[i][j];
00178                 if (std::abs(ref_val - comp_val) > reltol * std::abs(ref_val) &&
00179                     std::abs(ref_val - comp_val) > abstol) {
00180                     spdlog::debug("LUT value mismatch for '{}', index_1: {}, index_2: {}", arc_name,
00181                         ref_index_1[i], ref_index_2[j]);
00182                     table.add_row({related_pin + "->" + pin_name, std::to_string(ref_val),
00183                         std::to_string(comp_val), std::to_string(comp_val - ref_val),
00184                         std::to_string((comp_val - ref_val) / ref_val * 100), timing_type,
00185                         arc_name,
00186                         std::to_string(i + 1), std::to_string(ref_index_1[i]), std::to_string(j +
00187                             1),
00188                             std::to_string(ref_index_2[j]), "<"});
00189                     spdlog::debug("table size: {}", table.size());
00190                 } else {
00191                     spdlog::debug("LUT value match for '{}', index_1: {}, index_2: {}", arc_name,
00192                         ref_index_1[i],
00193                         ref_index_2[j]);
00194                 }
00195             }
00196         }
00197     } else {
00198         spdlog::error(
00199             "Empty or invalid 'values' array found for cell: '{}', pin: '{}', related_pin: '{}', "
00200             "timing_type: '{}', arc: '{}'",
00201             cell_name, pin_name, related_pin, timing_type, arc_name);

```

```

00202
00221 void LibraryComparator::compareTimingArc(const std::string &cell_name, const std::string &pin_name,
00222                                           const std::string &timing_type, const json &ref_timing_arc,
00223                                           const json &comp_timing_arc, Table &table) {
00224     spdlog::info("Comparing timing type '{}'. ...", timing_type);
00225
00226     std::string related_pin = ref_timing_arc["related_pin"].get<std::string>();
00227     spdlog::debug("Related pin: '{}'", related_pin);
00228
00229     std::vector<std::string> arc_names = {"cell_rise", "cell_fall", "rise_transition",
"fall_transition"};
00230     for (auto arc_name : arc_names) {
00231         if (comp_timing_arc.contains(arc_name)) {
00232             spdlog::debug("Comparing timing arc: '{}'", arc_name);
00233             // Look for the arc in the reference JSON
00234             if (ref_timing_arc.contains(arc_name)) {
00235                 // Found this arc
00236                 spdlog::debug("Found timing arc: '{}'", arc_name);
00237                 compareLut(cell_name, pin_name, timing_type, related_pin, arc_name, ref_timing_arc[arc_name],
00238                           comp_timing_arc[arc_name], table);
00239             } else {
00240                 // arc not found in reference JSON
00241                 spdlog::warn("Timing arc: '{}'. not found in reference JSON", arc_name);
00242             }
00243         }
00244     }
00245 }
00246
00261 void LibraryComparator::comparePin(const std::string &cell_name, const std::string &pin_name,
00262                                    const json &ref_pin, const json &comp_pin, Table &table) {
00263     spdlog::info("Comparing pin '{}'. ...", pin_name);
00264
00265     if (comp_pin.contains("timing_arcs")) {
00266         for (const auto &comp_arc : comp_pin["timing_arcs"]) {
00267             std::string timing_type = comp_arc["timing_type"].get<std::string>();
00268             spdlog::debug("Comparing timing type: '{}'", timing_type);
00269             // Look for the timing type in the reference JSON
00270             auto ref_arc_it = std::find_if(
00271                 ref_pin["timing_arcs"].begin(), ref_pin["timing_arcs"].end(),
00272                 [&timing_type](const json &ref_arc) { return ref_arc["timing_type"] == timing_type; });
00273             if (ref_arc_it != ref_pin["timing_arcs"].end()) {
00274                 // Found this timing type
00275                 spdlog::debug("Found timing type: '{}'", timing_type);
00276                 compareTimingArc(cell_name, pin_name, timing_type, *ref_arc_it, comp_arc, table);
00277             } else {
00278                 // timing arc not found in reference JSON
00279                 spdlog::warn("Timing arc: '{}'. not found in reference JSON", timing_type);
00280             }
00281         }
00282     } else {
00283         spdlog::info("No timing arcs found for pin: '{}'", comp_pin["pin_name"].get<std::string>());
00284     }
00285 }
00286
00300 void LibraryComparator::compareCell(const std::string &cell_name, const json &ref_cell,
00301                                     const json &comp_cell, Table &table) {
00302     spdlog::info("Comparing cell '{}'. ...", cell_name);
00303
00304     if (comp_cell.contains("output_pins")) {
00305         for (const auto &comp_pin : comp_cell["output_pins"]) {
00306             std::string pin_name = comp_pin["pin_name"].get<std::string>();
00307             spdlog::debug("Comparing pin: '{}'", pin_name);
00308             // Look for the pin in the reference JSON
00309             auto ref_pin_it =
00310                 std::find_if(ref_cell["output_pins"].begin(), ref_cell["output_pins"].end(),
00311                             [&pin_name](const json &ref_pin) { return ref_pin["pin_name"] == pin_name; });
00312             if (ref_pin_it != ref_cell["output_pins"].end()) {
00313                 // Found this pin
00314                 spdlog::debug("Found pin: '{}'", pin_name);
00315                 comparePin(cell_name, pin_name, *ref_pin_it, comp_pin, table);
00316             } else {
00317                 // pin not found in reference JSON
00318                 spdlog::warn("Pin: '{}'. not found in reference JSON", pin_name);
00319             }
00320         }
00321     } else {
00322         spdlog::info("No output pins found for cell: '{}'", cell_name);
00323     }
00324 }
00325
00342 void LibraryComparator::generateReport(const std::string &output_file) {
00343     std::ofstream outfile(output_file);
00344     outfile << "# LIBRARY comparison\n" << std::endl;
00345     outfile << "***Reference library: " << ref_lib_path_ << "***\n" << std::endl;
00346     outfile << "***Comparison library: " << comp_lib_path_ << "***\n" << std::endl;
00347     outfile << "***Absolute tolerance: " << abstol_ << "***\n" << std::endl;
00348     outfile << "***Relative tolerance: " << reltol_ << "***\n" << std::endl;

```

```

00349 outfile << "***Performed by " << APP_NAME << " v" << APP_VERSION << " from " << APP_AUTHOR;
00350 auto now = std::chrono::system_clock::now();
00351 std::time_t now_time_t = std::chrono::system_clock::to_time_t(now);
00352 std::tm *now_tm = std::localtime(&now_time_t);
00353 outfile << ". on: " << std::put_time(now_tm, "%c") << "**\n" << std::endl;
00354 outfile << "> Legend: < outlier, * scaled, ! indices switched, ^ slews extrapolated, ~ loads "
00355         "extrapolated,\n";
00356 outfile << "> \n";
00357 outfile
00358     << "> Legend: + padding added, /0 divide by zero, / slews interpolated, # loads interpolated\n";
00359 outfile << "> \n";
00360 outfile << "> Legend: < value is less but unknown, > value is more but unknown\n\n";
00361
00362 spdlog::info("Starting comparison report generation ...");
00363 for (const auto &comp_cell : comp_json["cells"]) {
00364     std::string cell_name = comp_cell["cell_name"].get<std::string>();
00365     spdlog::debug("Comparing cell: '{}'", cell_name);
00366     // Look for the cell in the reference JSON
00367     auto ref_cell_it =
00368         std::find_if(ref_json["cells"].begin(), ref_json["cells"].end(),
00369             [&cell_name](const json &ref_cell) { return ref_cell["cell_name"] == cell_name;
00370 });
00371 if (ref_cell_it != ref_json["cells"].end()) {
00372     // Found this cell
00373     spdlog::debug("Found cell: '{}'", cell_name);
00374     Table table;
00375     table.add_row({"Pin Name", "Reference", "Comparison", "Diff", "Diff %", "Type", "Arc Name",
00376         "Row #", "Index_1", "Col #", "Index_2", "Note"});
00377     // Header formatting
00378     for (size_t i = 0; i < table[0].size(); ++i) {
00379         table[0][i].format().font_color(Color::yellow).font_style({FontStyle::bold});
00380     }
00381     compareCell(cell_name, *ref_cell_it, comp_cell, table);
00382
00383     if (table.size() > 1) {
00384         // Data starts from row 1, row 0 is header
00385         size_t failed_count = table.size() - 1;
00386         double sum_diff = 0.0;
00387         double sum_diff_percent = 0.0;
00388         size_t outlier_count = 0;
00389         double max_diff = -1e9;
00390         double max_diff_percent = -1e9;
00391         std::vector<double> diff_values; // Store diff values for std deviation calculation
00392
00393         // Index of columns
00394         const int diff_index = 3;
00395         const int diff_percent_index = 4;
00396         const int note_index = 11;
00397
00398         for (size_t i = 1; i < table.size(); ++i) {
00399             try {
00400                 double diff = std::stod(table[i][diff_index].get_text());
00401                 double diff_percent = std::stod(table[i][diff_percent_index].get_text());
00402
00403                 sum_diff += diff;
00404                 sum_diff_percent += diff_percent;
00405                 diff_values.push_back(diff_percent);
00406
00407                 if (table[i][note_index].get_text() == "<") {
00408                     outlier_count++;
00409                 }
00410                 max_diff = std::max(max_diff, diff);
00411                 max_diff_percent = std::max(max_diff_percent, diff_percent);
00412             } catch (const std::invalid_argument &e) {
00413                 spdlog::error("Could not convert value to double: {}", e.what());
00414                 continue; // Skip this row if there's an error
00415             }
00416         }
00417
00418         double avg_diff = sum_diff / failed_count;
00419         double avg_diff_percent = sum_diff_percent / failed_count;
00420
00421         outfile << "## " << cell_name << "\n\n";
00422         outfile << "Delay Comparison in ns\n\n";
00423         if (output_file.substr(output_file.size() - 3) == ".md") {
00424             MarkdownExporter exporter;
00425             auto markdown = exporter.dump(table);
00426             outfile << markdown << std::endl;
00427         } else {
00428             outfile << table << std::endl;
00429         }
00430         outfile << std::endl;
00431         std::cout << table << std::endl;
00432         outfile << cell_name << " delay SUMMARY (abstol: " << abstol_ << ", reltol: " << reltol_
00433             << ")\n";
00434

```

```

00435         // Create summary table
00436         Table summary_table;
00437         summary_table.add_row({"Cell Name", "Data Type", "Failed Count", "Avg Diff", "Avg Diff%",
00438                               "Max Diff", "Max Diff%", "Outliers"});
00439         summary_table.add_row({cell_name, "delay(ns)", std::to_string(failed_count),
00440                               std::to_string(avg_diff), std::to_string(avg_diff_percent) + "%",
00441                               std::to_string(max_diff), std::to_string(max_diff_percent) + "%",
00442                               std::to_string(outlier_count)});
00443
00444         // Output summary table
00445         if (output_file.substr(output_file.size() - 3) == ".md") {
00446             MarkdownExporter exporter;
00447             auto markdown = exporter.dump(summary_table);
00448             outfile << markdown << std::endl;
00449         } else {
00450             outfile << summary_table << std::endl;
00451         }
00452         outfile << "Worst delay outlier: Max Abs: " << max_diff << ", Max Rel: " << max_diff_percent
00453                 << "%, Outliers: " << outlier_count << "\n\n";
00454     }
00455 } else {
00456     // cell not found in reference JSON
00457     spdlog::warn("Cell: '{}' not found in reference JSON", cell_name);
00458 }
00459 }
00460
00461 outfile.close();
00462 }

```

6.22 src/iterators.cpp File Reference

#include "iterators.hpp"

Include dependency graph for iterators.cpp:

6.23 iterators.cpp

[Go to the documentation of this file.](#)

```

00001 #include "iterators.hpp"
00002
00003 GroupsIterator::GroupsIterator(si2drGroupsIdT groups, si2drErrorT &err) : groups_(groups), err_(err) {
00004     group_ = si2drIterNextGroup(groups_, &err_);
00005 }
00006 GroupsIterator::~GroupsIterator() { si2drIterQuit(groups_, &err_); }
00007
00008 void GroupsIterator::next() { group_ = si2drIterNextGroup(groups_, &err_); }
00009 bool GroupsIterator::end() { return si2drObjectIsNull(group_, &err_); }
00010
00011 LibGroup GroupsIterator::get() { return LibGroup(group_, err_); }
00012
00013 AttributesIterator::AttributesIterator(si2drAttrsIdT attrs, si2drErrorT &err)
00014     : attrs_(attrs), err_(err) {
00015     attr_ = si2drIterNextAttr(attrs_, &err_);
00016 }
00017 AttributesIterator::~AttributesIterator() { si2drIterQuit(attrs_, &err_); }
00018
00019 void AttributesIterator::next() { attr_ = si2drIterNextAttr(attrs_, &err_); }
00020 bool AttributesIterator::end() { return si2drObjectIsNull(attr_, &err_); }
00021
00022 LibAttribute AttributesIterator::get() { return LibAttribute(attr_, err_); }
00023
00024 ValuesIterator::ValuesIterator(si2drValuesIdT values, si2drErrorT &err) : values_(values), err_(err) {
00025     si2drIterNextComplexValue(values_, &vtype_, &int_, &float_, &str_, &bool_, &exprp_, &err_);
00026 }
00027 ValuesIterator::~ValuesIterator() { si2drIterQuit(values_, &err_); }
00028
00029 void ValuesIterator::next() {
00030     si2drIterNextComplexValue(values_, &vtype_, &int_, &float_, &str_, &bool_, &exprp_, &err_);
00031 }
00032 bool ValuesIterator::end() { return vtype_ == SI2DR_UNDEFINED_VALUETYPE; }

```

6.24 src/json_utils.cpp File Reference

```
#include "spdlog/spdlog.h"
#include "iterators.hpp"
#include "json_utils.hpp"
Include dependency graph for json_utils.cpp:
```

Functions

- `std::vector< double > parseStringToVector` (const std::string &str)
Parses a string representation of a vector of doubles into a vector of doubles.
- `json generateLutJson` (LibGroup &lib_lut_group, si2drErrorT &err)
Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.
- `json generateTimingJson` (LibGroup &lib_timing_group, si2drErrorT &err)
Generates a JSON representation of timing information from a library timing group.
- `json generatePowerJson` (LibGroup &lib_power_group, si2drErrorT &err)
Converts a Liberty power group into a JSON representation.
- `std::pair< std::string, json > generatePinJson` (LibGroup &lib_pin_group, si2drErrorT &err)
Generates a JSON representation of a Liberty pin group.
- `json generateCellJson` (LibGroup &lib_cell_group, si2drErrorT &err)
Generates a JSON representation of a cell from a [LibGroup](#) object.

6.24.1 Function Documentation

6.24.1.1 generateCellJson()

```
json generateCellJson (
    LibGroup & lib_cell_group,
    si2drErrorT & err )
```

Generates a JSON representation of a cell from a [LibGroup](#) object.

This function processes a [LibGroup](#) representing a cell and converts it into a JSON object. It extracts the cell name and processes specific attributes such as area, cell_leakage_power, and cell_footprint. It also processes pins within the cell by categorizing them based on their direction (input, output, internal, inout).

Parameters

<i>lib_cell_group</i>	The LibGroup object representing the cell
<i>err</i>	Reference to a si2drErrorT object for error handling

Returns

json A JSON object containing the cell's information with the following structure:

- "cell_name": string - name of the cell

- "area": float (optional) - cell area if present
- "cell_leakage_power": float (optional) - cell leakage power if present
- "cell_footprint": string (optional) - cell footprint if present
- "input_pins": array - list of input pins
- "output_pins": array - list of output pins
- "internal_pins": array - list of internal pins
- "inout_pins": array - list of inout pins

Definition at line 272 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.24.1.2 generateLutJson()

```
json generateLutJson (
    LibGroup & lib_lut_group,
    si2drErrorT & err )
```

Generates a JSON object representation of a look-up table (LUT) from a [LibGroup](#) object.

This function iterates through the attributes of the provided [LibGroup](#) object representing a LUT and converts them into a JSON structure. It specifically handles the following attributes:

- "index_1": Converted to a vector and stored in the JSON
- "index_2": Converted to a vector and stored in the JSON
- "values": Each value is parsed into a vector and added to an array in the JSON

Any other attributes encountered will trigger a warning message.

Parameters

<i>lib_lut_group</i>	The LibGroup object containing the LUT data to be converted
<i>err</i>	Reference to an si2drErrorT object to track any errors during processing

Returns

json A JSON object representing the LUT data with keys for "index_1", "index_2", and "values"

Definition at line 62 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.24.1.3 generatePinJson()

```
std::pair< std::string, json > generatePinJson (
    LibGroup & lib_pin_group,
    si2drErrorT & err )
```

Generates a JSON representation of a Liberty pin group.

This function traverses a Liberty pin group, extracts relevant attributes and sub-groups, and converts them into a JSON object. It also determines the pin's direction.

Processes the following pin attributes:

- direction
- max_transition, capacitance, rise_capacitance, fall_capacitance, max_capacitance (float types)
- function, power_down_function, related_ground_pin, related_power_pin, three_state (string types)
- clock (boolean type)

Handles the following sub-groups:

- internal_power: Converted using [generatePowerJson\(\)](#)
- timing: Converted using [generateTimingJson\(\)](#)

Parameters

<i>lib_pin_group</i>	The Liberty pin group to process
<i>err</i>	Reference to an error object for tracking Liberty API errors

Returns

A pair containing the pin direction (string) and the JSON representation of the pin

Definition at line 206 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.24.1.4 generatePowerJson()

```
json generatePowerJson (
    LibGroup & lib_power_group,
    si2drErrorT & err )
```

Converts a Liberty power group into a JSON representation.

This function processes a Liberty power group and converts its attributes and subgroups into a JSON object. It handles attributes like "when", "related_pin", and "related_pg_pin", as well as "rise_power" and "fall_power" subgroups.

Parameters

<i>lib_power_group</i>	The Liberty power group to convert
<i>err</i>	Reference to an si2drErrorT object for error handling

Returns

json A JSON object representing the power group data

The function:

- Extracts string attributes (when, related_pin, related_pg_pin)
- Processes rise_power and fall_power subgroups by converting them to LUT JSON format
- Logs warnings for unknown power subgroup types

Definition at line 154 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.24.1.5 generateTimingJson()

```
json generateTimingJson (
    LibGroup & lib_timing_group,
    si2drErrorT & err )
```

Generates a JSON representation of timing information from a library timing group.

This function extracts timing attributes and sub-groups from a Liberty timing group and organizes them into a JSON object. It processes standard timing attributes like 'related_pin', 'timing_type', 'timing_sense', and 'when', as well as timing tables such as 'cell_fall', 'cell_rise', 'fall_transition', 'rise_transition', 'fall_constraint', and 'rise_constraint'.

Parameters

<i>lib_timing_group</i>	The Liberty timing group to process
<i>err</i>	Reference to an si2drErrorT object for error reporting

Returns

json A JSON object containing the extracted timing information

Definition at line 106 of file [json_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.24.1.6 parseStringToVector()

```
std::vector< double > parseStringToVector (
    const std::string & str )
```

Parses a string representation of a vector of doubles into a vector of doubles.

This function takes a string containing comma-separated numbers, cleans it by removing backslashes and newline characters, and then converts each comma-separated value into a double that is added to the resulting vector.

Parameters

<i>str</i>	The input string to be parsed, containing comma-separated numbers
------------	---

Returns

`std::vector<double>` A vector of doubles parsed from the input string

Exceptions

<code>std::invalid_argument</code>	If the string contains values that cannot be converted to double
<code>std::out_of_range</code>	If the string contains values that are out of range for double

Definition at line 28 of file `json_utils.cpp`.

Here is the caller graph for this function:

6.25 json_utils.cpp

[Go to the documentation of this file.](#)

```

00001 #include "spdlog/spdlog.h"
00002
00003 #include "iterators.hpp"
00004 #include "json_utils.hpp"
00005
00006 /*
00007 json
00008 []
00009 JSON null
00010 {}
00011 JSON
00012 null
00013 */
00014
00028 std::vector<double> parseStringToVector(const std::string &str) {
00029     std::vector<double> result;
00030     std::string cleaned_str;
00031
00032     // Remove backslashes and newline characters
00033     for (char c : str) {
00034         if (c != '\\' && c != '\n') {
00035             cleaned_str += c;
00036         }
00037     }
00038
00039     std::stringstream ss(cleaned_str);
00040     std::string item;
00041     while (std::getline(ss, item, ',')) {
00042         result.push_back(std::stod(item));
00043     }
00044     return result;
00045 }
00046
00062 json generateLutJson(LibGroup &lib_lut_group, si2drErrorT &err) {
00063     json lut_json;
00064
00065     AttributesIterator attr_iter(lib_lut_group.getAttrs(), err);
00066     for (; !attr_iter.end(); attr_iter.next()) {
00067         LibAttribute lib_attr = attr_iter.get();
00068         std::string lut_attr_name = lib_attr.getName();
00069
00070         // spdlog::debug("LUT Attribute Name: {}", lib_attr.getName());
00071         // spdlog::debug("Is Complex? {}", lib_attr.isComplex());
00072
00073         if (lut_attr_name == "index_1" || lut_attr_name == "index_2") {
00074             ValuesIterator values_iter(lib_attr.getValues(), err);
00075             for (; !values_iter.end(); values_iter.next()) {
00076                 // spdlog::debug("Type: {}", int(values_iter.vtype_)); // 5 is string

```

```

00077         // spdlog::debug("Str: {}", values_iter.str_);
00078         lut_json[lut_attr_name] = parseStringToVector(std::string(values_iter.str_));
00079     }
00080     } else if (lut_attr_name == "values") {
00081         ValuesIterator values_iter(lib_attr.getValues(), err);
00082         for (; !values_iter.end(); values_iter.next()) {
00083             // spdlog::debug("{} ", values_iter.str_);
00084             lut_json["values"].push_back(parseStringToVector(std::string(values_iter.str_)));
00085         }
00086     } else {
00087         spdlog::warn("Unknown LUT attribute name: {}", lut_attr_name);
00088     }
00089 }
00090 return lut_json;
00091 }
00092
00106 json generateTimingJson(LibGroup &lib_timing_group, si2drErrorT &err) {
00107     json timing_json;
00108
00109     AttributesIterator attr_iter(lib_timing_group.getAttrs(), err);
00110     for (; !attr_iter.end(); attr_iter.next()) {
00111         LibAttribute lib_attr = attr_iter.get();
00112
00113         std::string attr_name = lib_attr.getName();
00114         if (attr_name == "related_pin" || attr_name == "timing_type" || attr_name == "timing_sense" ||
00115             attr_name == "when") {
00116             timing_json[attr_name] = lib_attr.getString();
00117         }
00118         // More timing attributes can be added here
00119     }
00120
00121     GroupsIterator timing_sub_group_iter(lib_timing_group.getGroups(), err);
00122     for (; !timing_sub_group_iter.end(); timing_sub_group_iter.next()) {
00123         LibGroup lib_timing_sub_group = timing_sub_group_iter.get();
00124
00125         std::string timing_sub_group_type = lib_timing_sub_group.getType();
00126         // std::string timing_sub_group_name = lib_timing_sub_group.getName();
00127         if (timing_sub_group_type == "cell_fall" || timing_sub_group_type == "cell_rise" ||
00128             timing_sub_group_type == "fall_transition" || timing_sub_group_type == "rise_transition" ||
00129             timing_sub_group_type == "fall_constraint" || timing_sub_group_type == "rise_constraint") {
00130             timing_json[timing_sub_group_type] = generateLutJson(lib_timing_sub_group, err);
00131         } else {
00132             spdlog::warn("Unknown timing sub group type: {}", timing_sub_group_type);
00133         }
00134     }
00135     return timing_json;
00136 }
00137
00154 json generatePowerJson(LibGroup &lib_power_group, si2drErrorT &err) {
00155     json power_json;
00156
00157     AttributesIterator attr_iter(lib_power_group.getAttrs(), err);
00158     for (; !attr_iter.end(); attr_iter.next()) {
00159         LibAttribute lib_attr = attr_iter.get();
00160
00161         std::string attr_name = lib_attr.getName();
00162         if (attr_name == "when" || attr_name == "related_pin" || attr_name == "related_pg_pin") {
00163             power_json[attr_name] = lib_attr.getString();
00164         }
00165         // More power attributes can be added here
00166     }
00167
00168     GroupsIterator power_sub_group_iter(lib_power_group.getGroups(), err);
00169     for (; !power_sub_group_iter.end(); power_sub_group_iter.next()) {
00170         LibGroup lib_power_sub_group = power_sub_group_iter.get();
00171
00172         std::string power_sub_group_type = lib_power_sub_group.getType();
00173         // std::string power_sub_group_name = lib_power_sub_group.getName();
00174         if (power_sub_group_type == "rise_power") {
00175             // spdlog::debug("Rise Power: {}", power_sub_group_name);
00176             power_json["rise_power"] = generateLutJson(lib_power_sub_group, err);
00177         } else if (power_sub_group_type == "fall_power") {
00178             power_json["fall_power"] = generateLutJson(lib_power_sub_group, err);
00179         } else {
00180             spdlog::warn("Unknown power sub group type: {}", power_sub_group_type);
00181         }
00182     }
00183     return power_json;
00184 }
00185
00206 std::pair<std::string, json> generatePinJson(LibGroup &lib_pin_group, si2drErrorT &err) {
00207     json pin_json;
00208     std::string direction;
00209     pin_json["pin_name"] = lib_pin_group.getName();
00210
00211     AttributesIterator attr_iter(lib_pin_group.getAttrs(), err);
00212     for (; !attr_iter.end(); attr_iter.next()) {

```

```

00213     LibAttribute lib_attr = attr_iter.get();
00214
00215     std::string attr_name = lib_attr.getName();
00216     if (attr_name == "direction") {
00217         direction = lib_attr.getString();
00218     } else if (attr_name == "max_transition" || attr_name == "capacitance" ||
00219         attr_name == "rise_capacitance" || attr_name == "fall_capacitance" ||
00220         attr_name == "max_capacitance") {
00221         pin_json[attr_name] = lib_attr.getFloat();
00222     } else if (attr_name == "function" || attr_name == "power_down_function" ||
00223         attr_name == "related_ground_pin" || attr_name == "related_power_pin" ||
00224         attr_name == "three_state") {
00225         pin_json[attr_name] = lib_attr.getString();
00226     } else if (attr_name == "clock") {
00227         pin_json[attr_name] = lib_attr.getBoolean();
00228     }
00229     // More pin attributes can be added here
00230 }
00231
00232 GroupsIterator pin_sub_group_iter(lib_pin_group.getGroups(), err);
00233 for (; !pin_sub_group_iter.end(); pin_sub_group_iter.next()) {
00234     LibGroup lib_pin_sub_group = pin_sub_group_iter.get();
00235
00236     std::string pin_sub_group_type = lib_pin_sub_group.getType();
00237     // std::string pin_sub_group_name = lib_pin_sub_group.getName();
00238     if (pin_sub_group_type == "internal_power") {
00239         json power_json = generatePowerJson(lib_pin_sub_group, err);
00240         pin_json["power_arcs"].push_back(power_json);
00241     } else if (pin_sub_group_type == "timing") {
00242         // spdlog::debug("Has Timing: {}", lib_pin_group.getName());
00243         json timing_json = generateTimingJson(lib_pin_sub_group, err);
00244         pin_json["timing_arcs"].push_back(timing_json);
00245     } else {
00246         spdlog::warn("Unknown pin sub group type: {}", pin_sub_group_type);
00247     }
00248 }
00249 return std::make_pair(direction, pin_json);
00250 }
00251
00272 json generateCellJson(LibGroup &lib_cell_group, si2drErrorT &err) {
00273     json cell_json;
00274     cell_json["cell_name"] = lib_cell_group.getName();
00275
00276     AttributesIterator attr_iter(lib_cell_group.getAttrs(), err);
00277     for (; !attr_iter.end(); attr_iter.next()) {
00278         LibAttribute lib_attr = attr_iter.get();
00279
00280         std::string attr_name = lib_attr.getName();
00281         if (attr_name == "area" || attr_name == "cell_leakage_power") {
00282             cell_json[attr_name] = lib_attr.getFloat();
00283         } else if (attr_name == "cell_footprint") {
00284             cell_json[attr_name] = lib_attr.getString();
00285         }
00286         // More cell attributes can be added here
00287     }
00288
00289     GroupsIterator cell_sub_group_iter(lib_cell_group.getGroups(), err);
00290     for (; !cell_sub_group_iter.end(); cell_sub_group_iter.next()) {
00291         LibGroup lib_cell_sub_group = cell_sub_group_iter.get();
00292
00293         std::string cell_sub_group_type = lib_cell_sub_group.getType();
00294         // std::string cell_sub_group_name = lib_cell_sub_group.getName();
00295
00296         // spdlog::debug("Cell Sub Group Type: {}", cell_sub_group_type);
00297         // spdlog::debug("Cell Sub Group Name: {}", cell_sub_group_name);
00298
00299         if (cell_sub_group_type == "pin") {
00300             auto [direction, pin_json] = generatePinJson(lib_cell_sub_group, err);
00301             if (direction == "input") {
00302                 cell_json["input_pins"].push_back(pin_json);
00303             } else if (direction == "output") {
00304                 cell_json["output_pins"].push_back(pin_json);
00305             } else if (direction == "internal") {
00306                 cell_json["internal_pins"].push_back(pin_json);
00307             } else if (direction == "inout") {
00308                 cell_json["inout_pins"].push_back(pin_json);
00309             } else {
00310                 spdlog::warn("Unknown direction: {}", direction);
00311             }
00312         }
00313     }
00314     return cell_json;
00315 }

```

6.26 src/lib_attribute.cpp File Reference

#include "lib_attribute.hpp"
 Include dependency graph for lib_attribute.cpp:

6.27 lib_attribute.cpp

[Go to the documentation of this file.](#)

```
00001 #include "lib_attribute.hpp"
00002
00003 LibAttribute::LibAttribute(si2drAttrIdT attr, si2drErrorT &err) : attr_(attr), err_(err) {}
00004
00005 LibAttribute::~LibAttribute() {}
00006
00007 std::string LibAttribute::getName() {
00008     si2drStringT name = si2drAttrGetName(attr_, &err_);
00009     return name ? std::string(name) : std::string();
00010 }
00011
00012 bool LibAttribute::isComplex() { return si2drAttrGetAttrType(attr_, &err_) ? 1 : 0; }
00013
00014 si2drValuesIdT LibAttribute::getValues() { return si2drComplexAttrGetValues(attr_, &err_); }
00015
00016 long int LibAttribute::getInt() { return si2drSimpleAttrGetInt32Value(attr_, &err_); }
00017
00018 double LibAttribute::getFloat() { return si2drSimpleAttrGetFloat64Value(attr_, &err_); }
00019
00020 std::string LibAttribute::getString() {
00021     si2drStringT str = si2drSimpleAttrGetStringValue(attr_, &err_);
00022     return str ? std::string(str) : std::string();
00023 }
00024
00025 bool LibAttribute::getBoolean() { return si2drSimpleAttrGetBooleanValue(attr_, &err_) ? 1 : 0; }
```

6.28 src/lib_file.cpp File Reference

```
#include <chrono>
#include <filesystem>
#include <fstream>
#include <sstream>
#include <unordered_set>
#include "spdlog/sinks/basic_file_sink.h"
#include "spdlog/sinks/stdout_color_sinks.h"
#include "spdlog/spdlog.h"
#include "iterators.hpp"
#include "json_utils.hpp"
#include "lib_file.hpp"
#include "verilog_utils.hpp"
Include dependency graph for lib_file.cpp:
```

6.29 lib_file.cpp

[Go to the documentation of this file.](#)

```
00001 #include <chrono>
00002 #include <filesystem>
00003 #include <fstream>
00004 #include <sstream>
00005 #include <unordered_set>
00006
00007 #include "spdlog/sinks/basic_file_sink.h"
```

```

00008 #include "spdlog/sinks/stdout_color_sinks.h"
00009 #include "spdlog/spdlog.h"
00010
00011 #include "iterators.hpp"
00012 #include "json_utils.hpp"
00013 #include "lib_file.hpp"
00014 #include "verilog_utils.hpp"
00015
00028 LibFile::LibFile(const std::string &filepath, const std::string &loggername)
00029 : filepath_(filepath), loggername_(loggername) {
00030     filename_ = filepath_.string();
00031     basename_ = filepath_.stem().string();
00032     jsonname_ = basename_ + ".json";
00033
00034     auto console_sink = std::make_shared<spdlog::sinks::stdout_color_sink_mt>();
00035     console_sink->set_level(spdlog::level::info);
00036
00037     auto file_sink = std::make_shared<spdlog::sinks::basic_file_sink_mt>(loggername_, true);
00038     file_sink->set_level(spdlog::level::debug);
00039
00040     std::vector<spdlog::sink_ptr> sinks{console_sink, file_sink};
00041     logger_ = std::make_shared<spdlog::logger>(loggername_, sinks.begin(), sinks.end());
00042     logger_->set_level(spdlog::level::debug);
00043
00044     logger_->info("Created LibFile object for '{}'", filename_);
00045     logger_->info("Debug log in: '{}'", loggername_);
00046 }
00047
00048 LibFile::~LibFile() { logger_->info("Closing file: '{}'", filename_); }
00049
00060 void LibFile::writeJsonToFile() {
00061     std::ofstream out(jsonname_);
00062     if (!out.is_open()) {
00063         logger_->error("Could not open file '{}' for writing", jsonname_);
00064         return;
00065     }
00066     out << lib_json_.dump(2);
00067     out.close();
00068     logger_->info("JSON data written to '{}'", jsonname_);
00069 }
00070
00091 void LibFile::read() {
00092     logger_->info("Reading '{}' ...", filename_);
00093
00094     auto start = std::chrono::high_resolution_clock::now();
00095     si2drReadLibertyFile(const_cast<char *>(filepath_.c_str()), &err_);
00096     auto end = std::chrono::high_resolution_clock::now();
00097     std::chrono::duration<double> duration = end - start;
00098
00099     if (err_ == SI2DR_INVALID_NAME) {
00100         logger_->error("Could not open file '{}' for parsing, quitting...", filename_);
00101         exit(301);
00102     } else if (err_ == SI2DR_SYNTAX_ERROR) {
00103         logger_->error("Syntax Errors were detected in the input file!");
00104         exit(401);
00105     } else {
00106         logger_->info("Done. Read time: {:.2f} seconds", duration.count());
00107     }
00108 }
00109
00130 void LibFile::parse() {
00131     si2drPIInit(&err_); // Initialize private error handler
00132     this->read();        // Read the Liberty file
00133
00134     logger_->info("Parsing '{}' ...", filename_);
00135     // Create a scope for the top-level groups iteration
00136     {
00137         GroupsIterator group_iter(si2drPIGetGroups(&err_), err_);
00138         for (; !group_iter.end(); group_iter.next()) {
00139             LibGroup lib_group = group_iter.get();
00140
00141             if (lib_group.getName() != "") {
00142                 libname_ = lib_group.getName();
00143                 lib_json_["library_name"] = this->libname_;
00144                 logger_->info("Library Name: {}", libname_);
00145             } else {
00146                 logger_->warn("Library Name: <NONAME>");
00147             }
00148             // Second level groups
00149             GroupsIterator sub_group_iter(lib_group.getGroups(), err_);
00150             for (; !sub_group_iter.end(); sub_group_iter.next()) {
00151                 LibGroup lib_sub_group = sub_group_iter.get();
00152
00153                 std::string sub_group_type = lib_sub_group.getType();
00154                 std::string sub_group_name = lib_sub_group.getName();
00155
00156                 if (sub_group_type == "cell") {

```



```

00157         logger_>debug("Cell Name: {}", sub_group_name);
00158         // if (sub_group_name != "AN2D0") {
00159         //     continue;
00160         // }
00161         // Handle each cell
00162         json cell_json = generateCellJson(lib_sub_group, err_);
00163         lib_json["cells"].push_back(cell_json);
00164     } else if (sub_group_type == "operating_conditions") {
00165         logger_>info("Operating Conditions: {}", sub_group_name);
00166         // Get PVT from Operating Conditions
00167         AttributesIterator attr_iter(lib_sub_group.getAttrs(), err_);
00168         for (; !attr_iter.end(); attr_iter.next()) {
00169             LibAttribute lib_attr = attr_iter.get();
00170             logger_>debug("Attribute Name: {}", lib_attr.getName());
00171             logger_>debug("Int Value: {}", lib_attr.getInt());
00172             logger_>debug("Float Value: {}", lib_attr.getFloat());
00173             logger_>debug("String Value: {}", lib_attr.getString());
00174             if (lib_attr.getName() == "process") {
00175                 process_ = lib_attr.getInt();
00176                 lib_json["process"] = process_;
00177             } else if (lib_attr.getName() == "voltage") {
00178                 voltage_ = lib_attr.getFloat();
00179                 // Round to 2 decimal places to avoid floating-point precision issues
00180                 lib_json["voltage"] = std::round(voltage_ * 100) / 100.0;
00181             } else if (lib_attr.getName() == "temperature") {
00182                 temperature_ = lib_attr.getInt();
00183                 lib_json["temperature"] = temperature_;
00184             }
00185         }
00186         logger_>info("P: {}, V: {}, T: {}", process_, voltage_, temperature_);
00187     }
00188     // sub_group_iter's lifetime ends here, and the destructor is called
00189 }
00190 }
00191 // group_iter's lifetime ends here, and the destructor is called
00192 }
00193 si2drPIQuit(&err_);
00194 }
00195
00196 void LibFile::modify() { logger_>info("Modifying the file..."); }
00197
00219 bool LibFile::checkTimingArcMonotonicity(const json &cell, const json &pin, const json &arc,
00220     const std::string &timing_arc_name, const bool is_slew) {
00221     if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00222         logger_>debug("Checking cell: '{}', pin: '{}', related_pin: '{}', timing_arc: '{}', when: '{}'",
00223             cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00224             arc["related_pin"].get<std::string>(), timing_arc_name,
00225             arc["when"].get<std::string>());
00226     } else {
00227         logger_>debug("Checking cell: '{}', pin: '{}', related_pin: '{}', timing_arc: '{}',
00228             cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00229             arc["related_pin"].get<std::string>(), timing_arc_name);
00230     }
00231     bool is_monotonic = true; // Assume the values are monotonic initially
00232     if (arc.contains(timing_arc_name) && arc[timing_arc_name].contains("values")) {
00233         std::vector<std::vector<double>> value_matrix;
00234         // Generate a matrix of values from the JSON array
00235         for (const auto &row_val : arc[timing_arc_name]["values"]) {
00236             std::vector<double> row_data;
00237             // Check if the value is an array
00238             if (!row_val.is_array()) {
00239                 logger_>error("Invalid format: '{}' values should be an array of arrays in cell '{}', pin "
00240                     "'{}', related_pin '{}'",
00241                     timing_arc_name, cell["cell_name"].get<std::string>(),
00242                     pin["pin_name"].get<std::string>(), arc["related_pin"].get<std::string>());
00243                 return false;
00244             }
00245             // Check for non-numeric values
00246             for (const auto &val : row_val) {
00247                 if (val.is_number()) {
00248                     row_data.push_back(val.get<double>());
00249                 } else {
00250                     logger_>warn("Non-numeric value found in '{}'.values, skipping value: {} in cell '{}', "
00251                         "pin '{}', related_pin '{}'",
00252                         timing_arc_name, val.dump(), cell["cell_name"].get<std::string>(),
00253                         pin["pin_name"].get<std::string>(), arc["related_pin"].get<std::string>());
00254                 }
00255             }
00256             value_matrix.push_back(row_data);
00257         }
00258         // Check the matrix for monotonicity
00259         if (!value_matrix.empty() && !value_matrix[0].empty()) {
00260             // Check the monotonic incrementalilty of rows (by output load capacitance, index 2)
00261             for (size_t i = 0; i < value_matrix.size(); ++i) {
00262                 for (size_t j = 1; j < value_matrix[i].size(); ++j) {
00263                     if ((value_matrix[i][j] < value_matrix[i][j - 1] && pin["pin_name"] != arc["related_pin"])

```

```

00264         (value_matrix[i][j] == 0 && value_matrix[i][j - 1] == 0)) {
00265             // If contains "when" key, log the when value
00266             if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00267                 logger_>warn("Non-monotonic (by load) '{}' values: ({}, {}) {} < ({}, {}) {} "
00268                     "for Cell: {} Pin: {}->{} when: \"{}\"",
00269                     timing_arc_name, i, j, value_matrix[i][j], i, j - 1, value_matrix[i][j -
1],
00270                     cell["cell_name"].get<std::string>(),
00271                     arc["related_pin"].get<std::string>(),
00272                     pin["pin_name"].get<std::string>(), arc["when"].get<std::string>());
00273             } else {
00274                 logger_>warn("Non-monotonic (by load) '{}' values: ({}, {}) {} < ({}, {}) {} "
00275                     "for Cell: {} Pin: {}->{}",
00276                     timing_arc_name, i, j, value_matrix[i][j], i, j - 1, value_matrix[i][j -
1],
00277                     cell["cell_name"].get<std::string>(),
00278                     arc["related_pin"].get<std::string>(),
00279                     pin["pin_name"].get<std::string>());
00280             }
00281             is_monotonic = false;
00282         }
00283     }
00284     if (is_slew) {
00285         // Check the monotonic incrementality of columns (by input slew, index 1)
00286         for (size_t j = 0; j < value_matrix[0].size(); ++j) {
00287             for (size_t i = 1; i < value_matrix.size(); ++i) {
00288                 if ((value_matrix[i][j] < value_matrix[i - 1][j] && pin["pin_name"] != arc["related_pin"])
00289                     ||
00290                     (value_matrix[i][j] == 0 && value_matrix[i - 1][j] == 0)) {
00291                         // If contains "when" key, log the when value
00292                         if (arc.contains("when") && !arc["when"].get<std::string>().empty()) {
00293                             logger_>warn("Non-monotonic (by slew) '{}' values: ({}, {}) {} < ({}, {}) {} "
00294                                 "for Cell: {} Pin: {}->{} when: \"{}\"",
00295                                 timing_arc_name, i, j, value_matrix[i][j], i - 1, j,
00296                                 value_matrix[i - 1][j], cell["cell_name"].get<std::string>(),
00297                                 arc["related_pin"].get<std::string>(),
00298                                 pin["pin_name"].get<std::string>(),
00299                                 arc["when"].get<std::string>());
00300                         } else {
00301                             logger_>warn("Non-monotonic (by slew) '{}' values: ({}, {}) {} < ({}, {}) {} "
00302                                 "for Cell: {} Pin: {}->{}",
00303                                 timing_arc_name, i, j, value_matrix[i][j], i - 1, j,
00304                                 value_matrix[i - 1][j], cell["cell_name"].get<std::string>(),
00305                                 arc["related_pin"].get<std::string>(),
00306                                 pin["pin_name"].get<std::string>());
00307                         }
00308                         is_monotonic = false;
00309                     }
00310             }
00311         } else {
00312             logger_>warn(
00313                 "Empty or invalid 'values' array found for cell: '{}', pin: '{}', related_pin: '{}', "
00314                 "timing_arc: '{}",
00315                 cell["cell_name"].get<std::string>(), pin["pin_name"].get<std::string>(),
00316                 arc["related_pin"].get<std::string>(), timing_arc_name);
00317         }
00318     }
00319     return is_monotonic;
00320 }
00321
00322 void LibFile::mono(const bool is_slew) {
00323     /*
00324     * Use this command to check the following data in the current library to ensure
00325     * the tables are monotonically increasing with respect to output load:
00326     * cell_rise retaining_rise
00327     * cell_fall retaining_fall
00328     * rise_transition retain_rise_slew
00329     * fall_transition retain_fall_slew
00330     * mpw
00331     */
00332     logger_>info("Monotonicity check of '{}' starting ...", filename_);
00333
00334     if (!std::filesystem::exists(jsonname_)) {
00335         logger_>info("JSON file not found. Parsing Liberty file first.");
00336         this->parse();
00337         this->writeJsonToFile();
00338     } else {
00339         // Read the JSON file into json object
00340         std::ifstream in(jsonname_);
00341         if (!in.is_open()) {
00342             logger_>error("Could not open file '{}' for reading", jsonname_);
00343             return;
00344         }
00345         try {
00346

```

```

00364     lib_json_ = json::parse(in);
00365 } catch (const json::parse_error &e) {
00366     logger_>error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00367     in.close();
00368     return;
00369 }
00370 in.close();
00371 }
00372
00373 std::map<std::string, bool> cell_monotonicity_status; // Track pass/fail status for each cell
00374 std::vector<std::string> failed_cells;                // List of failed cell names
00375 int total_cells = 0;
00376 int passed_cells = 0;
00377
00378 // Check the monotonicity of delay values
00379 // The index 1 (time values) must be monotonically increasing and >= 0
00380 for (const auto &cell : lib_json_["cells"]) {
00381     total_cells++;
00382     bool cell_is_monotonic = true; // Assume cell is monotonic initially
00383     std::string cell_name = cell["cell_name"].get<std::string>();
00384     cell_monotonicity_status[cell_name] =
00385         true; // Initialize to pass, will be set to false if any check fails
00386
00387     if (cell.contains("output_pins")) {
00388         for (const auto &pin : cell["output_pins"]) {
00389             if (pin.contains("timing_arcs")) {
00390                 for (const auto &arc : pin["timing_arcs"]) {
00391                     // Check 4 timing arc names and accumulate monotonicity status
00392                     for (const auto &name : {"cell_rise", "cell_fall", "rise_transition", "fall_transition"})
00393 {
00394                         if (!checkTimingArcMonotonicity(cell, pin, arc, name, is_slew)) {
00395                             cell_is_monotonic = false;
00396                         }
00397                     }
00398                 }
00399             }
00400         }
00401     if (cell.contains("input_pins")) {
00402         for (const auto &pin : cell["input_pins"]) {
00403             if (!pin.contains("clock")) {
00404                 continue; // Skip non-clock pins
00405             }
00406             // logger_>debug("Clock pin: {}", pin["pin_name"].get<std::string>());
00407             if (pin.contains("timing_arcs")) {
00408                 for (const auto &arc : pin["timing_arcs"]) {
00409                     if (arc.contains("timing_type")) {
00410                         // logger_>debug("Timing type: {}", arc["timing_type"].get<std::string>());
00411                         if (arc["timing_type"].get<std::string>() == "min_pulse_width") {
00412                             for (const auto &name : {"rise_constraint", "fall_constraint"}) {
00413                                 if (!checkTimingArcMonotonicity(cell, pin, arc, name, is_slew)) {
00414                                     cell_is_monotonic = false;
00415                                 }
00416                             }
00417                         }
00418                     }
00419                 }
00420             }
00421         }
00422     }
00423     // Update cell status based on all checks
00424     cell_monotonicity_status[cell_name] = cell_is_monotonic;
00425     if (cell_is_monotonic) {
00426         passed_cells++;
00427     } else {
00428         failed_cells.push_back(cell_name);
00429     }
00430 }
00431 logger_>info("Monotonicity check of '{}' completed.", filename_);
00432
00433 // Output summary statistics
00434 logger_>info("validate_monotonicity : {} out of {} cells passed", passed_cells, total_cells);
00435 logger_>info("validate_monotonicity : {} out of {} cells failed", total_cells - passed_cells,
00436             total_cells);
00437 if (!failed_cells.empty()) {
00438     std::stringstream failed_cells_ss;
00439     for (size_t i = 0; i < failed_cells.size(); ++i) {
00440         failed_cells_ss << failed_cells[i];
00441         if (i < failed_cells.size() - 1) {
00442             failed_cells_ss << ", "; // Add comma if not the last element
00443         }
00444     }
00445     logger_>info("Failed cell list : {}", failed_cells_ss.str());
00446 }
00447 }
00448
00467 void LibFile::supercell(const int chain_length, const std::vector<std::string> &cell_names) {

```

```

00468  /*
00469  * Supercells are named as follows:
00470  * <cellname>__X<chain_length>__<input_pin>__<output_pin>
00471  */
00472  logger_>info("Creating supercells for '{}'", filename_);
00473
00474  if (!std::filesystem::exists(jsonname_)) {
00475      logger_>info("JSON file not found. Parsing Liberty file first.");
00476      this->parse();
00477      this->writeJsonToFile();
00478  } else {
00479      // Read the JSON file into json object
00480      std::ifstream in(jsonname_);
00481      if (!in.is_open()) {
00482          logger_>error("Could not open file '{}' for reading", jsonname_);
00483          return;
00484      }
00485      try {
00486          lib_json_ = json::parse(in);
00487      } catch (const json::parse_error &e) {
00488          logger_>error("JSON parsing error in file '{}': {}", jsonname_, e.what());
00489          in.close();
00490          return;
00491      }
00492      in.close();
00493  }
00494
00495  // write to .map file
00496  std::ofstream out(basename_ + ".map");
00497  if (!out.is_open()) {
00498      logger_>error("Could not open file '{}' for writing", basename_ + ".map");
00499      return;
00500  }
00501  // if cell_names is empty, create supercells for all cells
00502  // else create supercells for only the specified cells
00503
00504  // Check if we're processing specific cells or all cells
00505  bool process_all_cells = cell_names.empty();
00506
00507  if (process_all_cells) {
00508      logger_>info("Creating supercells for ALL cells in '{}'", filename_);
00509  } else {
00510      logger_>info("Creating supercells for {} specified cells in '{}'", cell_names.size(), filename_);
00511  }
00512
00513  // Create a set for faster lookups if we have specific cell names
00514  std::unordered_set<std::string> cell_set;
00515  std::unordered_set<std::string> found_cells;
00516  if (!process_all_cells) {
00517      cell_set.insert(cell_names.begin(), cell_names.end());
00518  }
00519
00520  for (const auto &cell : lib_json_["cells"]) {
00521      bool is_sequential = false;
00522      std::string cell_name = cell["cell_name"].get<std::string>();
00523
00524      // Skip cells not in the specified list
00525      if (!process_all_cells && cell_set.find(cell_name) == cell_set.end()) {
00526          continue;
00527      }
00528
00529      // Mark this cell as found
00530      if (!process_all_cells) {
00531          found_cells.insert(cell_name);
00532      }
00533
00534      logger_>debug("Creating supercells for cell: '{}'", cell_name);
00535
00536      std::vector<std::string> output_pins;
00537      std::vector<std::string> input_pins;
00538      // Extract input and output pins
00539      if (cell.contains("output_pins")) {
00540          for (const auto &pin : cell["output_pins"]) {
00541              output_pins.push_back(pin["pin_name"].get<std::string>());
00542          }
00543      }
00544      if (cell.contains("input_pins")) {
00545          for (const auto &pin : cell["input_pins"]) {
00546              if (pin.contains("clock")) {
00547                  is_sequential = true;
00548                  // clock pin not use for supercell
00549                  continue;
00550              }
00551              input_pins.push_back(pin["pin_name"].get<std::string>());
00552          }
00553      }
00554  }

```

```

00555 // create supercells for all combinations of input and output pins
00556 for (const auto &output_pin : output_pins) {
00557     for (const auto &input_pin : input_pins) {
00558         if (is_sequential) {
00559             const int chain_len = 1;
00560             std::string supercell_name =
00561                 cell_name + "__X" + std::to_string(chain_len) + "__" + input_pin + "__" + output_pin;
00562             out << cell_name << " " << supercell_name << std::endl;
00563         } else {
00564             std::string supercell_name =
00565                 cell_name + "__X" + std::to_string(chain_length) + "__" + input_pin + "__" + output_pin;
00566             out << cell_name << " " << supercell_name << std::endl;
00567         }
00568     }
00569 }
00570 }
00571
00572 // Check for cells that were specified but not found
00573 if (!process_all_cells) {
00574     std::vector<std::string> not_found_cells;
00575     for (const auto &requested_cell : cell_names) {
00576         if (found_cells.find(requested_cell) == found_cells.end()) {
00577             not_found_cells.push_back(requested_cell);
00578         }
00579     }
00580
00581     // Output warning for cells that weren't found
00582     if (!not_found_cells.empty()) {
00583         std::string missing_cells = not_found_cells[0];
00584         for (size_t i = 1; i < not_found_cells.size(); ++i) {
00585             missing_cells += ", " + not_found_cells[i];
00586         }
00587         logger_>warn("{} specified cell{} not found in the library: {}", not_found_cells.size(),
00588             not_found_cells.size() > 1 ? "s were" : " was", missing_cells);
00589     }
00590 }
00591
00592 out.close();
00593 logger_>info("Supercell creation complete in '{}', basename_ + ".map");
00594 }
00595
00596 void LibFile::verilog(const int chain_length, const std::vector<std::string> &cell_names) {
00597     logger_>info("Creating Verilog for '{}', filename_);
00598
00599     this->supercell(chain_length, cell_names);
00600
00601     // write to .v file
00602     std::ofstream out(basename_ + ".v");
00603     if (!out.is_open()) {
00604         logger_>error("Could not open file '{}' for writing", basename_ + ".v");
00605         return;
00606     }
00607
00608     // Read the .map file into a vector to preserve all entries and their order
00609     std::vector<std::pair<std::string, std::string>> supercell_entries;
00610     std::ifstream in(basename_ + ".map");
00611     if (!in.is_open()) {
00612         logger_>error("Could not open file '{}' for reading", basename_ + ".map");
00613         return;
00614     }
00615
00616     std::string line;
00617     while (std::getline(in, line)) {
00618         std::istringstream iss(line);
00619         std::string cell_name, supercell_name;
00620         if (!(iss >> cell_name >> supercell_name)) {
00621             logger_>warn("Invalid line format in map file: '{}', line);
00622             continue;
00623         }
00624         supercell_entries.push_back({cell_name, supercell_name});
00625     }
00626     in.close();
00627     logger_>info("Read {} supercells from '{}', supercell_entries.size(), basename_ + ".map");
00628
00629     // Generate verilog module for each supercell
00630     for (const auto &supercell_entry : supercell_entries) {
00631         // Check if the cell is sequential
00632         bool is_sequential = false;
00633         // Count the number of instances will be created in verilog
00634         int instance_count = 0;
00635         // Get original cell name from pair
00636         const std::string &cell_name = supercell_entry.first;
00637         // Get supercell name(as well as module name) from pair
00638         const std::string &module_name = supercell_entry.second;
00639
00640         logger_>info("Creating Module: '{}' from Cell '{}', module_name, cell_name);
00641

```

```

00642 // Get the cell JSON object
00643 json cell_json;
00644 for (const auto &cell : lib_json["cells"]) {
00645     if (cell["cell_name"].get<std::string>() == cell_name) {
00646         cell_json = cell;
00647         break;
00648     }
00649 }
00650
00651 // Get input/output pins from the cell JSON object
00652 std::vector<std::string> input_pins;
00653 std::vector<std::string> output_pins;
00654 std::stringstream input_pins_ss;
00655 std::stringstream output_pins_ss;
00656 if (cell_json.contains("input_pins")) {
00657     for (const auto &pin : cell_json["input_pins"]) {
00658         if (pin.contains("clock")) {
00659             is_sequential = true;
00660         }
00661         input_pins.push_back(pin["pin_name"].get<std::string>());
00662         input_pins_ss << pin["pin_name"].get<std::string>() << ", ";
00663     }
00664 }
00665 if (cell_json.contains("output_pins")) {
00666     for (const auto &pin : cell_json["output_pins"]) {
00667         output_pins.push_back(pin["pin_name"].get<std::string>());
00668         output_pins_ss << pin["pin_name"].get<std::string>() << ", ";
00669     }
00670 }
00671 logger_>debug("Input pins set: {}", input_pins_ss.str());
00672 logger_>debug("Output pins set: {}", output_pins_ss.str());
00673
00674 // Sequential cells will have only one instance
00675 // Combinational cells will have instances equal to chain length
00676 if (is_sequential) {
00677     instance_count = 1;
00678 } else {
00679     instance_count = chain_length;
00680 }
00681
00682 // Create ANSI port list
00683 std::string port_list_text = "(";
00684 // Add input ports
00685 bool isFirstPort = true;
00686 for (const auto &input_pin : input_pins) {
00687     if (!isFirstPort) {
00688         port_list_text += ", ";
00689     }
00690     port_list_text += "input " + input_pin;
00691     isFirstPort = false;
00692 }
00693 // Add output ports
00694 for (const auto &output_pin : output_pins) {
00695     if (!isFirstPort) {
00696         port_list_text += ", ";
00697     }
00698     port_list_text += "output " + output_pin;
00699     isFirstPort = false;
00700 }
00701 port_list_text += ")";
00702 logger_>debug("ANSI Port list: {}", port_list_text);
00703
00704 // Create full module header text
00705 std::string fullModuleText = "module " + module_name + port_list_text + ";\nendmodule";
00706 logger_>debug("Full module text: {}", fullModuleText);
00707
00708 // Generate the syntax tree for the module using slang library
00709 auto tree = slang::syntax::SyntaxTree::fromText(fullModuleText);
00710 if (tree) {
00711     // Add ports to the syntax tree
00712     ModuleRewriter rewriter(input_pins, output_pins, supercell_entry, instance_count, logger_);
00713
00714     tree = rewriter.transform(tree);
00715     rewriter.visit(tree->root());
00716
00717     // Output the transformed syntax tree
00718     out << "\timescale lns/lps" << std::endl;
00719     out << slang::syntax::SyntaxPrinter::printFile(*tree) << std::endl << std::endl;
00720 } else {
00721     logger_>error("Failed to create syntax tree for module '{}'", module_name);
00722     continue;
00723 }
00724 }
00725
00726 out.close();
00727 logger_>info("Verilog creation complete in '{}', basename_ + ".v");
00728 }

```

6.30 src/lib_group.cpp File Reference

```
#include "lib_group.hpp"
```

Include dependency graph for lib_group.cpp:

6.31 lib_group.cpp

[Go to the documentation of this file.](#)

```
00001 #include "lib_group.hpp"
00002
00003 LibGroup::LibGroup(si2drGroupIdT group, si2drErrorT &err) : group_(group), err_(err) {}
00004
00005 LibGroup::~LibGroup() {}
00006
00007 std::string LibGroup::getName() {
00008     si2drNamesIdT names = si2drGroupGetNames(group_, &err_);
00009     si2drStringT name = si2drIterNextName(names, &err_);
00010     si2drIterQuit(names, &err_);
00011     return name ? std::string(name) : std::string();
00012 }
00013
00014 std::string LibGroup::getType() {
00015     si2drStringT type = si2drGroupGetGroupType(group_, &err_);
00016     return type ? std::string(type) : std::string();
00017 }
00018
00019 si2drAttrsIdT LibGroup::getAttrs() { return si2drGroupGetAttrs(group_, &err_); }
00020
00021 si2drGroupsIdT LibGroup::getGroups() { return si2drGroupGetGroups(group_, &err_); }
```

6.32 src/main.cpp File Reference

```
#include <filesystem>
#include <iostream>
#include <string>
#include <thread>
#include <vector>
#include "CLI/CLI.hpp"
#include "si2dr_liberty.h"
#include "spdlog/sinks/basic_file_sink.h"
#include "spdlog/sinks/stdout_color_sinks.h"
#include "spdlog/spdlog.h"
#include "compare.hpp"
#include "lib_file.hpp"
#include "verilog_utils.hpp"
#include "version.h"
```

Include dependency graph for main.cpp:

Functions

- void [printlnInfo](#) ()
Sets up and configures the global logger and prints application information.
- void [parseLibFile](#) (const std::string &library_path, const std::string log_file_name)
Parses a library file and generates corresponding output.
- void [monoCheckLibFile](#) (const std::string &library_path, const std::string log_file_name, bool is_slew)
Performs a monotonicity check on a library file.

- void [supercellLibFile](#) (const std::string &library_path, const std::string &log_file_name, int chain_length, const std::vector< std::string > &cell_names)
Creates supercell map structures from a Liberty library file.
- void [verilogLibFile](#) (const std::string &library_path, const std::string &log_file_name, int chain_length, const std::vector< std::string > &cell_names)
Generates Verilog files from a library file for specified cells.
- void [compareLibFiles](#) (const std::string &ref_lib, const std::string &comp_lib, const double reltol, const double abstol, std::string &report_file_name)
Compares two library files and generates a detailed comparison report.
- void [funcLibFile](#) (const std::string &ref_file, const std::string &comp_file, const std::vector< std::string > &cell_names)
- int [main](#) (int argc, char *argv[])

6.32.1 Function Documentation

6.32.1.1 compareLibFiles()

```
void compareLibFiles (
    const std::string & ref_lib,
    const std::string & comp_lib,
    const double reltol,
    const double abstol,
    std::string & report_file_name )
```

Compares two library files and generates a detailed comparison report.

This function compares a reference library file with another library file, performing validation checks based on specified tolerance parameters. The comparison results are written to a report file in markdown or text format.

Parameters

<i>ref_lib</i>	Path to the reference library file to use as the baseline
<i>comp_lib</i>	Path to the library file to compare against the reference
<i>reltol</i>	Relative tolerance for numerical comparisons (must be ≥ 0.0)
<i>abstol</i>	Absolute tolerance for numerical comparisons
<i>report_file_name</i>	[in,out] Name of the file to write the comparison report to. If empty, defaults to [comp_lib_basename].cmp.md. If provided but doesn't end with .txt or .md, .md will be appended.

Note

The function will log an error and return without comparing if reltol is invalid.

Log files will be created with the naming pattern [library_basename].cmp.log

The function uses the [LibraryComparator](#) class to perform the actual comparison.

Definition at line 216 of file [main.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.32.1.2 funcLibFile()

```
void funcLibFile (
    const std::string & ref_file,
    const std::string & comp_file,
    const std::vector< std::string > & cell_names )
```

Definition at line 242 of file [main.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.32.1.3 main()

```
int main (
    int argc,
    char * argv[] )
```

Definition at line 260 of file [main.cpp](#).

Here is the call graph for this function:

6.32.1.4 monoCheckLibFile()

```
void monoCheckLibFile (
    const std::string & library_path,
    const std::string log_file_name,
    bool is_slew )
```

Performs a monotonicity check on a library file.

This function loads a library file and performs monotonicity validation on it. Results are logged to a file. If no log file name is provided, it generates one based on the library file name.

Parameters

<i>library_path</i>	Path to the library file to check
<i>log_file_name</i>	Name of the log file (optional - if empty, generates a name based on library file)
<i>is_slew</i>	Boolean flag indicating whether to check slew monotonicity (true) or other parameters (false)

Exceptions

<i>Any</i>	exceptions from the monotonicity check are caught and logged
------------	--

Definition at line 97 of file [main.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.32.1.5 parseLibFile()

```
void parseLibFile (
```

```
const std::string & library_path,
const std::string log_file_name )
```

Parses a library file and generates corresponding output.

This function processes the given library file, parsing its contents and generating a JSON output. It also logs the parsing process to a specified log file or creates a default log file if none is provided.

Parameters

<i>library_path</i>	Path to the library file that needs to be parsed
<i>log_file_name</i>	Optional name for the log file. If empty, a default name is generated based on the library filename with ".parse.log" extension

Exceptions

<i>The</i>	function catches and logs any exceptions but doesn't propagate them
------------	---

Definition at line 65 of file [main.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.32.1.6 printInfo()

```
void printInfo ( )
```

Sets up and configures the global logger and prints application information.

This function performs the following operations:

1. Creates a console sink for logging with level set to INFO
2. Creates a file sink for logging with level set to DEBUG, saving to [APP_NAME].log
3. Configures a logger with both sinks and sets it as the default logger
4. Outputs basic application information:
 - Version and build timestamp
 - Author information
 - Log file location

Note

Uses spdlog library for logging functionality

Depends on APP_NAME, APP_VERSION, BUILD_TIMESTAMP, APP_AUTHOR, and APP_CONTACT macros

Definition at line 34 of file [main.cpp](#).

Here is the caller graph for this function:

6.32.1.7 supercellLibFile()

```
void supercellLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
```

Creates supercell map structures from a Liberty library file.

This function reads a Liberty file, creates supercell structures based on the specified chain length and cell names, and logs the process to a file.

Parameters

<i>library_path</i>	Path to the Liberty file to process
<i>log_file_name</i>	Name of the log file (if empty, defaults to "[library_name].supercell.log")
<i>chain_length</i>	The length of chains to create (must be ≥ 1)
<i>cell_names</i>	Vector of cell names to process for supercell generation

Exceptions

May	pass through exceptions from the LibFile::supercell method
-----	--

Note

The function validates the chain length and logs all activities including errors that might occur during processing

Definition at line 130 of file [main.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.32.1.8 verilogLibFile()

```
void verilogLibFile (
    const std::string & library_path,
    const std::string & log_file_name,
    int chain_length,
    const std::vector< std::string > & cell_names )
```

Generates Verilog files from a library file for specified cells.

This function processes a library file and generates Verilog representation for the specified cell names with a given chain length. The operation results are logged to a specified or default log file.

Parameters

<i>library_path</i>	Path to the library file to process
<i>log_file_name</i>	Name for the log file (if empty, a default name will be generated)
<i>chain_length</i>	Number of cells to chain together, must be ≥ 1
<i>cell_names</i>	Vector of cell names to generate Verilog for

Exceptions

<i>Catches</i>	any exceptions from the verilog generation process and logs them
----------------	--

Definition at line 171 of file [main.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.33 main.cpp

[Go to the documentation of this file.](#)

```

00001 // ./src/main.cpp
00002 #include <filesystem>
00003 #include <iostream>
00004 #include <string>
00005 #include <thread>
00006 #include <vector>
00007
00008 #include "CLI/CLI.hpp"
00009 #include "si2dr_liberty.h"
00010 #include "spdlog/sinks/basic_file_sink.h"
00011 #include "spdlog/sinks/stdout_color_sinks.h"
00012 #include "spdlog/spdlog.h"
00013
00014 #include "compare.hpp"
00015 #include "lib_file.hpp"
00016 #include "verilog_utils.hpp"
00017 #include "version.h"
00018
00034 void printInfo() {
00035     // Set Global Logger
00036     auto console_sink = std::make_shared<spdlog::sinks::stdout_color_sink_mt>();
00037     console_sink->set_level(spdlog::level::info);
00038     std::string app_name = APP_NAME;
00039     auto file_sink = std::make_shared<spdlog::sinks::basic_file_sink_mt>(app_name + ".log", true);
00040     file_sink->set_level(spdlog::level::debug);
00041
00042     std::vector<spdlog::sink_ptr> sinks{console_sink, file_sink};
00043     auto logger = std::make_shared<spdlog::logger>(APP_NAME, sinks.begin(), sinks.end());
00044     logger->set_level(spdlog::level::debug);
00045     spdlog::set_default_logger(logger);
00046
00047     spdlog::info("Version {}, Built: {}", APP_VERSION, BUILD_TIMESTAMP);
00048     spdlog::info("Author: {}, Email: {}", APP_AUTHOR, APP_CONTACT);
00049     spdlog::info("Global log file in: '{}'", app_name + ".log");
00050 }
00051
00065 void parseLibFile(const std::string &library_path, const std::string log_file_name) {
00066     std::string logname = log_file_name.empty()
00067         ? std::filesystem::path(library_path).stem().string() + ".parse.log"
00068         : log_file_name;
00069     LibFile libfile(library_path, logname);
00070
00071     libfile.logger_->info("Starting parse for file: '{} ...", library_path);
00072
00073     try {
00074         libfile.parse();
00075         libfile.writeJsonToFile();
00076         libfile.logger_->info("Successfully parsed file: '{}'", library_path);
00077     } catch (const std::exception &e) {
00078         libfile.logger_->error("Error parsing file '{}': {}", library_path, e.what());
00079     }
00080 }
00081
00097 void monoCheckLibFile(const std::string &library_path, const std::string log_file_name, bool is_slew)
00098 {
00099     std::string logname = log_file_name.empty()
00099         ? std::filesystem::path(library_path).stem().string() + ".mono.log"
00100         : log_file_name;
00101     LibFile libfile(library_path, logname);
00102
00103     libfile.logger_->info("Starting monotonicity check for file: '{}', input slew: {}", library_path,
00104         is_slew);
00105
00106     try {
00107         libfile.mono(is_slew);

```

```

00108     libfile.logger_>info("Successfully completed monotonicity check for file: '{}'", library_path);
00109 } catch (const std::exception &e) {
00110     libfile.logger_>error("Error during monotonicity check for file '{}': {}", library_path,
00111         e.what());
00112 }
00113
00130 void supercellLibFile(const std::string &library_path, const std::string &log_file_name,
00131     int chain_length, const std::vector<std::string> &cell_names) {
00132     std::string logname = log_file_name.empty()
00133         ? std::filesystem::path(library_path).stem().string() + ".supercell.log"
00134         : log_file_name;
00135     LibFile libfile(library_path, logname);
00136
00137     // Check chain length validity
00138     if (chain_length < 1) {
00139         libfile.logger_>error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00140         return;
00141     }
00142     std::stringstream ss;
00143     for (const auto &cell : cell_names) {
00144         ss << cell << ", ";
00145     }
00146     libfile.logger_>info("Starting supercell generation for file: '{}', chain length: {}, cells: {}",
00147         library_path, chain_length, ss.str());
00148     try {
00149         libfile.supercell(chain_length, cell_names);
00150         libfile.logger_>info("Successfully generated supercells for file: '{}'", library_path);
00151     } catch (const std::exception &e) {
00152         libfile.logger_>error("Error during supercell generation for file '{}': {}", library_path,
00153             e.what());
00154     }
00155 }
00156
00171 void verilogLibFile(const std::string &library_path, const std::string &log_file_name, int
00172     chain_length,
00173     const std::vector<std::string> &cell_names) {
00174     std::string logname = log_file_name.empty()
00175         ? std::filesystem::path(library_path).stem().string() + ".verilog.log"
00176         : log_file_name;
00177     LibFile libfile(library_path, logname);
00178
00179     // Check chain length validity
00180     if (chain_length < 1) {
00181         libfile.logger_>error("Invalid chain length: {}. Chain length must be >= 1.", chain_length);
00182         return;
00183     }
00184     std::stringstream ss;
00185     for (const auto &cell : cell_names) {
00186         ss << cell << ", ";
00187     }
00188     libfile.logger_>info("Starting Verilog generation for file: '{}', chain length: {}, cells: {}",
00189         library_path, chain_length, ss.str());
00190     try {
00191         libfile.verilog(chain_length, cell_names);
00192         libfile.logger_>info("Successfully generated Verilog for file: '{}'", library_path);
00193     } catch (const std::exception &e) {
00194         libfile.logger_>error("Error during Verilog generation for file '{}': {}", library_path,
00195             e.what());
00196     }
00197
00216 void compareLibFiles(const std::string &ref_lib, const std::string &comp_lib, const double reltol,
00217     const double abstol, std::string &report_file_name) {
00218     std::string ref_logname = std::filesystem::path(ref_lib).stem().string() + ".cmp.log";
00219     std::string comp_logname = std::filesystem::path(comp_lib).stem().string() + ".cmp.log";
00220     LibFile ref_libfile(ref_lib, ref_logname), comp_libfile(comp_lib, comp_logname);
00221
00222     // Check relative tolerance validity
00223     if (reltol < 0.0) {
00224         spdlog::error("Invalid relative tolerance: {}. Relative tolerance must be >= 0.0.", reltol);
00225         return;
00226     }
00227     if (report_file_name.empty()) {
00228         report_file_name = comp_libfile.basename_ + ".cmp.md";
00229     } else if (std::filesystem::path(report_file_name).extension() != ".txt" &&
00230         std::filesystem::path(report_file_name).extension() != ".md") {
00231         // report file name not end in .txt or .md, warn user and append .md
00232         report_file_name = report_file_name + ".md";
00233         spdlog::warn("Report file name does not end in .txt or .md. Report will be written to '{}'",
00234             report_file_name);
00235     }
00236
00237     LibraryComparator comparator(ref_libfile, comp_libfile, reltol, abstol);
00238     comparator.generateReport(report_file_name);
00239     spdlog::info("Comparison completed. Report written to: '{}'", report_file_name);
00240 }

```

```

00241
00242 void funcLibFile(const std::string &ref_file, const std::string &comp_file,
00243                 const std::vector<std::string> &cell_names) {
00244     // Check if the files are Liberty or Verilog
00245     std::string ref_ext = std::filesystem::path(ref_file).extension();
00246     std::string comp_ext = std::filesystem::path(comp_file).extension();
00247     for (const auto &cell : cell_names) {
00248         spdlog::info("Checking functional equivalence for cell: '{}'", cell);
00249         if (ref_ext == ".v") {
00250             spdlog::info("Reference file in Verilog format.");
00251             getAST(ref_file, cell);
00252         }
00253         if (comp_ext == ".v") {
00254             spdlog::info("Comparison file in Verilog format.");
00255             getAST(comp_file, cell);
00256         }
00257     }
00258 }
00259
00260 int main(int argc, char *argv[]) {
00261     // Parse command line arguments
00262     std::vector<std::string> library_paths; // Support multiple files
00263     std::string log_file_name = "";
00264
00265     // Command line parameter parsing using CLI11
00266     CLI::App app{APP_NAME};
00267     app.set_version_flag("-v,--version", APP_VERSION);
00268
00269     // Add subcommand for parse mode
00270     CLI::App *parse_cmd = app.add_subcommand("parse", "Parse the Liberty file and write JSON to a
file");
00271     parse_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00272         ->check(CLI::ExistingFile)
00273         ->required();
00274     parse_cmd->add_option("-l,--log", log_file_name,
00275         "Specify the log file name. Default: <basename>.parse.log");
00276     parse_cmd->callback([&] {
00277         printInfo();
00278         // Check if multi files
00279         if (library_paths.size() > 1) {
00280             spdlog::info("Running sequential parsing for {} files.", library_paths.size());
00281             spdlog::info("Each library will write to its own log file.");
00282             // Sequential parsing
00283             for (const auto &library_path : library_paths) {
00284                 parseLibFile(library_path, log_file_name = "");
00285             }
00286         } else {
00287             parseLibFile(library_paths[0], log_file_name);
00288         }
00289     });
00290
00291     // Add subcommand for mono check mode
00292     bool is_slew = false;
00293     CLI::App *mono_cmd = app.add_subcommand("mono", "Check the monotonicity of timing arc values");
00294     mono_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00295         ->check(CLI::ExistingFile)
00296         ->required();
00297     mono_cmd->add_option("-l,--log", log_file_name,
00298         "Specify the log file name. Default: <basename>.mono.log");
00299     mono_cmd->add_flag("-s,--slew", is_slew, "Specify that monotonicity checks also include input
slew.");
00300     mono_cmd->callback([&] {
00301         printInfo();
00302         // Check if multi files
00303         if (library_paths.size() > 1) {
00304             spdlog::info("Running multi-threaded monotonicity check for {} files.", library_paths.size());
00305             spdlog::info("Each thread will write to its own log file.");
00306
00307             // Sequential json file check
00308             for (const auto &library_path : library_paths) {
00309                 if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() + ".json")) {
00310                     spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00311                     parseLibFile(library_path, log_file_name = "");
00312                 }
00313             }
00314             spdlog::info("All JSON files prepared.");
00315             // Parallel monotonicity check
00316             std::vector<std::thread> threads;
00317             for (const auto &library_path : library_paths) {
00318                 threads.emplace_back(monoCheckLibFile, library_path, log_file_name = "", is_slew);
00319             }
00320             for (auto &thread : threads) {
00321                 thread.join();
00322             }
00323             spdlog::info("All threads completed.");
00324         } else {
00325             monoCheckLibFile(library_paths[0], log_file_name, is_slew);

```

```

00326     }
00327 });
00328
00329 // Add subcommand for compare mode
00330 std::string ref_lib, comp_lib, report_file_name;
00331 CLI::App *compare_cmd = app.add_subcommand(
00332     "compare", "Compare the comparison library against the reference library and report
differences");
00333 compare_cmd->add_option("--ref", ref_lib, "Specify the reference library file")
00334     ->check(CLI::ExistingFile)
00335     ->required();
00336 compare_cmd->add_option("--comp", comp_lib, "Specify the comparison library file")
00337     ->check(CLI::ExistingFile)
00338     ->required();
00339 double abstol = 0.002;
00340 compare_cmd->add_option("--abstol", abstol,
00341     "Specify the absolute tolerance for comparison. Default: 0.002ns");
00342 double reltol = 0.02;
00343 compare_cmd->add_option("--reltol", reltol,
00344     "Specify the relative tolerance for comparison. Default: 0.02/2.0%");
00345 compare_cmd->add_option("--report", report_file_name,
00346     "Specify the report file name. Default: <comp_lib>.cmp.md");
00347 compare_cmd->callback([&] {
00348     printInfo();
00349     compareLibFiles(ref_lib, comp_lib, reltol, abstol, report_file_name);
00350 });
00351
00352 // Add subcommand for supercell generation
00353 int chain_length = 1;
00354 CLI::App *supercell_cmd =
00355     app.add_subcommand("supercell", "Generate supercells for the given Liberty file");
00356 supercell_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00357     ->check(CLI::ExistingFile)
00358     ->required();
00359 supercell_cmd->add_option("-l,--log", log_file_name,
00360     "Specify the log file name. Default: <basename>.supercell.log");
00361 supercell_cmd->add_option("-c,--chain", chain_length,
00362     "Specify the chain length for supercell generation. Default: 1");
00363 std::vector<std::string> cell_names = {}; // "CMPE42D1" "AN2D0", "DFQD1"
00364 supercell_cmd->add_option("--cells", cell_names, "Specify the cell names to generate supercells
for");
00365 supercell_cmd->callback([&] {
00366     printInfo();
00367     // Check if multi files
00368     if (library_paths.size() > 1) {
00369         spdlog::info("Running multi-threaded supercell generation for {} files.", library_paths.size());
00370         spdlog::info("Each library will write to its own log file.");
00371     }
00372     // Sequential json file check
00373     for (const auto &library_path : library_paths) {
00374         if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() + ".json")) {
00375             spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00376             parseLibFile(library_path, log_file_name = "");
00377         }
00378     }
00379     spdlog::info("All JSON files prepared.");
00380     // Parallel supercell generation
00381     std::vector<std::thread> threads;
00382     for (const auto &library_path : library_paths) {
00383         threads.emplace_back(supercellLibFile, library_path, log_file_name = "", chain_length,
00384             cell_names);
00385     }
00386     for (auto &thread : threads) {
00387         thread.join();
00388     }
00389     spdlog::info("All threads completed.");
00390 } else {
00391     supercellLibFile(library_paths[0], log_file_name, chain_length, cell_names);
00392 }
00393 });
00394
00395 // Add subcommand for zlibboost
00396 CLI::App *zlibboost_cmd =
00397     app.add_subcommand("zlibboost", "ZlibBoost - Multi-threaded Library Processing Tool");
00398 std::string config_dir = "/home/songzx/Projects/zlibboost/config.tcl";
00399 std::string python_dir = "/home/guocj/anaconda3/envs/myenv/bin/python";
00400 std::string main_py_dir = "/home/songzx/Projects/zlibboost/zlibboost.py";
00401 zlibboost_cmd->add_option(
00402     "-c, --config", config_dir,
00403     "Specify the configuration TCL file. Default: /home/songzx/Projects/zlibboost/config.tcl");
00404 zlibboost_cmd->add_option(
00405     "--python", python_dir,
00406     "Specify the python directory. Default: /home/guocj/anaconda3/envs/myenv/bin/python");
00407 zlibboost_cmd->add_option(
00408     "--main", main_py_dir,
00409     "Specify the main python script directory. Default:
/home/songzx/Projects/zlibboost/zlibboost.py");

```

```

00410 zlibboost_cmd->callback([&] {
00411     printInfo();
00412     // Run the ZlibBoost tool
00413     std::string command = python_dir + " " + main_py_dir + " -c " + config_dir;
00414     spdlog::info("Running ZlibBoost with command: '{}'", command);
00415     int ret = std::system(command.c_str());
00416     if (ret == 0) {
00417         spdlog::info("ZlibBoost completed successfully.");
00418     } else {
00419         spdlog::error("ZlibBoost failed with return code: {}", ret);
00420     }
00421 });
00422
00423 // Add subcommand for clear
00424 CLI::App *clear_cmd =
00425     app.add_subcommand("clear", "Clear the log, JSON, map, markdown files in this directory");
00426 clear_cmd->callback([&] {
00427     printInfo();
00428     std::filesystem::path current_dir = std::filesystem::current_path();
00429     for (const auto &entry : std::filesystem::directory_iterator(current_dir)) {
00430         if (entry.path().extension() == ".log" || entry.path().extension() == ".json" ||
00431             entry.path().extension() == ".map" || entry.path().extension() == ".md") {
00432             spdlog::info("Removing file: '{}', entry.path().string()");
00433             std::filesystem::remove(entry.path());
00434         }
00435     }
00436     spdlog::info("All log, JSON, map, markdown files cleared.");
00437 });
00438
00439 // Add subcommand for Verilog generation
00440 CLI::App *verilog_cmd = app.add_subcommand("verilog", "Generate Verilog file for given Liberty
file");
00441 verilog_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00442     ->check(CLI::ExistingFile)
00443     ->required();
00444 verilog_cmd->add_option("-l,--log", log_file_name,
00445     "Specify the log file name. Default: <basename>.verilog.log");
00446 verilog_cmd->add_option("-c,--chain", chain_length,
00447     "Specify the chain length for verilog generation. Default: 1");
00448 verilog_cmd->add_option("--cells", cell_names, "Specify the cell names to generate Verilog for");
00449 verilog_cmd->callback([&] {
00450     printInfo();
00451     // Check if multi files
00452     if (library_paths.size() > 1) {
00453         spdlog::info("Running multi-threaded Verilog generation for {} files.", library_paths.size());
00454         spdlog::info("Each library will write to its own log file.");
00455
00456         // Sequential json file check
00457         for (const auto &library_path : library_paths) {
00458             if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() + ".json")) {
00459                 spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00460                 parseLibFile(library_path, log_file_name = "");
00461             }
00462         }
00463         spdlog::info("All JSON files prepared.");
00464         // Parallel Verilog generation
00465         std::vector<std::thread> threads;
00466         for (const auto &library_path : library_paths) {
00467             threads.emplace_back(verilogLibFile, library_path, log_file_name = "", chain_length,
cell_names);
00468         }
00469         for (auto &thread : threads) {
00470             thread.join();
00471         }
00472         spdlog::info("All threads completed.");
00473     } else {
00474         verilogLibFile(library_paths[0], log_file_name, chain_length, cell_names);
00475     }
00476 });
00477
00478 // Add subcommand for SPICE generation
00479 CLI::App *spice_cmd = app.add_subcommand("spice", "Generate SPICE file for given Liberty file");
00480 spice_cmd->add_option("library_path", library_paths, "Specify the library file to process")
00481     ->check(CLI::ExistingFile)
00482     ->required();
00483 spice_cmd->add_option("-l,--log", log_file_name,
00484     "Specify the log file name. Default: <basename>.spice.log");
00485 spice_cmd->add_option("--cells", cell_names, "Specify the cell names to generate SPICE for");
00486 spice_cmd->callback([&] {
00487     printInfo();
00488     // Check if multi files
00489     if (library_paths.size() > 1) {
00490         spdlog::info("Running multi-threaded SPICE generation for {} files.", library_paths.size());
00491         spdlog::info("Each library will write to its own log file.");
00492
00493         // Sequential json file check
00494         for (const auto &library_path : library_paths) {

```



```

00495         if (!std::filesystem::exists(std::filesystem::path(library_path).stem().string() + ".json")) {
00496             spdlog::info("JSON file not found for '{}'. Parsing Liberty file first.", library_path);
00497             parseLibFile(library_path, log_file_name = "");
00498         }
00499     }
00500     spdlog::info("All JSON files prepared.");
00501     // Parallel SPICE generation
00502     std::vector<std::thread> threads;
00503     for (const auto &library_path : library_paths) {
00504         // threads.emplace_back(spiceLibFile, library_path, log_file_name = "", cell_names);
00505     }
00506     for (auto &thread : threads) {
00507         thread.join();
00508     }
00509     spdlog::info("All threads completed.");
00510 } else {
00511     // spiceLibFile(library_paths[0], log_file_name, cell_names);
00512 }
00513 });
00514
00515 // Add subcommand for funtional equivalence check
00516 CLI::App *func_cmd =
00517     app.add_subcommand("func", "Check functional equivalence of two Liberty files or Verilog
files");
00518     std::string ref_file, comp_file;
00519     func_cmd->add_option("--ref", ref_file, "Specify the reference Liberty or Verilog file")
00520         ->check(CLI::ExistingFile)
00521         ->required();
00522     func_cmd->add_option("--comp", comp_file, "Specify the comparison Liberty or Verilog file")
00523         ->check(CLI::ExistingFile)
00524         ->required();
00525     func_cmd->add_option("--cells", cell_names,
00526         "Specify the cell names to check functional equivalence for");
00527     func_cmd->callback([&] {
00528         printInfo();
00529         funcLibFile(ref_file, comp_file, cell_names);
00530     });
00531
00532     CLI11_PARSE(app, argc, argv);
00533
00534     // End of program
00535     char hostname[256];
00536     gethostname(hostname, sizeof(hostname));
00537
00538     auto now = std::chrono::system_clock::now();
00539     auto time_now = std::chrono::system_clock::to_time_t(now);
00540     std::stringstream ss;
00541     ss << std::put_time(std::localtime(&time_now), "%c");
00542
00543     spdlog::info("ZlibValidation exited on '{}' at {}", hostname, ss.str());
00544     return 0;
00545 }

```

6.34 src/verilog_utils.cpp File Reference

```

#include <unordered_set>
#include "spdlog/spdlog.h"
#include "verilog_utils.hpp"
Include dependency graph for verilog_utils.cpp:

```

Functions

- void [getAST](#) (const std::string &verilog_file, const std::string &cell)

6.34.1 Function Documentation

6.34.1.1 getAST()

```
void getAST (
    const std::string & verilog_file,
    const std::string & cell )
```

Definition at line 408 of file [verilog_utils.cpp](#).

Here is the call graph for this function: Here is the caller graph for this function:

6.35 verilog_utils.cpp

[Go to the documentation of this file.](#)

```
00001 #include <unordered_set>
00002
00003 #include "spdlog/spdlog.h"
00004
00005 #include "verilog_utils.hpp"
00006
00007 // Adding a generic handler method to record all visited nodes and increment the depth counter
00008 void VerilogVisitor::handle(const slang::syntax::SyntaxNode &node) {
00009     std::string indent(depth_ * 2, ' ');
00010     spdlog::debug("{}Node: {}", indent, slang::syntax::toString(node.kind));
00011
00012     // Increase depth, visit child nodes, then decrease depth
00013     depth_++;
00014     visitDefault(node);
00015     depth_--;
00016 }
00017
00018 // Handle module declarations
00019 void VerilogVisitor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00020     std::string indent(depth_ * 2, ' ');
00021
00022     if (module.header && module.header->name.valid()) {
00023         std::string_view moduleName = module.header->name.valueText();
00024         spdlog::info("{}Module Name: {}", indent, moduleName);
00025
00026         inTargetModule_ = !targetCell_.empty() && moduleName == targetCell_;
00027         if (!inTargetModule_) {
00028             return;
00029         }
00030
00031         // If a specific module name is specified, check if it matches
00032         if (inTargetModule_) {
00033             spdlog::info("{}Found target module: {}", indent, targetCell_);
00034             // Print the module ports
00035             if (module.header->ports) {
00036                 spdlog::info("{}Ports: {}", indent, module.header->ports->toString());
00037             }
00038         }
00039     }
00040
00041     // Continue processing child nodes
00042     depth_++;
00043     visitDefault(module);
00044     depth_--;
00045 }
00046
00047 // Handle port declarations
00048 void VerilogVisitor::handle(const slang::syntax::PortDeclarationSyntax &portDecl) {
00049     if (!inTargetModule_) {
00050         return;
00051     }
00052
00053     std::string indent = std::string(depth_ * 2, ' ');
00054
00055     // Get port direction
00056     std::string direction = "unknown";
00057     if (portDecl.header) {
00058         // Try to convert the header to different types of port headers
00059         if (auto *varPort = portDecl.header->as_if<slang::syntax::VariablePortHeaderSyntax>()) {
00060             if (varPort->direction) {
00061                 direction = varPort->direction.valueText();
00062             }
00063         } else if (auto *netPort = portDecl.header->as_if<slang::syntax::NetPortHeaderSyntax>()) {
```

```

00064         if (netPort->direction) {
00065             direction = netPort->direction.valueText();
00066         }
00067     }
00068
00069     // Get port name
00070     for (const auto &declarator : portDecl.declarators) {
00071         if (declarator->name.valid()) {
00072             std::string_view portName = declarator->name.valueText();
00073             spdlog::info("{}Port Name: {} ({})", indent, portName, direction);
00074         }
00075     }
00076 }
00077 }
00078
00079 // Handle hierarchical instantiation (module instance)
00080 void VerilogVisitor::handle(const slang::syntax::HierarchyInstantiationSyntax &hierarchyInst) {
00081     if (!inTargetModule_) {
00082         return;
00083     }
00084
00085     std::string indent(depth_ * 2, ' ');
00086
00087     if (hierarchyInst.type.valid()) {
00088         std::string_view instType = hierarchyInst.type.valueText();
00089         spdlog::info("{}Hierarchy Instance Type: {}", indent, instType);
00090
00091         // Safely print parameter values (if any)
00092         try {
00093             if (hierarchyInst.parameters) {
00094                 spdlog::info("{}Parameters:", indent);
00095                 for (const auto &param : hierarchyInst.parameters->parameters) {
00096                     if (!param)
00097                         continue; // Null pointer check
00098
00099                     if (auto *orderedParam = param->as_if<slang::syntax::OrderedParamAssignmentSyntax>()) {
00100                         if (orderedParam->expr) {
00101                             spdlog::info("{} Parameter: {}", indent, orderedParam->expr->toString());
00102                         }
00103                     } else if (auto *namedParam = param->as_if<slang::syntax::NamedParamAssignmentSyntax>()) {
00104                         if (namedParam->name.valid() && namedParam->expr) {
00105                             spdlog::info("{} Parameter {}: {}", indent, namedParam->name.valueText(),
00106                                     namedParam->expr->toString());
00107                         }
00108                     }
00109                 }
00110             }
00111         } catch (const std::exception &e) {
00112             spdlog::warn("{}Exception while processing parameters: {}", indent, e.what());
00113         }
00114
00115         // Safely handle instances
00116         try {
00117             for (const auto &instance : hierarchyInst.instances) {
00118                 if (!instance) {
00119                     spdlog::warn("{}Invalid instance", indent);
00120                     continue;
00121                 } else if (!instance->decl) {
00122                     spdlog::warn("{}Invalid instance declaration", indent);
00123                     // continue;
00124                 } else {
00125                     try {
00126                         if (instance->decl->name.valid()) {
00127                             std::string_view instName = instance->decl->name.valueText();
00128                             spdlog::info("{}Instance Name: {}", indent, instName);
00129                         }
00130                     } catch (...) {
00131                         spdlog::debug("{}Error printing name", indent);
00132                     }
00133
00134                     // Safely print dimensions
00135                     try {
00136                         if (!instance->decl->dimensions.empty()) {
00137                             spdlog::info("{} Dimensions: {}", indent, instance->decl->dimensions.toString());
00138                         }
00139                     } catch (...) {
00140                         spdlog::debug("{}Error printing dimensions", indent);
00141                     }
00142                 }
00143             }
00144
00145             // Safely print port connections
00146             spdlog::info("{} Port connections:", indent);
00147             for (const auto &conn : instance->connections) {
00148                 if (!conn) {
00149                     spdlog::warn("{} Null connection", indent);
00150                     continue; // Null pointer check
00151                 }
00152             }
00153         }
00154     }

```

```

00151
00152     try {
00153         // Use as_if method for safe type conversion
00154         if (auto *ordered = conn->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00155             if (ordered->expr) {
00156                 spdlog::info("{}    Ordered connection: {}", indent, ordered->expr->toString());
00157             } else {
00158                 spdlog::info("{}    Ordered connection: <empty>", indent);
00159             }
00160         } else if (auto *named = conn->as_if<slang::syntax::NamedPortConnectionSyntax>()) {
00161             if (named->name.valid()) {
00162                 std::string exprStr = named->expr ? named->expr->toString() : "<empty>";
00163                 spdlog::info("{}    .{}({})", indent, named->name.valueText(), exprStr);
00164             }
00165         } else if (conn->as_if<slang::syntax::EmptyPortConnectionSyntax>()) {
00166             spdlog::info("{}    <empty connection>", indent);
00167         } else if (conn->as_if<slang::syntax::WildcardPortConnectionSyntax>()) {
00168             spdlog::info("{}    .* (wildcard connection)", indent);
00169         } else {
00170             spdlog::info("{}    Unknown connection type: {}", indent,
00171                 slang::syntax::toString(conn->kind));
00172         }
00173     } catch (const std::exception &e) {
00174         spdlog::warn("{}Exception processing connection: {}", indent, e.what());
00175     } catch (...) {
00176         spdlog::warn("{}Unknown exception processing connection", indent);
00177     }
00178 }
00179 }
00180 } catch (const std::exception &e) {
00181     spdlog::warn("{}Exception while processing instances: {}", indent, e.what());
00182 }
00183 }
00184 // Continue processing child nodes
00185 depth_++;
00186 visitDefault(hierarchyInst);
00187 depth--;
00188 }
00189 }
00190
00191 // Handle primitive gate instantiation
00192 void VerilogVisitor::handle(const slang::syntax::PrimitiveInstantiationSyntax &primitiveInst) {
00193     if (!inTargetModule_) {
00194         return;
00195     }
00196
00197     std::string indent(depth_ * 2, ' ');
00198
00199     // Get primitive gate type
00200     if (primitiveInst.type.valid()) {
00201         std::string_view gateType = primitiveInst.type.valueText();
00202         spdlog::info("{}Primitive Gate: {}", indent, gateType);
00203
00204         // Print delay information (if any)
00205         if (primitiveInst.delay) {
00206             spdlog::info("{}Delay: {}", indent, primitiveInst.delay->toString());
00207         }
00208
00209         // Print strength information (if any)
00210         if (primitiveInst.strength) {
00211             spdlog::info("{}Strength: {}", indent, primitiveInst.strength->toString());
00212         }
00213
00214         // Print each instance
00215         for (const auto &instance : primitiveInst.instances) {
00216             // Primitive gates usually don't have names, but if they do, print them
00217             if (instance->decl && instance->decl->name.valid()) {
00218                 std::string_view instName = instance->decl->name.valueText();
00219                 spdlog::info("{} Gate instance: {}", indent, instName);
00220             }
00221
00222             // Print connections
00223             spdlog::info("{} Gate connections:", indent);
00224             for (size_t i = 0; i < instance->connections.size(); ++i) {
00225                 const auto &conn = instance->connections[i];
00226                 // For gate-level instantiation, connections are usually ordered
00227                 if (auto *ordered = conn->as_if<slang::syntax::OrderedPortConnectionSyntax>()) {
00228                     if (ordered->expr) {
00229                         // The first is usually the output, the rest are inputs
00230                         std::string portType = (i == 0) ? "output" : "input";
00231                         spdlog::info("{}    {} {}: {}", indent, portType, i, ordered->expr->toString());
00232                     }
00233                 }
00234             }
00235         }
00236     }
00237     // Continue to traverse deeper when specific standard gate type

```

```

00238 // Create a set of safe standard gate types
00239 static const std::unordered_set<std::string_view> safeGateTypes = {
00240     "and", "or", "nand", "nor", "xor", "xnor", "not", "buf",
"bufif0",
00241     "bufif1", "notif0", "notif1", "pullup", "pulldown", "cmos", "rcmos", "nmos", "pmos",
00242     "rnmos", "rpmos", "tran", "rtran", "tranif0", "tranif1", "rtranif0", "rtranif1"};
00243
00244 if (safeGateTypes.find(gateType) != safeGateTypes.end()) {
00245     // Continue processing child nodes
00246     depth_++;
00247     visitDefault(primitiveInst);
00248     depth_--;
00249 } else {
00250     spdlog::warn("{}Skipping deeper traversal of primitive: {}", indent, gateType);
00251 }
00252
00253 } else {
00254     spdlog::warn("{}Primitive instantiation missing gate type", indent);
00255 }
00256 }
00257
00258 // Handle specify block
00259 void VerilogVisitor::handle(const slang::syntax::SpecifyBlockSyntax &specifyBlock) {
00260     if (!inTargetModule_) {
00261         return;
00262     }
00263
00264     std::string indent(depth_ * 2, ' ');
00265     spdlog::info("{}Specify Block:", indent);
00266
00267     // Iterate through all path declarations
00268     for (const auto &item : specifyBlock.items) {
00269         if (auto *pathDecl = item->as_if<slang::syntax::PathDeclarationSyntax>()) {
00270             if (pathDecl->desc) {
00271                 std::string pathSrc;
00272                 std::string pathDst;
00273
00274                 // Get path source
00275                 if (!pathDecl->desc->inputs.empty()) {
00276                     // Get the first input
00277                     if (auto *identifier =
00278                         pathDecl->desc->inputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00279                         if (identifier->identifier.valid()) {
00280                             pathSrc = identifier->identifier.valueText();
00281                         }
00282                     }
00283                 }
00284
00285                 // Get path destination
00286                 if (pathDecl->desc->suffix) {
00287                     if (auto *simpleSuffix =
00288                         pathDecl->desc->suffix->as_if<slang::syntax::SimplePathSuffixSyntax>()) {
00289                         if (!simpleSuffix->outputs.empty()) {
00290                             if (auto *identifier =
00291                                 simpleSuffix->outputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00292                                 if (identifier->identifier.valid()) {
00293                                     pathDst = identifier->identifier.valueText();
00294                                 }
00295                             }
00296                         } else if (auto *edgeSuffix = pathDecl->desc->suffix
00297                             ->as_if<slang::syntax::EdgeSensitivePathSuffixSyntax>()) {
00298                             if (!edgeSuffix->outputs.empty()) {
00299                                 if (auto *identifier =
00300                                     edgeSuffix->outputs[0]->as_if<slang::syntax::IdentifierNameSyntax>()) {
00301                                     if (identifier->identifier.valid()) {
00302                                         pathDst = identifier->identifier.valueText();
00303                                     }
00304                                 }
00305                             }
00306                         }
00307                     }
00308                 }
00309
00310                 // Construct path string
00311                 std::string pathStr = pathSrc + " => " + pathDst;
00312
00313                 // Get path delay
00314                 std::string delayStr = "(";
00315                 for (size_t i = 0; i < pathDecl->delays.size(); ++i) {
00316                     if (i > 0)
00317                         delayStr += ", ";
00318                     delayStr += pathDecl->delays[i]->toString();
00319                 }
00320                 delayStr += ")";
00321
00322                 spdlog::info("{} Path: {} = {}", indent, pathStr, delayStr);
00323             }
00324         }
00325     }

```

```

00323     }
00324     // Can add handling for ConditionalPathDeclarationSyntax and IfNonePathDeclarationSyntax
00325     else if (auto *condPath = item->as_if<slang::syntax::ConditionalPathDeclarationSyntax>()) {
00326         if (condPath->path && condPath->predicate) {
00327             spdlog::info("{} Conditional Path (if {})", indent, condPath->predicate->toString());
00328             // Recursively handle internal path
00329             handle(*condPath->path);
00330         }
00331     } else if (auto *ifNonePath = item->as_if<slang::syntax::IfNonePathDeclarationSyntax>()) {
00332         if (ifNonePath->path) {
00333             spdlog::info("{} If-None Path", indent);
00334             // Recursively handle internal path
00335             handle(*ifNonePath->path);
00336         }
00337     }
00338 }
00339
00340 // Continue processing child nodes
00341 depth_++;
00342 visitDefault(specifyBlock);
00343 depth--;
00344 }
00345 }
00346
00347 // Handle module declarations, only keep the target module
00348 void CellExtractor::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00349     if (module.header && module.header->name.valid()) {
00350         std::string_view moduleName = module.header->name.valueText();
00351
00352         // If it is the target module, mark as found, otherwise remove it
00353         if (!targetCell_.empty() && moduleName == targetCell_) {
00354             foundTarget_ = true;
00355             // Do not modify, keep this module
00356         } else {
00357             // Remove non-target modules
00358             remove(module);
00359         }
00360     }
00361 }
00362
00363 // Get result
00364 bool CellExtractor::foundTargetCell() const { return foundTarget_; }
00365
00366 // Handle module declarations
00367 void CellPrinter::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00368     if (module.header && module.header->name.valid()) {
00369         std::string_view moduleName = module.header->name.valueText();
00370
00371         // Check if it is the target module
00372         if (!targetCell_.empty() && moduleName == targetCell_) {
00373             foundTarget_ = true;
00374
00375             // Print module definition
00376             out_ << "\timescale lns/10ps\n";
00377             out_ << module.toString();
00378
00379             return; // Do not traverse further
00380         }
00381     }
00382
00383     // Continue traversing other modules
00384     visitDefault(module);
00385 }
00386
00387 void ModuleRewriter::handle(const slang::syntax::SyntaxNode &node) {
00388     std::string indent(depth_ * 2, ' ');
00389     logger_>debug("{}Node: {}", indent, slang::syntax::toString(node.kind));
00390
00391     // Continue processing child nodes
00392     depth_++;
00393     visitDefault(node);
00394     depth--;
00395 }
00396
00397 void ModuleRewriter::handle(const slang::syntax::ModuleDeclarationSyntax &module) {
00398     logger_>info("Processing module: {}", module.header->name.valueText());
00399     // auto &newNode = parse("\n" + this->cellName_ + " I" + this->moduleName_ + " (");
00400     // logger_>debug("New node: {}", newNode.toString());
00401     for (int i = 0; i < instance_count_ - 1; i++) {
00402         auto &newDataNode = parse("\n wire OP_" + std::to_string(i) + ";");
00403         insertAtBack(module.members, newDataNode);
00404     }
00405 }
00406
00407 void getAST(const std::string &verilog_file, const std::string &cell) {
00408     try {

```

```

00410     spdlog::info("Starting get AST from Verilog file: '{}'", verilog_file);
00411     auto result = slang::syntax::SyntaxTree::fromFile(verilog_file);
00412     if (result) {
00413         spdlog::info("Successfully parsed Verilog file.");
00414
00415         try {
00416             // First, use the visitor to print basic information
00417             VerilogVisitor visitor(cell);
00418             visitor.visit(result.value()->root());
00419
00420             // If a target cell is specified, use the rewriter to extract the relevant code
00421             if (!cell.empty()) {
00422                 // Create a rewriter to only keep the target cell related code
00423                 CellExtractor extractor(cell);
00424                 auto extractedTree = extractor.transform(result.value());
00425
00426                 if (extractor.foundTargetCell()) {
00427                     // Use SyntaxPrinter to print the extracted code
00428                     std::string extractedCode = slang::syntax::SyntaxPrinter::printFile(*extractedTree);
00429
00430                     // Save to a file named after the cell name
00431                     std::string outputFile = cell + ".v";
00432                     std::ofstream cellOut(outputFile);
00433                     if (cellOut) {
00434                         cellOut << extractedCode;
00435                         cellOut.close();
00436                         spdlog::info("Extracted '{}' cell code to '{}'", cell, outputFile);
00437                     } else {
00438                         spdlog::error("Failed to write extracted cell code to '{}'", outputFile);
00439                     }
00440                 } else {
00441                     spdlog::warn("Target cell '{}' not found in the Verilog file", cell);
00442                 }
00443             }
00444
00445             spdlog::info("Print full source code to 'full_source_code.v'");
00446             // Optionally save the entire syntax tree
00447             std::string fullOutput = slang::syntax::SyntaxPrinter::printFile(*result.value());
00448             std::ofstream out("full_source_code.v");
00449             out << fullOutput;
00450             out.close();
00451
00452             spdlog::info("Print target cell code to 'cell_code.v'");
00453             // Use CellPrinter to print the target cell's code
00454             std::ofstream cellOut("cell_code.v");
00455             CellPrinter cellPrinter(cell, cellOut);
00456             cellPrinter.visit(result.value()->root());
00457             cellOut.close();
00458
00459             } catch (const std::exception &e) {
00460                 spdlog::error("Exception during AST traversal: {}", e.what());
00461             } catch (...) {
00462                 spdlog::error("Unknown exception during AST traversal");
00463             }
00464         } else {
00465             spdlog::error("Error parsing Verilog file.");
00466         }
00467     } catch (const std::exception &e) {
00468         spdlog::error("Exception during Verilog parsing: {}", e.what());
00469     } catch (...) {
00470         spdlog::error("Unknown exception during Verilog parsing");
00471     }
00472 }

```


Index

- ~GroupsIterator
 - GroupsIterator, [19](#)
- ~LibAttribute
 - LibAttribute, [22](#)
- ~LibFile
 - LibFile, [26](#)
- ~LibGroup
 - LibGroup, [34](#)
- ~ValuesIterator
 - ValuesIterator, [48](#)
- abstol_
 - LibraryComparator, [40](#)
- APP_AUTHOR
 - version.h, [93](#)
- APP_CONTACT
 - version.h, [93](#)
- APP_NAME
 - version.h, [93](#)
- APP_VERSION
 - version.h, [93](#)
- APP_VERSION_MAJOR
 - version.h, [93](#)
- APP_VERSION_MINOR
 - version.h, [93](#)
- APP_VERSION_PATCH
 - version.h, [94](#)
- attr_
 - LibAttribute, [23](#)
- AttributesIterator, [15](#)
- b
 - si2drExprT, [44](#)
- basename_
 - LibFile, [31](#)
- bool_
 - ValuesIterator, [48](#)
- BUILD_TIMESTAMP
 - version.h, [94](#)
- CellExtractor, [15](#)
 - CellExtractor, [16](#)
 - foundTarget_, [16](#)
 - foundTargetCell, [16](#)
 - handle, [16](#)
 - targetCell_, [16](#)
- cellName_
 - ModuleRewriter, [42](#)
- CellPrinter, [17](#)
 - CellPrinter, [17](#)
- foundTarget_, [18](#)
- handle, [18](#)
- out_, [18](#)
- targetCell_, [18](#)
- checkTimingArcMonotonicity
 - LibFile, [27](#)
- comp_json_
 - LibraryComparator, [40](#)
- comp_lib_path_
 - LibraryComparator, [40](#)
- compare.hpp
 - json, [53](#)
- compareCell
 - LibraryComparator, [37](#)
- compareLibFiles
 - main.cpp, [116](#)
- compareLut
 - LibraryComparator, [37](#)
- comparePin
 - LibraryComparator, [38](#)
- compareTimingArc
 - LibraryComparator, [39](#)
- d
 - si2drExprT, [44](#)
- depth_
 - ModuleRewriter, [43](#)
 - VerilogVisitor, [52](#)
- dim_sizes
 - liberty_value_data, [24](#)
- dimensions
 - liberty_value_data, [24](#)
- end
 - GroupsIterator, [19](#)
 - ValuesIterator, [48](#)
- err_
 - GroupsIterator, [20](#)
 - LibAttribute, [23](#)
 - LibFile, [31](#)
 - LibGroup, [35](#)
 - ValuesIterator, [48](#)
- exprp_
 - ValuesIterator, [49](#)
- filename_
 - LibFile, [31](#)
- filepath_
 - LibFile, [31](#)
- float_

- ValuesIterator, 49
- foundTarget_
 - CellExtractor, 16
 - CellPrinter, 18
- foundTargetCell
 - CellExtractor, 16
- funcLibFile
 - main.cpp, 116
- generateCellJson
 - json_utils.cpp, 100
 - json_utils.hpp, 56
- generateLutJson
 - json_utils.cpp, 101
 - json_utils.hpp, 56
- generatePinJson
 - json_utils.cpp, 101
 - json_utils.hpp, 57
- generatePowerJson
 - json_utils.cpp, 102
 - json_utils.hpp, 58
- generateReport
 - LibraryComparator, 39
- generateTimingJson
 - json_utils.cpp, 103
- get
 - GroupsIterator, 20
- get_function
 - si2dr_liberty.h, 75
- getAST
 - verilog_utils.cpp, 125
 - verilog_utils.hpp, 91
- getAttrs
 - LibGroup, 34
- getBoolean
 - LibAttribute, 22
- getFloat
 - LibAttribute, 22
- getGroups
 - LibGroup, 34
- getInt
 - LibAttribute, 22
- getName
 - LibAttribute, 22
 - LibGroup, 34
- getString
 - LibAttribute, 23
- getType
 - LibGroup, 34
- getValues
 - LibAttribute, 23
- group_
 - GroupsIterator, 20
 - LibGroup, 35
- groups_
 - GroupsIterator, 20
- GroupsIterator, 18
 - ~GroupsIterator, 19
 - end, 19
- err_, 20
- get, 20
- group_, 20
- groups_, 20
- GroupsIterator, 19
- next, 20
- handle
 - CellExtractor, 16
 - CellPrinter, 18
 - ModuleRewriter, 42
 - VerilogVisitor, 51, 52
- i
 - si2drExprT, 45
- include/compare.hpp, 53, 54
- include/iterators.hpp, 54
- include/json_utils.hpp, 55, 59
- include/lib_attribute.hpp, 59
- include/lib_file.hpp, 60
- include/lib_group.hpp, 61
- include/si2dr_liberty.h, 61, 83
- include/verilog_utils.hpp, 90, 91
- include/version.h, 92, 94
- index_info
 - liberty_value_data, 24
- inputPins_
 - ModuleRewriter, 43
- instance_count_
 - ModuleRewriter, 43
- int_
 - ValuesIterator, 49
- inTargetModule_
 - VerilogVisitor, 52
- isComplex
 - LibAttribute, 23
- json
 - compare.hpp, 53
 - json_utils.hpp, 56
 - lib_file.hpp, 60
- json_utils.cpp
 - generateCellJson, 100
 - generateLutJson, 101
 - generatePinJson, 101
 - generatePowerJson, 102
 - generateTimingJson, 103
 - parseStringToVector, 103
- json_utils.hpp
 - generateCellJson, 56
 - generateLutJson, 56
 - generatePinJson, 57
 - generatePowerJson, 58
 - json, 56
- jsonname_
 - LibFile, 31
- left
 - si2drExprT, 45

- lib_file.hpp
 - json, [60](#)
- lib_json_
 - LibFile, [32](#)
- LibAttribute, [21](#)
 - ~LibAttribute, [22](#)
 - attr_, [23](#)
 - err_, [23](#)
 - getBoolean, [22](#)
 - getFloat, [22](#)
 - getInt, [22](#)
 - getName, [22](#)
 - getString, [23](#)
 - getValues, [23](#)
 - isComplex, [23](#)
 - LibAttribute, [21](#)
- liberty_destroy_value_data
 - si2dr_liberty.h, [75](#)
- liberty_get_element
 - si2dr_liberty.h, [76](#)
- liberty_get_values_data
 - si2dr_liberty.h, [76](#)
- liberty_value_data, [24](#)
 - dim_sizes, [24](#)
 - dimensions, [24](#)
 - index_info, [24](#)
 - values, [25](#)
- LibFile, [25](#)
 - ~LibFile, [26](#)
 - basename_, [31](#)
 - checkTimingArcMonotonicity, [27](#)
 - err_, [31](#)
 - filename_, [31](#)
 - filepath_, [31](#)
 - jsonname_, [31](#)
 - lib_json_, [32](#)
 - LibFile, [26](#)
 - libname_, [32](#)
 - logger_, [32](#)
 - loggername_, [32](#)
 - modify, [28](#)
 - mono, [28](#)
 - parse, [28](#)
 - process_, [32](#)
 - read, [29](#)
 - supercell, [30](#)
 - temperature_, [32](#)
 - verilog, [30](#)
 - voltage_, [33](#)
 - writeJsonToFile, [30](#)
- LibGroup, [33](#)
 - ~LibGroup, [34](#)
 - err_, [35](#)
 - getAttrs, [34](#)
 - getGroups, [34](#)
 - getName, [34](#)
 - getType, [34](#)
 - group_, [35](#)
 - LibGroup, [33](#)
- libname_
 - LibFile, [32](#)
- LibraryComparator, [35](#)
 - abstol_, [40](#)
 - comp_json_, [40](#)
 - comp_lib_path_, [40](#)
 - compareCell, [37](#)
 - compareLut, [37](#)
 - comparePin, [38](#)
 - compareTimingArc, [39](#)
 - generateReport, [39](#)
 - LibraryComparator, [36](#)
 - ref_json_, [40](#)
 - ref_lib_path_, [40](#)
 - reitol_, [41](#)
- logger_
 - LibFile, [32](#)
 - ModuleRewriter, [43](#)
- loggername_
 - LibFile, [32](#)
- LONG_DOUBLE
 - si2dr_liberty.h, [64](#)
- main
 - main.cpp, [117](#)
- main.cpp
 - compareLibFiles, [116](#)
 - funcLibFile, [116](#)
 - main, [117](#)
 - monoCheckLibFile, [117](#)
 - parseLibFile, [117](#)
 - printInfo, [118](#)
 - supercellLibFile, [118](#)
 - verilogLibFile, [119](#)
- main_liberty_parser
 - si2dr_liberty.h, [76](#)
- modify
 - LibFile, [28](#)
- moduleName_
 - ModuleRewriter, [43](#)
- ModuleRewriter, [41](#)
 - cellName_, [42](#)
 - depth_, [43](#)
 - handle, [42](#)
 - inputPins_, [43](#)
 - instance_count_, [43](#)
 - logger_, [43](#)
 - moduleName_, [43](#)
 - ModuleRewriter, [42](#)
 - outputPins_, [43](#)
- mono
 - LibFile, [28](#)
- monoCheckLibFile
 - main.cpp, [117](#)
- next
 - GroupsIterator, [20](#)
 - ValuesIterator, [48](#)

- out_
 - CellPrinter, 18
- outputPins_
 - ModuleRewriter, 43
- parse
 - LibFile, 28
- parseLibFile
 - main.cpp, 117
- parseStringToVector
 - json_utils.cpp, 103
- print_version
 - si2dr_liberty.h, 76
- printlnfo
 - main.cpp, 118
- process_
 - LibFile, 32
- read
 - LibFile, 29
- README.md, 94
- ref_json_
 - LibraryComparator, 40
- ref_lib_path_
 - LibraryComparator, 40
- reltol_
 - LibraryComparator, 41
- right
 - si2drExprT, 45
- s
 - si2drExprT, 45
- SI2_ARGS
 - si2dr_liberty.h, 65, 76–82
- SI2_BOOLEAN
 - si2dr_liberty.h, 75
- SI2_DOUBLE
 - si2dr_liberty.h, 75
- SI2_EXPR
 - si2dr_liberty.h, 75
- SI2_FALSE
 - si2dr_liberty.h, 73
- SI2_FLOAT
 - si2dr_liberty.h, 75
- SI2_ITER
 - si2dr_liberty.h, 75
- SI2_LONG
 - si2dr_liberty.h, 75
- SI2_LONG_MAX
 - si2dr_liberty.h, 65
- SI2_LONG_MIN
 - si2dr_liberty.h, 65
- SI2_MAX_VALUETYPE
 - si2dr_liberty.h, 75
- SI2_OID
 - si2dr_liberty.h, 75
- SI2_STRING
 - si2dr_liberty.h, 75
- SI2_TRUE
 - si2dr_liberty.h, 73
- SI2_ULONG_MAX
 - si2dr_liberty.h, 65
- SI2_UNDEFINED_VALUETYPE
 - si2dr_liberty.h, 75
- SI2_VOID
 - si2dr_liberty.h, 75
- si2BooleanEnum
 - si2dr_liberty.h, 72
- si2BooleanT
 - si2dr_liberty.h, 67
- si2DoubleT
 - si2dr_liberty.h, 67
- SI2DR_ATTR
 - si2dr_liberty.h, 74
- SI2DR_BOOLEAN
 - si2dr_liberty.h, 75
- SI2DR_COMPLEX
 - si2dr_liberty.h, 73
- SI2DR_COMPLEXVAL
 - si2dr_liberty.h, 74
- SI2DR_DEFINE
 - si2dr_liberty.h, 74
- SI2DR_EXPR
 - si2dr_liberty.h, 75
- SI2DR_EXPR_OP_ADD
 - si2dr_liberty.h, 74
- SI2DR_EXPR_OP_DIV
 - si2dr_liberty.h, 74
- SI2DR_EXPR_OP_EXP
 - si2dr_liberty.h, 74
- SI2DR_EXPR_OP_LOG10
 - si2dr_liberty.h, 74
- SI2DR_EXPR_OP_LOG2
 - si2dr_liberty.h, 74
- SI2DR_EXPR_OP_MUL
 - si2dr_liberty.h, 74
- SI2DR_EXPR_OP_PAREN
 - si2dr_liberty.h, 74
- SI2DR_EXPR_OP_SUB
 - si2dr_liberty.h, 74
- SI2DR_EXPR_VAL
 - si2dr_liberty.h, 74
- SI2DR_FALSE
 - si2dr_liberty.h, 65
- SI2DR_FLOAT64
 - si2dr_liberty.h, 75
- SI2DR_GROUP
 - si2dr_liberty.h, 74
- SI2DR_INT32
 - si2dr_liberty.h, 75
- SI2DR_INTERNAL_SYSTEM_ERROR
 - si2dr_liberty.h, 73
- SI2DR_INVALID_ATTRTYPE
 - si2dr_liberty.h, 73
- SI2DR_INVALID_NAME
 - si2dr_liberty.h, 73
- SI2DR_INVALID_OBJECTTYPE

- si2dr_liberty.h, 73
- SI2DR_INVALID_VALUE
 - si2dr_liberty.h, 73
- si2dr_liberty.h
 - get_function, 75
 - liberty_destroy_value_data, 75
 - liberty_get_element, 76
 - liberty_get_values_data, 76
 - LONG_DOUBLE, 64
 - main_liberty_parser, 76
 - print_version, 76
 - SI2_ARGS, 65, 76–82
 - SI2_BOOLEAN, 75
 - SI2_DOUBLE, 75
 - SI2_EXPR, 75
 - SI2_FALSE, 73
 - SI2_FLOAT, 75
 - SI2_ITER, 75
 - SI2_LONG, 75
 - SI2_LONG_MAX, 65
 - SI2_LONG_MIN, 65
 - SI2_MAX_VALUETYPE, 75
 - SI2_OID, 75
 - SI2_STRING, 75
 - SI2_TRUE, 73
 - SI2_ULONG_MAX, 65
 - SI2_UNDEFINED_VALUETYPE, 75
 - SI2_VOID, 75
 - si2BooleanEnum, 72
 - si2BooleanT, 67
 - si2DoubleT, 67
 - SI2DR_ATTR, 74
 - SI2DR_BOOLEAN, 75
 - SI2DR_COMPLEX, 73
 - SI2DR_COMPLEXVAL, 74
 - SI2DR_DEFINE, 74
 - SI2DR_EXPR, 75
 - SI2DR_EXPR_OP_ADD, 74
 - SI2DR_EXPR_OP_DIV, 74
 - SI2DR_EXPR_OP_EXP, 74
 - SI2DR_EXPR_OP_LOG10, 74
 - SI2DR_EXPR_OP_LOG2, 74
 - SI2DR_EXPR_OP_MUL, 74
 - SI2DR_EXPR_OP_PAREN, 74
 - SI2DR_EXPR_OP_SUB, 74
 - SI2DR_EXPR_VAL, 74
 - SI2DR_FALSE, 65
 - SI2DR_FLOAT64, 75
 - SI2DR_GROUP, 74
 - SI2DR_INT32, 75
 - SI2DR_INTERNAL_SYSTEM_ERROR, 73
 - SI2DR_INVALID_ATTRTYPE, 73
 - SI2DR_INVALID_NAME, 73
 - SI2DR_INVALID_OBJECTTYPE, 73
 - SI2DR_INVALID_VALUE, 73
 - SI2DR_MAX_ERROR, 73
 - SI2DR_MAX_FLOAT32, 65
 - SI2DR_MAX_FLOAT64, 66
 - SI2DR_MAX_INT32, 66
 - SI2DR_MAX_OBJECTTYPE, 74
 - SI2DR_MAX_STRING_LEN, 66
 - SI2DR_MAX_VALUETYPE, 75
 - SI2DR_MIN_FLOAT32, 66
 - SI2DR_MIN_FLOAT64, 66
 - SI2DR_MIN_INT32, 66
 - SI2DR_NO_ERROR, 73
 - SI2DR_OBJECT_ALREADY_EXISTS, 73
 - SI2DR_OBJECT_NOT_FOUND, 73
 - SI2DR_PIINIT_NOT_CALLED, 73
 - SI2DR_REFERENCE_ERROR, 73
 - SI2DR_SEMANTIC_ERROR, 73
 - SI2DR_SEVERITY_ERR, 74
 - SI2DR_SEVERITY_NOTE, 74
 - SI2DR_SEVERITY_WARN, 74
 - SI2DR_SIMPLE, 73
 - SI2DR_STRING, 75
 - SI2DR_SYNTAX_ERROR, 73
 - SI2DR_TRACE_FILES_CANNOT_BE_OPENED, 73
 - SI2DR_TRUE, 67
 - SI2DR_UNDEFINED_OBJECTTYPE, 74
 - SI2DR_UNDEFINED_VALUETYPE, 75
 - SI2DR_UNUSABLE_OID, 73
 - si2drAttrComplexValIdT, 67
 - si2drAttrIdT, 67
 - si2drAttrSIdT, 67
 - si2drAttrTypeT, 68, 73
 - si2drBooleanT, 68
 - si2drDefaultMessageHandler, 82
 - si2drDefinIdT, 68
 - si2drDefinesIdT, 68
 - si2drErrorT, 68, 73
 - si2drExprT, 68
 - si2drExprTypeT, 68, 73
 - si2drFloat32T, 69
 - si2drFloat64T, 69
 - si2drGroupIdT, 69
 - si2drGroupsIdT, 69
 - si2drInt32T, 69
 - si2drIterIdT, 69
 - si2drMessageHandlerT, 70
 - si2drNamesIdT, 70
 - si2drObjectIdT, 70
 - si2drObjectsIdT, 70
 - si2drObjectTypeT, 70, 74
 - si2drPIGetMessageHandler, 82
 - si2drPISetMessageHandler, 82
 - si2drSeverityT, 70, 74
 - si2drStringT, 71
 - si2drValuesIdT, 71
 - si2drValueTypeT, 71, 75
 - si2drVoidT, 71
 - si2FloatT, 71
 - si2IteratorIdT, 71
 - si2LongT, 72
 - si2ObjectIdT, 72

- si2StringT, [72](#)
- si2TypeEnum, [75](#)
- si2TypeT, [72](#)
- si2VoidT, [72](#)
- usage, [83](#)
- SI2DR_MAX_ERROR
 - si2dr_liberty.h, [73](#)
- SI2DR_MAX_FLOAT32
 - si2dr_liberty.h, [65](#)
- SI2DR_MAX_FLOAT64
 - si2dr_liberty.h, [66](#)
- SI2DR_MAX_INT32
 - si2dr_liberty.h, [66](#)
- SI2DR_MAX_OBJECTTYPE
 - si2dr_liberty.h, [74](#)
- SI2DR_MAX_STRING_LEN
 - si2dr_liberty.h, [66](#)
- SI2DR_MAX_VALUETYPE
 - si2dr_liberty.h, [75](#)
- SI2DR_MIN_FLOAT32
 - si2dr_liberty.h, [66](#)
- SI2DR_MIN_FLOAT64
 - si2dr_liberty.h, [66](#)
- SI2DR_MIN_INT32
 - si2dr_liberty.h, [66](#)
- SI2DR_NO_ERROR
 - si2dr_liberty.h, [73](#)
- SI2DR_OBJECT_ALREADY_EXISTS
 - si2dr_liberty.h, [73](#)
- SI2DR_OBJECT_NOT_FOUND
 - si2dr_liberty.h, [73](#)
- SI2DR_PIINIT_NOT_CALLED
 - si2dr_liberty.h, [73](#)
- SI2DR_REFERENCE_ERROR
 - si2dr_liberty.h, [73](#)
- SI2DR_SEMANTIC_ERROR
 - si2dr_liberty.h, [73](#)
- SI2DR_SEVERITY_ERR
 - si2dr_liberty.h, [74](#)
- SI2DR_SEVERITY_NOTE
 - si2dr_liberty.h, [74](#)
- SI2DR_SEVERITY_WARN
 - si2dr_liberty.h, [74](#)
- SI2DR_SIMPLE
 - si2dr_liberty.h, [73](#)
- SI2DR_STRING
 - si2dr_liberty.h, [75](#)
- SI2DR_SYNTAX_ERROR
 - si2dr_liberty.h, [73](#)
- SI2DR_TRACE_FILES_CANNOT_BE_OPENED
 - si2dr_liberty.h, [73](#)
- SI2DR_TRUE
 - si2dr_liberty.h, [67](#)
- SI2DR_UNDEFINED_OBJECTTYPE
 - si2dr_liberty.h, [74](#)
- SI2DR_UNDEFINED_VALUETYPE
 - si2dr_liberty.h, [75](#)
- SI2DR_UNUSABLE_OID
 - si2dr_liberty.h, [73](#)
- si2drAttrComplexValIdT
 - si2dr_liberty.h, [67](#)
- si2drAttrIdT
 - si2dr_liberty.h, [67](#)
- si2drAttrIdT
 - si2dr_liberty.h, [67](#)
- si2drAttrTypeT
 - si2dr_liberty.h, [68](#), [73](#)
- si2drBooleanT
 - si2dr_liberty.h, [68](#)
- si2drDefaultMessageHandler
 - si2dr_liberty.h, [82](#)
- si2drDefinIdT
 - si2dr_liberty.h, [68](#)
- si2drDefinesIdT
 - si2dr_liberty.h, [68](#)
- si2drErrorT
 - si2dr_liberty.h, [68](#), [73](#)
- si2drExprT, [44](#)
 - b, [44](#)
 - d, [44](#)
 - i, [45](#)
 - left, [45](#)
 - right, [45](#)
 - s, [45](#)
 - si2dr_liberty.h, [68](#)
 - type, [45](#)
 - u, [45](#)
 - valuetype, [46](#)
- si2drExprTypeT
 - si2dr_liberty.h, [68](#), [73](#)
- si2drFloat32T
 - si2dr_liberty.h, [69](#)
- si2drFloat64T
 - si2dr_liberty.h, [69](#)
- si2drGroupIdT
 - si2dr_liberty.h, [69](#)
- si2drGroupsIdT
 - si2dr_liberty.h, [69](#)
- si2drInt32T
 - si2dr_liberty.h, [69](#)
- si2drIterIdT
 - si2dr_liberty.h, [69](#)
- si2drMessageHandlerT
 - si2dr_liberty.h, [70](#)
- si2drNamesIdT
 - si2dr_liberty.h, [70](#)
- si2drObjectIdT
 - si2dr_liberty.h, [70](#)
- si2drObjectsIdT
 - si2dr_liberty.h, [70](#)
- si2drObjectTypeT
 - si2dr_liberty.h, [70](#), [74](#)
- si2drPIGetMessageHandler
 - si2dr_liberty.h, [82](#)
- si2drPISetMessageHandler
 - si2dr_liberty.h, [82](#)

- si2drSeverityT
 - si2dr_liberty.h, [70](#), [74](#)
- si2drStringT
 - si2dr_liberty.h, [71](#)
- si2drValuesIdT
 - si2dr_liberty.h, [71](#)
- si2drValueTypeT
 - si2dr_liberty.h, [71](#), [75](#)
- si2drVoidT
 - si2dr_liberty.h, [71](#)
- si2FloatT
 - si2dr_liberty.h, [71](#)
- si2IteratorIdT
 - si2dr_liberty.h, [71](#)
- si2LongT
 - si2dr_liberty.h, [72](#)
- si2ObjectIdT
 - si2dr_liberty.h, [72](#)
- si2OID, [46](#)
 - v1, [46](#)
 - v2, [46](#)
- si2StringT
 - si2dr_liberty.h, [72](#)
- si2TypeEnum
 - si2dr_liberty.h, [75](#)
- si2TypeT
 - si2dr_liberty.h, [72](#)
- si2VoidT
 - si2dr_liberty.h, [72](#)
- src/compare.cpp, [94](#), [95](#)
- src/iterators.cpp, [99](#)
- src/json_utils.cpp, [100](#), [104](#)
- src/lib_attribute.cpp, [107](#)
- src/lib_file.cpp, [107](#)
- src/lib_group.cpp, [115](#)
- src/main.cpp, [115](#), [120](#)
- src/verilog_utils.cpp, [125](#), [126](#)
- str_
 - ValuesIterator, [49](#)
- supercell
 - LibFile, [30](#)
- supercellLibFile
 - main.cpp, [118](#)
- targetCell_
 - CellExtractor, [16](#)
 - CellPrinter, [18](#)
 - VerilogVisitor, [52](#)
- temperature_
 - LibFile, [32](#)
- type
 - si2drExprT, [45](#)
- u
 - si2drExprT, [45](#)
- usage
 - si2dr_liberty.h, [83](#)
- v1
 - si2OID, [46](#)
- v2
 - si2OID, [46](#)
- values
 - liberty_value_data, [25](#)
- values_
 - ValuesIterator, [49](#)
- ValuesIterator, [47](#)
 - ~ValuesIterator, [48](#)
 - bool_, [48](#)
 - end, [48](#)
 - err_, [48](#)
 - exprp_, [49](#)
 - float_, [49](#)
 - int_, [49](#)
 - next, [48](#)
 - str_, [49](#)
 - values_, [49](#)
 - ValuesIterator, [47](#)
 - vtype_, [49](#)
- valuetype
 - si2drExprT, [46](#)
- verilog
 - LibFile, [30](#)
- verilog_utils.cpp
 - getAST, [125](#)
- verilog_utils.hpp
 - getAST, [91](#)
- verilogLibFile
 - main.cpp, [119](#)
- VerilogVisitor, [50](#)
 - depth_, [52](#)
 - handle, [51](#), [52](#)
 - inTargetModule_, [52](#)
 - targetCell_, [52](#)
 - VerilogVisitor, [50](#)
- version.h
 - APP_AUTHOR, [93](#)
 - APP_CONTACT, [93](#)
 - APP_NAME, [93](#)
 - APP_VERSION, [93](#)
 - APP_VERSION_MAJOR, [93](#)
 - APP_VERSION_MINOR, [93](#)
 - APP_VERSION_PATCH, [94](#)
 - BUILD_TIMESTAMP, [94](#)
- voltage_
 - LibFile, [33](#)
- vtype_
 - ValuesIterator, [49](#)
- writeJsonToFile
 - LibFile, [30](#)